



सत्यमेव जयते
भारत सरकार
Government of India
विद्युत मंत्रालय
Ministry of Power
पूर्वी क्षेत्रीय विद्युत समिति



Eastern Regional Power Committee

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सं/NO. ERPC/EE/OPERATION/2025/1711

दिनांक/DATE: 07.01.2025

सेवा में /To
संलग्न सूची के अनुसार /As per list enclosed.

विषय : 23 दिसंबर 2024 (सोमवार) को कोलकाता में भौतिक रूप से आयोजित 222वीं OCC बैठक का कार्यवृत्त - संबंध में।

Sub: Minutes of 222nd OCC Meeting held on 23.12.2024 (Monday) physically at Kolkata – reg

महोदय/महोदया,
Sir(s)/Madam,

कृपया अपनी जानकारी और आवश्यक कार्रवाई के लिए **23 दिसंबर 2024 (सोमवार) को कोलकाता में 11:00 बजे भौतिक रूप से आयोजित 222वीं ओसीसी बैठक** के संलग्न कार्यवृत्त देखें। यह ईआरपीसी वेबसाइट (www.erpc.gov.in) पर भी उपलब्ध है।

Please find enclosed **Minutes of 222nd OCC Meeting** held on **23.12.2024 (Monday) physically at Kolkata** at 11:00 hrs for your **kind information and necessary action**. The same is also available at ERPC website (www.erpc.gov.in).

टिप्पणियाँ, यदि कोई हों, कृपया यथाशीघ्र इस कार्यालय को अग्रेषित करें।
Observations, if any, may please be forwarded to this office at the earliest.

भवदीय /Yours faithfully

(S.Kejriwal)
SE(Operation)
एसई (ऑपरेशन)

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ERPC:: Kolkata

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मुख्य अभियंता, ओपीएम, सीईए	मुख्य अभियंता, एनपीसी, सीईए	सहायक सचिव,ईआरपीसी
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ईआरपीसी:: कोलकाता



**MINUTES
OF
222nd OCC MEETING**

EASTERN REGIONAL POWER COMMITTEE

Date: 23.12.2024

Time: 11:00Hrs

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EASTERN REGIONAL POWER COMMITTEE

MINUTES OF 222nd OCC MEETING HELD ON 23.12.2024 (MONDAY) AT 11:00 HRS

Member Secretary, ERPC chaired the 222nd OCC meeting. On welcoming all the participants, he outlined the performance of ER grid during **November -2024** and highlighted the following points:

- ❖ In **November-2024**, energy consumption of ER was **13562 MU** which is **almost same as November-2023**.
- ❖ In **November-2024**, Peak demand met of ER was **25883 MW** which is **11.5% more than November-2023**.
- ❖ During **November-2024**, **80.8 %** of time, the grid frequency was in IEGC Band (49.90Hz-50.05Hz).
- ❖ Thermal **PLF** of ER during **November-2024** was **73%**.
- ❖ Some thermal generating units were lauded for maintaining PLF more than **90%** during **November-2024** that are listed below:

Utility	Generating Stations	PLF %
WBPDC	Sagardighi TPS	93
NTPC	North karanpura TPP	95
OPGC	IB valley TPS	93
JITPL	Derang TPP	98
GMR	Kamalanga TPS	93

❖ **Coal stock position:**

☐ Coal stock position (As on 17.12.2024) is as follows:

SL.	Name of States/Power Stns.	% of Actual Stock vis-à-vis Normative Stock
1.	Jharkhand (TVNL)	18%
2.	Odisha/IBTPS	70%
3.	WBPDC	20%(Min.Bandel TPS -13.9%, Max. Sagardighi TPS – 78.5 %)
4.	D.P.L. TPS	0%
5.	DVC	75%(Min. Durgapur Steel TPS-46%, Max Bokaro TPS `A` exp - 135%)
6.	NTPC	75%(Max. Muzaffarpur TPS-114%,North karanpura TPP & Kahalgaon TPS -97% &Min.Talcher STPS - 35%)

- ❖ WBPDC and DPL were advised to focus on building coal stock as per their normative requirement. WBPDC assured of increasing their coal stock from January 2025 onwards. DPL was advised to submit action plan on enhancing their coal stock position.

Transmission line (132 kV & above) commissioned during November-2024:

220 kV-BANKA (PG)-Havelikharagpur-1(92.8 ckt Km) (ACSR Zebra) commissioned on 05.11.2024.

- ❖ He further highlighted the following:

- ✓ Ministry of Power vide letter dated 08.11.2024 has issued comprehensive guidelines titled "Workforce Adequacy Guidelines for Load Despatch Centres" to enhance the operational efficiency of Load Despatch Centres (LDCs) across India. This is to ensure that LDCs, including National Load Despatch Centre (NLDC), Regional Load Despatch Centres (RLDCs), and State Load Despatch Centres (SLDCs), are staffed with an adequate number of skilled professionals to manage the complexities of modern power systems.

- ❖ **ED, ERLDC** at the outset underlined the following facts and issues:

- W.r.t national grid in November 2024:
207 GW peak demand met in solar hours while 200 GW met in non-solar hours .
- W.r.t pan-India Power Market in November 2024:
213 MU exchanged in DAM, 100 MU in RTM and 791 MU transacted in G-DAM..
- W.r.t reserves and Ancillary services in November 2024:
178 MU (up) and 250 MU(down) in AGC,
450 MU(up) and 280 MU (down) in TRAS.
- Underfrequency (typically < 49.5Hz) is being observed in the morning hours. High RE penetration with intermittent nature is one of the major driving factors.
- AGC integration is necessary for major generators in ER to ensure adequate reserves in the system.

1. PART-A: CONFIRMATION OF MINUTES

1.1. Confirmation of Minutes of 221st OCC Meeting held on 27th November 2024 physically at ERPC Secretariat, Kolkata

The minutes of 221st Operation Coordination Sub-Committee meeting held on 27.11.2024 was circulated vide letter dated 03.12.2024.

Members may confirm the minutes of 221st OCC meeting.

Deliberation in the meeting

All members confirmed the Minutes of 221st OCC meeting.

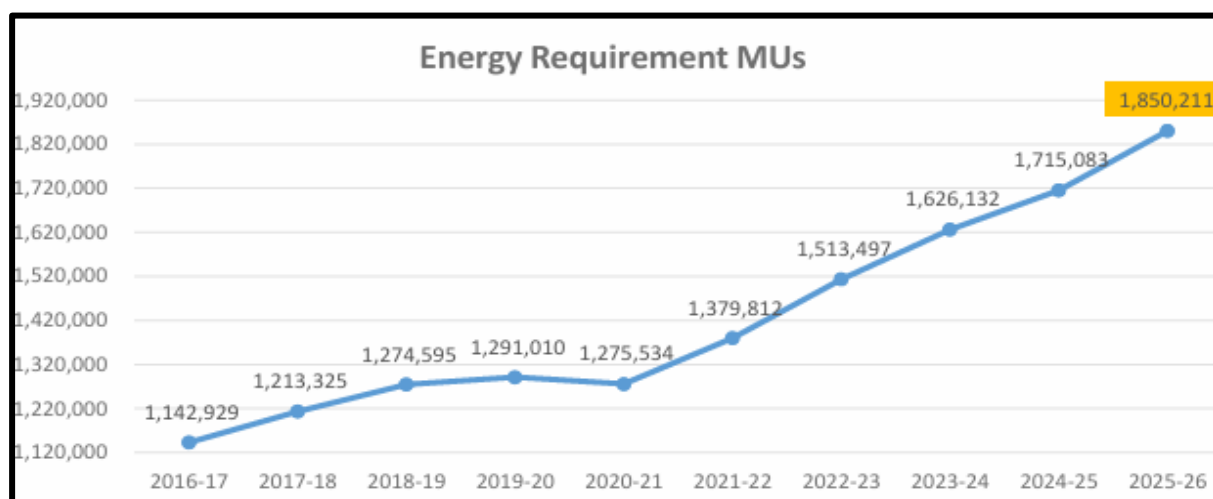
2. PART-B: ITEMS FOR DISCUSSION

2.1 Anticipated demand of electricity during FY 2025-26: CEA

The details of All India Power Supply Position in terms of Energy (MU) during the period 2021-22 to 2024-25 with the respective growth on year-to-year basis, have been furnished below. It may be mentioned that the figures cited herein are Actual for the years 2021-22 to 2023-24, Actual cum Anticipated Energy Requirement for 2024-25 (Actual upto October, 2024 and anticipated thereafter).

Energy						
Year	Energy Requirement		Energy Supplied		Energy Not Supplied	
	(MU)	% Growth	(MU)	% Growth	(MU)	% Growth
2021-22	13,79,812	8.2	13,74,024	8.1	5,787	0.4
2022-23	15,13,497	9.7	15,05,914	9.6	7,583	0.5
2023-24	16,26,132	7.4	16,22,020	7.7	4,112	0.3
2024-25	17,15,083	5.5	—	—	—	—

*data from October onwards are anticipated



The All India Anticipated Energy Requirement for FY 2025-26 furnished as per 20th Electric

Power Survey (EPS) (Ex-bus) and as forecasted after optimization of data received from Regional Power Committees (RPCs) are hereunder:

Anticipated Energy Requirement for FY 2025 -26		Ex Bus Figures in MU	State periphery figures in MU	% Growth
		(A)	(B) [(A)-2% losses]	C
20th EPS	2024-25	16,99,041	16,65,060	-
	2025-26	17,96,627	17,60,694	5.74
	2025-26 (Revised)	18,04,067	17,67,986	6.18
GM Division	2024-25	-	17,15,083	7.88
	2025-26	-	18,50,211	7.88

It may be seen from above figures that as per Revised 20th EPS there is only growth of 6.18% in All India Energy Requirement for FY 2025-26 as compared to last year projected in Revised 20th EPS. But the average growth in Actual Energy Requirement during last 3 years has been around 8% as per All India Power Supply Position in term of Energy. Considering this trend also the forecasted figures of 1,850 BU from GM Division are indicating the growth of 7.88%.

The annual energy requirement and peak demand has been forecasted by GM Division as:

Region/ State	Total Energy Requirement in MUs for 2025-26 (As per GM Division)	Peak Demand in MW for 2025-26 (As per GM Division)
Bihar	49,068	9056
Damodar Valley Corporation	29,794	4146
Jharkhand	15,746	2619
Odisha	51,399	7623
West Bengal	74,430	13541
Sikkim	674	158
Eastern Region	2,21,568	35781

Month wise details of ER states (peak demand and energy requirement) attached at **Annex B.2.1**

All concerned states may update on the projected demand pattern. Members may discuss.

Deliberation in the meeting

- As per information received from Constituent states of ER for FY 2025-26:

Region/ State	Projected Energy Requirement in MUs for 2025-26	Projected Peak Demand in MW for 2025-26
Bihar	To be intimated within a week	8800
Damodar Valley Corporation	28800	3600
Jharkhand	15,746	2580
Odisha	To be intimated within a week	To be intimated within a week
West Bengal	65000	13785
Sikkim	-	-
Eastern Region		35781

Sikkim representative was not present in the meeting.

OCC Decision

OCC advised all ER states to review their respective demand portfolio for FY 2025-26 as mentioned above and share the demand(MW) as well as total energy (MU) (with necessary changes if any) with ERPC secretariat within a week.

2.2 Amendment in CERC approved methodology for computation of Average Monthly Frequency Response Performance in terms of Beta 'β' factor: NLDC

In compliance with the provisions of Regulation 62(5) and 65(4) of the CERC (Terms and Conditions of Tariff) Regulations, 2024, NLDC had prepared a draft methodology for computation of Average Monthly Frequency Response Performance, Beta 'β'.

CERC approved "Methodology for computation of Average Monthly Frequency Response Performance, Beta 'β'" was **received vide letter dated 23.10.2024**, The scope of the methodology was enlarged to cover intra-state generating stations, whose tariff is determined by CERC and are falling under the jurisdiction of SLDCs. *Clause 4.4 (b) of the CERC approved methodology states that "FRO of generating stations, whose tariff is determined by CERC and are falling under the jurisdiction of SLDCs (in accordance with the control area jurisdiction as per Regulation 43 of CERC (EGC) Regulations, 2023), as assessed by concerned SLDC shall be considered for computation of Beta by the SLDC."*

Clauses 4.7 and 4.9 mention about computation and certification of Beta. The clauses have been mentioned below:

Clause 4.7: *The concerned LDC shall compute Average Monthly Frequency Response Performance, Beta 'B' (truncated up to 2 decimal places)..."*

Clause 4.9: *The concerned LDC would furnish Average Monthly Frequency Response Performance, Beta 'β' computed for a billing month to respective RPC along with all relevant supporting documents latest by 15th day of the following month, The Beta 'β', furnished by concerned LDCs will be certified and issued by the RPC through publication on its website to make it a part of commercial accounting."*

The communication from NLDC attached at **Annexure B.2.2**

Members may discuss.

Deliberation in the meeting

- OCC was apprised of the amendment in the methodology for computation of Beta ' β ' factor for generating stations as proposed by SRPC.
- WBSLDC stated:
 - ✓ No provision of incentive for primary frequency response exists in WBERC regulations
 - ✓ FRO of generating stations needs to be made mandatory instead of incentive based.
- Odisha SLDC proposed that an online training could be conducted for the procedure for calculating Beta ' β factor'.
- DVC agreed to this proposal and stated that the calculation of Beta ' β ' factor is done by them.
- ERLDC informed:
Computation of Beta factor for Barauni TPS has been taken up with Bihar SLDC. SLDC Bihar has consented to carry out the same.

OCC Decision

- OCC agreed in-principle to the proposed amendment, i.e concerned SLDCs have to compute and certify Beta factor w.r.t generators in their jurisdiction whose tariff is determined by CERC.
- All SLDCs were also advised to start computing Beta factor for those intra-state generators whose tariff is not determined by CERC to gain insight on frequency performance of generators.
- ERLDC was advised to conduct online training for better understanding of the calculation procedure of Beta factor by all ER constituents.
- OCC further suggested to have a database of " β " Factor for all generating stations in order to utilize this data for future planning.

2.3 Establishment of Kunkuri 400/132 kV S/S by LILO of both circuits of 400 kV Sipat - Ranchi D/c: CSPTCL

A meeting convened by CEA on **13.11.2024** to discuss the establishment of Kunkuri 400/132 kV S/s by LILO of both circuits of **400 kV Sipat - Ranchi D/c line**.

▪ **Background:**

CSPTCL vide **letter dated 24.09.2024** has informed that it is planned to establish 400/132 kV S/s under Intra-state by LILO of both circuits of 400 kV Sipat - Ranchi D/c line which is an ISTS line. It was also informed that a meeting was held among CTUIL, Grid-India, WRLDC, ERLDC and CSPTCL to discuss the same.

In view of the above, CSPTCL has requested to grant the permission of establishment of Kunkuri 400/132 kV S/s by LILO of both circuits of 400 kV Sipat - Ranchi D/c line. As per deliberation

- ✓ CSPTCL informed that establishment of 220/132 kV S/s by connecting it to nearby 220 kV S/s involves transmission line passing through forest area and also the length of the 220 kV lines to proposed Kunkuri S/s is about 135 km, however, establishment of 400/132 kV S/s by LILO of both circuits of 400 kV Sipat - Ranchi D/c line involves no forest and the length of the proposed LILO is also about 8 km.
- ✓ CTUIL presented the study and informed that due to the proposed LILO of both circuits of 400 kV Sipat - Ranchi D/c line, there is no additional loading observed in the network in 2028-29 timeframe. Also, reconductoring of the 400 kV Sipat - Ranchi D/c line may be done, if overloading would be observed in future. Accordingly, CSPTCL's proposal for establishment of Kunkuri 400/132 kV S/s may be agreed with a condition that CSPTCL has to do reconductoring of the LILO portion in future (commensurate with the reconductoring of 400 kV Sipat - Ranchi D/c line), if needed. CSPTCL agreed for the same.
- ✓ Grid-India stated that as the angular difference between Sipat and Ranchi buses is observed to be on the higher side, an additional corridor might be planned instead of reconductoring of the existing line.
- ✓ CEA stated that since CSPTCL's proposal includes the LILO of an Inter-Regional line (WR-ER), the proposal should also be presented before both the RPCs i.e. WRPC and ERPC for their concurrence.

In the meeting, it was envisaged that possibility of establishment of 220/132 kV S/s (by interconnecting it with nearby 220 kV S/s) instead of 400/132 kV S/s at Kunkuri may be explored by CSPTCL. Accordingly, CSPTCL has carried out the survey and main findings of the same are given below:

S. No.	Particular	220 kV DCDS from existing 220 kV Gerwani S/s at proposed Kunkuri S/s	LILO of 220 kV Raigarh - Tamnar (JPL) line at proposed Kunkuri S/s	LILO of 400 kV Sipat - Ranchi line At proposed Kunkuri S/s
1.	Tentative Route length	150 km	135 km	08 km
2.	Forest involvement	30 km	29 km	NIL
3.	EHV line crossing	29 Nos.	12 Nos.	03 Nos.
4.	River crossing	04 Nos.	04 Nos.	01 Nos.

CSPTCL's proposal of LILO of establishment of Kunkuri 400/132 kV (Intra-state) S/s by LILO of both circuits of 400 kV Sipat - Ranchi D/c line (an ISTS line) was agreed with the condition that the proposal should be presented before both the RPCs i.e. WRPC and ERPC by CSPTCL.

The details have been attached in **Annexure B.2.3**

Members may discuss.

Deliberation in the meeting

OCC was apprised of the following:

- ✓ CEA(Power System Planning and Appraisal-I Division) has already agreed to the proposal of LILO of 400 kV Sipat - Ranchi D/C for establishing intra-state S/S at Kunkuri.
- ✓ The proposal has been deliberated and concurred in both CMETS-ER and CMETS-WR after relevant system study by CTU.

OCC Decision

OCC consented to the proposal of LILO of both circuits of 400 kV Sipat - Ranchi D/C for establishment of 400/132 kV S/S at Kunkuri.

2.4 Approval for HTLS Project implementation in Jharkhand: JUSNL

The DPR of the said project was first submitted and deliberated in 207th OCC dated 15.09.2023.

207th OCC deliberation:

- ✓ JUSNL representative apprised the forum about the proposal for reconductoring of existing 132kV Line by HTLS Conductor for relieving congestion in JUSNL.
- ✓ After detailed deliberation, OCC opined that a special meeting may be convened to look into the intricate technical aspects of the project. OCC further opined that Jharkhand may forward the agenda to PSPA division of CEA for further approval.

Following clarifications were sought by ERPC Secretariat in the special meeting:

Updated Power map of JUSNL system with clear demarcation of present and future/planned system shall be submitted.

The following information also need to be submitted:

1. Basis of demand data that was considered for load flow study in present as well as for future case. The data has to be certified by SLDC or any competent dept of JUSNL.
2. Present peak loading(in percentage) observed for the lines.
3. Commissioning time line for already approved projects/substations/lines of JUSNL system. Is the future systems/elements which are going to be commissioned have been incorporated in the load flow study results?
4. Base case file of the load flow study results.

Further NPC has made recommendations on the various lines of JUSNL regarding the HTLS project:

Sr no:	Name of Line	Suggestions/ Recommendations/ Queries	Reply to Suggestions/ Recommendations/ Queries
1	Adityapur(Gamariya) to Ramchandrapur 132 KV D/C	JUSNL to submit SCADA snapshot of single line diagram	Attached SCADA snapshot of Single Line Diagram of Adityapur GSS and

		from SLDC control room.	Ramchandrapur GSS from SLDC Control Room as Annexure - A (Page No. 1 & 2)
		Present MW flow in the lines along with its ampere capacity to be ascertained.	Present peak load during N-1 Contingency is 128MW (Approx. 500Amp). Refer Annexure - A (Page No. 3-6)
2	Namkum - Sikidri 132 KV	In the D/C line, JUSNL to explain why one ckt was upgraded to HTLS while the other was not.	There are 2 separate 132 kV Transmission Lines from Namkum GSS to Sikidri Substation via separate tower, i.e.: a) 132kV D/C Sikidri – Namkum Transmission line has already been reconducted through HTLS conductor in 2017. b) 132kV S/C Sikidri – Namkum Transmission line is being proposed for reconductoring to HTLS Conductor from ACSR Panther conductor. Refer Annexure - A (Page No. 7) Now, the said T/L may be withdrawl from reconductoring with HTLS due to availability of more T/Ls to the NAMKUM GSS .
		Previous DPR of the line commissioned in 2017 may be submitted.	This project was executed by PGCIL under Jharkhand Consultancy Project (JCP) and DPR was not submitted to JUSNL by PGCIL.
		Age of the transmission line and conductor need to be confirmed.	This Transmission Line was commissioned in the year of 2017 with ACSR Panther Conductor.
3	Hatia old - Hatia new 132 KV D/C	Line loading at the time of study as well as present loading in peak period to be submitted.	There is total 3 Circuit Transmission Line between Hatia (I) old GSS and Hatia (II) new GSS as follows:1. 132 kV D/C Transmission Line consisting of Circuit 1 & 2 of length (0.6 + 0.6 = 1.2 KM) presently connected with ACSR Panther Conductor with present load of (50 + 40 = 90MW), which is being proposed for reconductoring which will be overloaded in N-1 condition.2. 132 kV S/C Twin Zebra Transmission Line (3rd Circuit - Connected via Main Bus) with present load 160 MW.Refer Annexure - A (Page No. 8)
4	Dumka 220/132 to Dumka/Maharo 132KV DC	Detailed feeder loading pattern along with proper justification to be submitted.	At present, Average load of 132kV D/C Dumka Madanpur (220/132 kV) to Dumka Maharo (132/33 kV) Transmission Line is approx. 70 MW and Peak load is approx.

			90MW, which will be overloaded in N-1 condition.Refer Annexure - A (Page No. 9)
		Previous DPR of the line commissioned in 2015 to be submitted.	This project was executed by PGCIL under Jharkhand Consultancy Project (JCP) and DPR was not submitted to JUSNL by PGCIL.
5	Adityapur - Rajkharsawa 132 KV S/C	Since existing towers of the line are more than 50 years old, tower health assessment report carried out by experts needs to be submitted to ascertain healthiness of towers for reconductoring. The minimum residual life of the tower shall be ensured as 25 years before going for reconductoring.	Tower Health Assessment of the said Transmission Lines with total 112 Nos. of Towers was conducted by JUSNL on sampling basis, i.e. for 27 Nos. of Towers. As per the report, Some of the Towers needed few preventive measures before reconductoring. Also, All Towers needed Tower Health Assessment.
6	Rajkharsawa - Chandil line via kandra 132 KV S/C	Since existing towers of the line are more than 50 years old, tower health assessment report carried out by experts needs to be submitted to ascertain healthiness of towers for reconductoring. The minimum residual life of the tower shall be ensured as 25 years before going for reconductoring.	Tower Health Assessment of the said Transmission Lines with total 118 Nos. of Towers was conducted by JUSNL on sampling basis, i.e. for 28 Nos. of Towers. As per the report, Some of the Towers needed few preventive measures before reconductoring. Also, All Towers needed Tower Health Assessment.
7	Jadugoda - Ramchandra pur LILOed at Sundarnagar 132 KV D/C	JUSNL to justify power flow in the lines exceeding rated capacity within 5 years of commissioning and	Due to availability of alternate source for Dhalbhumgarh GSS, the proposal for reconductoring of LILO of 132 kV D/C Jadugoda - Ramchandrapur T/L at Sundarnagar GSS is no more required. Therefore, the said Jadugoda -

		reason behind sudden load growth.	Ramchandrapur LILOed at Sundarnagar 132 KV D/C T/L may be withdrawl from reconductoring with HTLS due to availabilty of alternate source.
		JUSNL to ensure reliability of supply to Jaduguda substation with proper justification substantiated with load flow study by PRDC. Alternate source to Jadugoda may be explored.	
		Previous DPR of the line commissioned in 2019 may be submitted.	
8	132kV D/C Chandil - Golmuri Transmission Line S/C LILOed at 132/33 kV Mango GSS (32 kM)		At present, Average load on 132kV S/C Chandil - Golmuri T/ Line (30kM), and 132kV S/C Chandil - Mango T/L (32kM) is approx. 100 MW and Peak load is approx. 128 MW, which will be overloaded in N-1 condition.Average Load of Golmuri GSS is approx 70 MW and Peak Load is approx 90 MW.And, Average Load of Mango GSS is approx 35 MW and Peak Load is approx 40 MW.As per current need of load in both GSS, recommendation for installation of 50MVA Power Transformer in both GSS has been submitted by Distribution Company of Jharkhand, i.e. JBVNL.In view of the above, There is need of reconductoring of ACSR Panther Conductor with HTLS Conductor in 132kV D/C Chandil - Golmuri Transmission Line S/C LILOed at 132/33 kV Mango GSS as this transmission line is at it's maximum capacity.Refer Annexure - A (Page No. 10)

SLD as well as Load flow study resul,.ts enclosed at **Annex_B.2.4**

JUSNL may update. Members may discuss.

Deliberation in the meeting

JUSNL updated:

- 132 KV Namkum - Sikidri and 132 kV Jaduguda-Ramchandrapur have been withdrawn from the proposal of HTLS reconductoring due to availability of more T/Ls to the NAMKUM GSS.

- 220 kV Maithon Dumka T/L (73.3 km) is also proposed for HTLS reconductoring as present loading pattern is violating N-1 contingency.
- 132kV D/C Chandil - Golmuri Transmission Line S/C was added to the existing list of HTLS Project T/L

OCC Decision

- OCC agreed to the proposal of HTLS reconductoring of transmission lines in Sl. No. 1, 3, 4, 5 & 6.
- For reconductoring of 220 kV Maithon - Dumka & 132kV D/C Chandil - Golmuri T/L, JUSNL was advised to share the relevant study reports within a month for consideration of the proposal.

2.5 Approval for WAMS Project implementation: JUSNL

The DPR of the said project was first submitted and deliberated in **207th OCC** dated 15.09.2023.

207th OCC deliberation:

- ✓ OCC was of the view that DPR should be self-certified by respective state rather than forwarding to RPC for examination of technical details.
- ✓ Representative of ERPC submitted that the DPR may be prepared in line with the latest PMU guidelines as finalized in the 13th NPC Meeting held on 05.07.2023 at Kolkata

Further, as per NPC guidelines on WAMS:

- **Adherence to NPC Guidelines regarding PMUs:** PMUs need to be installed at all locations to ensure standardization across the national grid according to NPC Guidelines finalized in the 13th Meeting of NPC held on 05.07.2023.
- **OPGW Necessity:** Many locations in Jharkhand lack Optical Ground Wire (OPGW) installations. Before considering PMU installations, it is essential that OPGW is installed and operational to support PMU data requirements.
- **Cost Implications:** PMU installations come with substantial capital costs for equipment, installation, and ongoing maintenance. Installing PMUs at the 132 kV level might not provide commensurate benefits relative to the expenditure.
- **Data Management Concerns:** Introducing PMUs at 21 nos. 132 kV substation would result in a vast amount of data Handling this influx can be challenging and might strain existing data management and analytics systems.
- **Focus on Higher Voltage Levels:** The higher voltage levels (220 kV and above) are more critical junction points in the grid, carrying significant power flows. Concentrating resources and efforts on enhancing the observability and stability of these levels should be prioritized.
- **Detailed Cost Implications:** Jharkhand was requested to provide a comprehensive breakdown of the costs associated with PMU installations, including equipment, installation, data management infrastructure, and ongoing operational costs.

JUSNL may update. Members may discuss.

Deliberation in the meeting

JUSNL submitted:

21 PMUs proposed for installation at 132 kV level. 11 PMUs and 1 PMU have also been planned for installation at 220kV and 400kV level respectively.

OCC Decision

- Though PMU installation has been envisaged mainly at 220kV level and above at NPC (for all regions excluding NER), OCC recommended that PMUs may be installed at some critical 132 kV locations in intra-state system of JUSNL. The requirement of 21 PMUs at 132 kV level may be reviewed by JUSNL and revised proposal may be submitted after review. However, OCC agreed to the proposal for installation of PMUs at 220kV & above voltage level.
- OCC opined that before the installation of PMUs, OPGW network for data transmission to be made fully operational. JUSNL was advised to complete OPGW installation in the next 6 months.

2.6 Agenda Points: PVUNL

2.6.1 Synchronization of Unit 01 of 3X 800MW Thermal Power project

Background:

1. Start-up Power of PVUNL was envisaged through 400kV Patratu-Patratu Line (PVUNL-JUSNL GSS Katia).
2. As above line was not made available Start up power was arranged through 400kV Tenughat-PVUNL line.
3. Synchronization of Unit#01 of PVUNL was also envisaged through 400kV Patratu-Patratu Line.

Current Scenario:

1. Synchronization of Unit#01 of PVUNL is scheduled in January 2025.
2. 400kV Patratu-Patratu Line or any other ATS line are not available as on date and it will take time for its readiness.

Consequences:

1. In view of unavailability of ATS, Synchronization of Unit#01 in Jan-2025 is planned through start up power line 400kV through TVNL(Tenughat)-PVUNL S/C line.

Appeal:

1. Technical, Commercial and other statutory requirements for Synchronization of Unit#01 in Jan-2025 is planned through start up power line 400kV through TVNL(Tenughat)-PVUNL S/C line are to be finalized.

2.6.2 ATS of 3X 800MW Thermal Power project

Background:

1. Designated ATS of PVUNL is as follows:
 - i) 400kV QM D/C PVUNL-Patratu Line
 - ii) 400kV QM D/C PVUNL-Koderma Line
 - iii) 400kV QM D/c PVUNL-Chandil Line
2. 400kV QM D/C PVUNL-Koderma Line and PVUNL-Chandil Line are at initial stage, and it will not be available till 2025.
3. PVUNL-Patratu Line –
 - Status – Foundation – 12/27, Tower Erection – 11/27
 - Work in Forest area is still to be taken up.

Current Scenario:

1. PVUNL-Patratu Line –
 - Stage I Forest Clearance granted on 16.05.2024
 - Stage I Forest Clearance is pending.
2. No ATS is available for Power Evacuation of PVUNL.
3. COD of Unit #01 of PVUNL is scheduled in Feb 2025.

Consequences:

1. In view of unavailability of any ATS line, Power Evacuation PVUNL is not possible.

Appeal:

1. Early Resolution for availability to be ensured for Power Evacuation PVUNL.

2.6.3 Sustainable Power Evacuation through Katia Substation for COD of Unit 1 and subsequent Continuous operation

Background:

- Power Transfer from PVUNL is via Patratu-Patratu 400 kV line connecting Katia Sub-Station.
- Power transfer capability of various lines connecting 400 kV Katia Sub-Station area as follows:
 - Two Nos of lines connecting Katia & Hatia at 220 kV level with a capacity of power transfer of 200 MW each.
 - Two Nos of lines connecting Katia & Burmu at 220kV level with a capacity of power transfer of 200 MW each.
 - Two nos of lines connecting Katia & Bero at 400kV level with a capacity of power transfer of 800MW each (ISTS lines).
 - Two nos of lines connecting Katia & Latehar at 400 kV level and power transfer capacity of 800MW each (lines are under construction).

Current Scenario:

1. Present loading of Katia 400 kV BUS is around 350 to 380 MW which includes loads of Hatia, & Burma.
2. COD of Unit #01 of PVUNL is scheduled in Feb 2025 and continuous running of unit thereafter.

Conclusion:

1. Power transfer of 800 MW from Unit#1 PVUNL to Katia via state network is not possible

Appeal:

1. Methodology for sustainable power evacuation through Katia sub-station via its connected state/ISTS network may be detailed.

2.6.4 Discrepancies in Start-up bill for the month of October 24

Background:

- Agreement was to provide start up power through Patratu-Patratu line that is ATS line of PVUNL, however, due to delay in ATS by JUSNL, the TVNL-PVUNL line was made with modification of Towers for 400kV line, and this was erected at additional cost of INR 13.72 Cr(approx.) which was borne by PVUNL.

- This line was dedicated line for PVUNL to provide Start up Power with an agreement of minimum contract demand of 20.05 MVA by JBVNL even though the actual requirement of power for commissioning was approx. 2-3 MVA.
- Every month PVUNL was paying approximately 60 lakhs towards contract demand of 20 MVA even though actual consumption was 2-3 MVA.
- Details of Energy Bill payment of Last six months:

Month	Actual kVAh Consumed	Actual kWh Consumed	Payment done(INR)	Remarks
May-2024	0		6030920.00	actual consumption =2 MVA(approx.)
Jun-2024	0		5895361.00	actual consumption =2 MVA(approx.)
Jul-2024	0		5899355.00	actual consumption =2 MVA(approx.)
Aug-2024	0		6030920.00	actual consumption =2 MVA(approx.)
Sept-2024	481245	408512	9035555.00	
Oct-2024	30366822	1126134		Bill of INR 27 Cr(approx..) received.

- JBVNL employs a billing methodology based on kVA and kVAh, instead of the traditional kW and kWh approach. Despite separate agreement for drawl of Start up power in other projects, JBVNL here insisted for agreement for HT Consumer basis which is yet to signed and pending at JBVNL end.
- Grid Code categorize reactive power support provided for grid stability and quality as an "Ancillary Service". Also, entities that draw reactive power when system voltage exceeds 103% are eligible for incentives. The current billing approach by JBVNL contradicts these established regulations and results in an exorbitant charge to maintain grid stability.

Events:

- During October 2024, due to light loading condition on the transmission line, the voltage at our end increased due to Ferranti effect causing the line to trip couple of times. With critical project commissioning milestones underway, this frequent tripping of the line was hampering availability of Start-up power.
- To ensure grid stability and prevent voltage fluctuations, 125MVAR Bus Reactor at PVUNL end was taken into service in consultation with ERLDC/ SLDC. The reactor was in service for over 11 days (from 17.10.2024 to 29.10.2024) during the month.

Consequences:

- PVUNL has been significantly penalized for the reactive power consumed by the bus reactor, a critical measure taken solely to maintain grid stability and power quality.

Appeal:

- Reactive power does not contribute to real power consumption, which is measured in kWh. By penalizing us for reactive power consumption, JBVNL is essentially penalizing PVUNL for providing a service that benefits the entire power system.

- Immediate intervention is required to address this issue and ensure a fair resolution as PVUNL believe that the billing methodology used in this case is not equitable and does not accurately reflect the actual energy consumption of our facility

PVUNL may explain. Members may discuss.

Deliberation in the meeting

PVUNL representative submitted:

Synchronisation of Unit 1 is planned by end of January 2025. Synchronisation is possible with a minimum injection of 150 MW power. Challenges are faced in this regard for power evacuation through PVUNL-TVNL line.

Concern was expressed on full power evacuation through 400 kV Katia-Latehar D/C line which has not yet been commissioned.

ERLDC stated:

Maximum 80-90 MW generation can be allowed for injection by Patratu units as per the study considering both units of TVNL in operation as well as N-1 contingency.

OCC Decision

- OCC opined to convene a special meeting for resolving the power evacuation issue of PVUNL. ERLDC was advised to devise logic of SPS for generation backdown of PVUNL units so that loading of 220 kV Tenughat-Biharsarif and 220 kV Tenughat-Govindpur lines remains within thermal limit.
- JUSNL was advised to expedite the commissioning of all intra-state transmission lines connecting Patratu generating units after resolving forest clearance issues.
- Connectivity agreement issues of PVUNL to resolved with SLDC Jharkhand at the earliest.
- As PVUNL units connected to intra-state system of Jharkhand and Jharkhand being the sole beneficiary, scheduling of PVUNL shall be done by SLDC Jharkhand for meeting intra-state demand.
- OCC referred the discrepancy in start-up power bill to CCM for deliberation.

2.7 Update on Patna Islanding scheme: ERPC

The Patna islanding scheme would be formed with Units of NPGCL along with loads of Patna city.

NTPC was entrusted for carrying out study of NPGC units and M/S Solvinia had submitted report on study of islanding scheme dated 08th May 2024. Thereafter based on comments received from ERLDC, replies were submitted by M/S Solvinia. NTPC had communicated the report to all concerned including SLDC Bihar.

Some further tests needed could not be carried out due to non-receipt of relevant data from Bihar.

221st OCC Decision:

- ERLDC was advised to form a joint committee with SLDC Bihar, NTPC and Bihar DISCOMs for regular monitoring of implementation of Patna islanding scheme. A joint study on feasibility of this islanding scheme may also be carried out. The status of the same may be updated in the subsequent OCC meetings.

- OCC also advised that Patna islanding scheme must get implemented before Summer 2025

In view of the 221st OCC Decision an **online meeting was conducted on 05-12-24** (Minutes attached with **Annexure B.2.7.**) to expedite the implementation of Patna islanding scheme, members from NTPC, SLDC BIHAR, ERPC, and ERLDC were present. The points discussed in this meeting are:

- NTPC apprised that nonlinearity test of governor has not been conducted.
- SLDC Bihar mentioned that there is cost implication for Bihar islanding implementation. It was advised to discuss the matter in OCC forum with estimated budgetary cost by SLDC Bihar. Based on the discussion following action points were finalized:
- SLDC Bihar to submit:
 - load details (Maximum and minimum) of Patna Island in current scenario.
 - Feeder list which will be tripped for island formation.
 - Feeders and load quantum in MW for installing UFR within Island.
 - NTPC to communicate within a week via letter that NPGC unit is capable of islanding operation and there is no requirement of non-linearity test, (Action point-NTPC). Following details to be submitted within a week, (Action Point- SLDC Bihar).
 - Submit load details (Peak and minimum) of Patna Island in current scenario.
 - Updating list of Feeders which will be tripped for island formation.
 - Identification of feeders and loads along with quantum in MW for installing UFR within Island.
- Post receipt of the above details further meeting will be conducted for finalizing Islanding logic, other operational issues. Accordingly, action plan will be finalized for implementation of scheme as early as possible.
- NTPC to confirm NPGC's islanding capability with no further requirement for a non-linearity test via letter within a week.

NTPC and SLDC Bihar may update the present status along with future action plan.

Deliberation in the meeting

NTPC ER-I informed:

- ✓ *As per Solvina report and observation of OEM M/s GE, turbine speed of NPGC units remains within permissible levels in all scenarios except for one. One limitation as pointed out by M/s GE is that the maximum time limit for which the turbine can be exposed to 103% overspend limit is 20 minutes for the complete lifecycle of turbine.*
- ✓ *The said condition is appearing for 5 seconds in one of the scenarios when generation is maximum and Patna load is minimum.*
- ✓ *Formal consent is given from NTPC for implementation of Patna islanding with NPGC units subject to the mentioned limitation as per study by M/s Solvina .*

OCC Decision

- *ERLDC was advised to finalize Patna islanding logic considering overspeed limitation of NPGC units. Bihar SLDC was advised to identify loads to be included in the islanding scheme.*

- Bihar SLDC was directed to finalize and submit the approved DPR of Patna islanding before next OCC. ERLDC was requested to assist Bihar SLDC in preparation of DPR.
- After finalization of DPR, cost implications along with funding options for Patna islanding scheme to be deliberated in next OCC.

2.8 Bus split operationalization at NTPC Kahalgaon: ERPC

As decided in 219th OCC Meeting, a committee comprising of members from ERPC and ERLDC visited NTPC Kahalgaon on 17-10-2024 to assess the status of Bus splitting at 400 kV level and way forward for operationalization of 400 KV Bus sectionalizer.

Following works need to be done to complete the installation of ICT 3 & 4:

1. Determination of underground cable conduit path for 400/132 kV ICT-3, 4 and 5 allocated for stage 2 supply.
 2. Excavating the existing cable and relaying from Stage-1 132kV to New Stage-2 132 kV switchyard, where ICT 3 & 4 will be connected.
 3. Laying of additional 22.8 ckt. km control cable for STs.
 4. Jumpering of ICTs in 132kV & 400kV level.
 5. Bay equipment testing.
- NTPC apprised that determination of underground power cables is one of the major challenges to proceed further with laying of cables between two 132kV switchyards. The tentative time to complete the ICT commissioning is **25th May 2025**.
 - Meanwhile in view of increased fault level of NTPC Kahalgaon and to facilitate interim arrangement of standby ISTS connectivity to Godda Thermal Power project of M/s Adani Power (Jharkhand) Ltd. (APJL) with Indian grid, Bus splitting at 400KV Kahalgaon needs to be done on priority.

In compliance to 221st OCC decision, NTPC has submitted status (as on 20th December 24) i.r.o implementation of Committee recommendations as follows:

As per **221st OCC** Deliberation

NTPC Submitted:

- Layout of the existing cables has been mapped and the interface points between the old and new cabling systems have been identified.
- Excavation process is presently being undertaken to route the new cables. Due to considerable angular difference between individual 400 kV sections upon opening of sectionalizer breakers and consequent threat to equipment safety in 11 kV auxiliary network, this proposed interim arrangement could not be explored.
- Procurement of new control cables or sourcing them from in-house spares is being explored.

221st OCC Decision:

- OCC advised NTPC to expedite the overall process of excavation, laying of new cables and procurement of control cables for timely completion of the bus splitting arrangement.
- NTPC was advised to share the monthly report on progress i.r.o bus splitting activities at 400 kV switchyard of NTPC Kahalgaon to ERPC/ERLDC.

❖ As per latest update received from NTPC vide mail dated **20.12.2024**:

✓ **Healthiness of Bus CVTs:**

Based on daily monitoring results , secondary side voltages (110VAC) are indicated within acceptable range.

Testing Progress:

- Testing of all 400KV Lightning Arresters (LAs) and Current Transformers (CTs) has been completed.
- Testing of 132KV Lightning Arresters (LAs) is completed.
- Testing of 400KV CBs-4 Nos., 132KV Circuit Breakers-10 Nos., 132KV Current Transformers (CTs- 22/30 completed), and Capacitive Voltage Transformers (CVTs)-12 Nos. have been planned and is in progress.

Isolator Status:

- All 400KV/132KV Siemens make HCB isolators (36 Sets) were found in jammed condition.
- Post-maintenance, some 132KV isolators were revived. CRM testing done for the serviced
- 132kV Isolators and the test results are found beyond acceptable limits.
- The issue has been escalated to M/s GE for urgent action.

Earthing Work:

- Earthing work for ICT-3, ICT-4 and associated bay equipment is in progress and is scheduled for completion by **15/01/2025**.

Control Cable Arrangement:

- NTPC is arranging the required control cables through internal resources and through the agency.

NTPC may confirm timeline and feasibility of Bus splitting with current configuration. Member may discuss.

Deliberation in the meeting

NTPC submitted :

- ✓ *The new ICT is ready for charging at 400 kV switchyard of Kahalgaon once the arrangement of control cables is made.*
- ✓ *132 kV power cable laying work is getting hampered due to persistent contractual issues..*

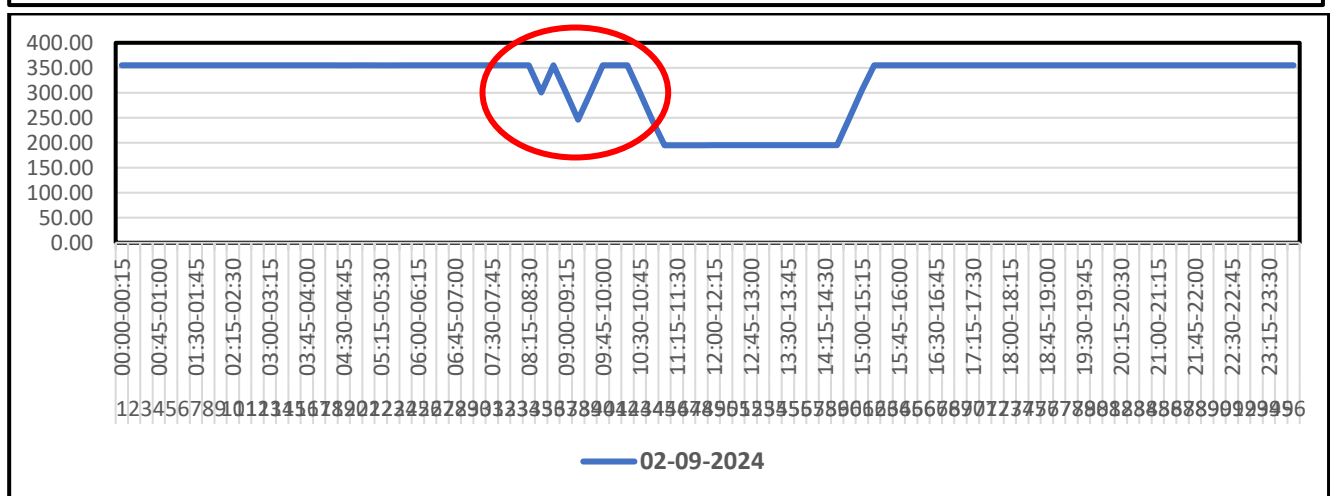
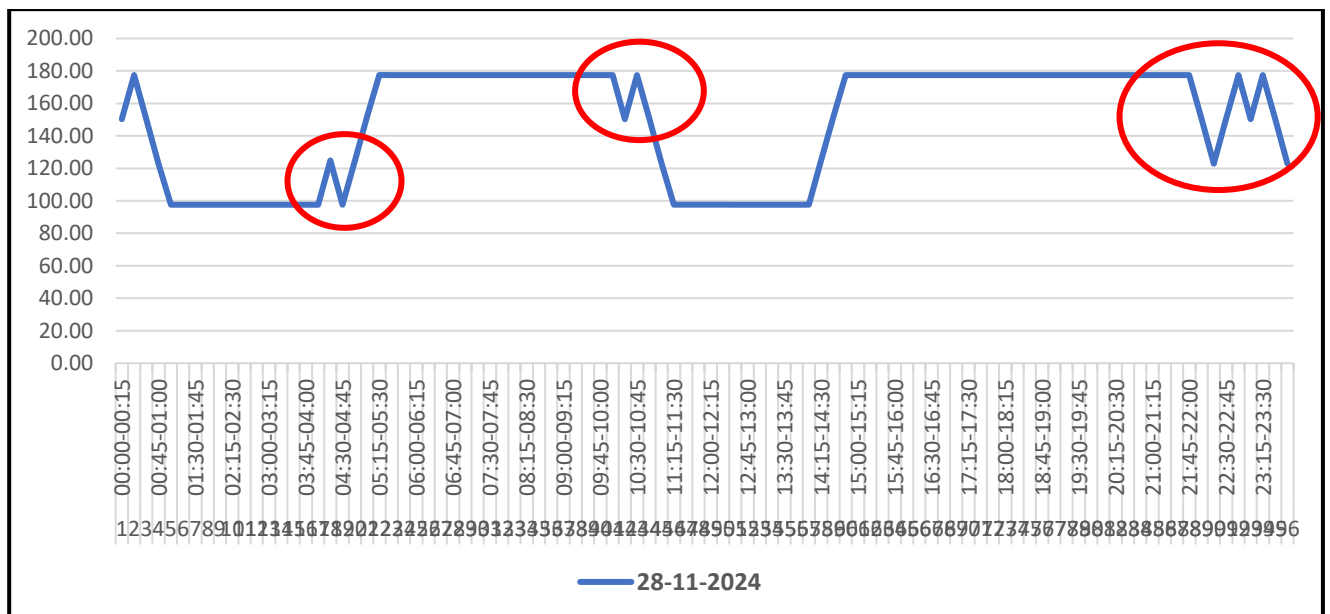
OCC Decision

In view of continued operation of 400 kV bus in synchronised mode at NTPC Kahalgaon at high fault level, OCC expressed serious concern over the lack of desired progress in 132 kV power cable laying and thereby suggested participation of representative from concerned site i.e NTPC Kahalgaon in next OCC meeting for clarity i.r.o progress in bus splitting activities at the site.

2.9 Frequent Cyclic Ramp up and Ramp down of MTPS-II units: NTPC ER-1

- It has been observed that **frequent cyclic ramp up and ramp down schedule is being given to MTPS-II on multiple occasions.**
- It is to mention that such frequent change in ramping direction is not desirable to generating machines barring some occasional emergency requirement.

- Often/block-to-block cyclic ramping is needlessly stressing our generating unit, as it is very difficult for mechanical systems of the unit to manage change in electrical system of the grid, this is severely increasing the stress on Boiler and Turbine.
 - Moreover, such frequent variations in schedule lead to huge losses for the station on account of DSM.
- Detailed cases of Sept'24 & Aug'24 have already been communicated to ERLDC (Copy of detailed cases & mail communications dt. **10.10.2024** & **20.08.2024** attached in **Annex B.2.9**).
- In these cases, frequent cyclic ramp up and ramp down schedule was given mainly on account of combination of AS & SCED. Similar instances have been observed in Nov'24 also (Detailed cases of Nov'24 also attached in **Annex B.2.9**).
- On **28.11.2024**, there were multiple cases observed of such cyclic scheduling pattern, details is as below :



Such frequent schedule variations to be avoided which are mainly due to combination of AS & SCED schedules.

NTPC ER-1 may please explain. Members may discuss.

Deliberation in the meeting

NTPC highlighted :

- ✓ *Frequent ramp up and ramp down schedule is provided to the MTPS-II plant especially due to frequent SCED Up and SCED Down despatch schedules.*
- ✓ *Although, the change in schedule is as per declared ramp rates, frequent ramp up and ramp down exerts stress on the generating units.*

ERLDC informed :

- ✓ *Variable charge (VC) of MTPS-II is Rs 2.595/- per unit and SMP (system marginal price) of SCED is presently varying between Rs. 2 to Rs. 5 (approximately) throughout the day. Thus, whenever SMP is more than VC, the plant gets SCED Up schedule and whenever SMP is less than VC, the plant gets SCED down despatch.*
- ✓ *Therefore, as VC of MTPS-II is in between the max and min SMP values, MTPS-II gets SCED despatch schedule in both directions.*

OCC Decision

OCC observed that there is no violation in ramping instructions issued to MTPS-II of NTPC as per extant guidelines

2.10 Scheduling related issues: NTPC ER-1

- *At present SG data is being fetched from ERLDC site every 5 minutes due to restriction by ERLDC after launching of new API. (Earlier in previous software frequency was 1 min). In a block, first revision data is fetched at 3 min 30 sec, considering that ERLDC posts its revision at 3rd minute after starting of block. The next ping to ERLDC website is placed at 8th min 30 sec.*
- *It is observed that from 14th Dec onwards, data is not being updated at ERLDC server before 3 min 30 sec of block starting. This is causing first ping to be unsuccessful and data is not updated till next ping.*
- *The above problem is causing non availability of data for monitoring in real time for plant operators.*

NTPC ER-1 may explain. Members may discuss.

Deliberation in the meeting

NTPC requested:

- ✓ *To enhance the limitation of API fetching in WBES from the present limit of 30 per hour.*
- ✓ *To provide additional API credentials for different stages of generating stations.*

ERLDC informed:

- ✓ *As the WBES is now a centralised software, high number of API fetching by multiple users may lead to slowness in working of the system and cause issues to all users including states and other generating stations..*

- ✓ *Different API credentials may be provided for different stages on basis of specific requirement.*

OCC Decision

OCC advised ERLDC to incorporate the additional features in WBES to the best feasible extent without compromising the reliability and efficiency of web based scheduling catering to ER constituents.

2.11 Report on No Load Charging of R-Y phase of 220kV Rajarhat-New Town-II Cable: ERLDC

220kV Rajarhat-New Town (AA-II)-2 has been out of service since 11:40hrs of 10th July'2024. The line was taken under emergency shutdown by WBSLDC for rectification of gas leakage problem in B-Ph breaker pole and charging attempt failed at the time (18:05hrs) of line charging on the same day. Thereafter, it was declared that the line is under breakdown since the said date.

As per **221st OCC** deliberation:

WBSLDC Apprised:

- Repair of the faulty cable (B phase) is getting delayed due to non-availability of the imported cable jointing kit.
- Neither charging the cable at low voltage nor charging from WBSETCL end is feasible.

Powergrid ER-II Apprised:

- Due to leading charging current drawn by the cable at no load, unbalance may be observed in Rajarhat ICTs between predominantly lagging load and cable charging current (leading).
- Earth fault setting of Rajarhat ICT may be required to be changed.

221st OCC Decision:

- OCC consented to the proposal of no-load charging of R and Y phases of the Rajarhat (PG)-New Town II C (circuit II) on interim basis. In this regard, relay settings of Rajarhat ICTs may be reviewed and modified based on the charging current in two healthy phases of the cable.
- OCC requested WBSETCL to expedite the cable repairing works and to share progress status of the same with ERPC.

The matter was discussed in 221st ERPC OCC meeting dated 25.11.24 and forum was advised to charge the line (two phases) as requested by WBSLDC to keep the UG cables in healthy condition. Subsequently, R-Y phase of the line was charged on 28.11.2024. Detail report is enclosed in **Annx-B.2.12**

ERLDC may respond. Members may discuss.

Deliberation in the meeting

- *ERLDC stated:*
- ✓ *No major unbalance as well as significant voltage change has been observed after charging R and Y phases of the faulty cable at no load. Flow of resultant unbalanced current was marginal after charging.*
- ✓ *Around 76 A charging current drawn by R and Y phases of the cable at leading p.f.*

- WB SLDC representative informed:
- ✓ Work order for cable jointing has been placed on 04.10.2024. The challenge of heterogenous cable jointing kit has been sorted out and requisite materials expected to be available by mid-May 2025.
- ✓ The cable shall be put into healthy operation by June 2025.

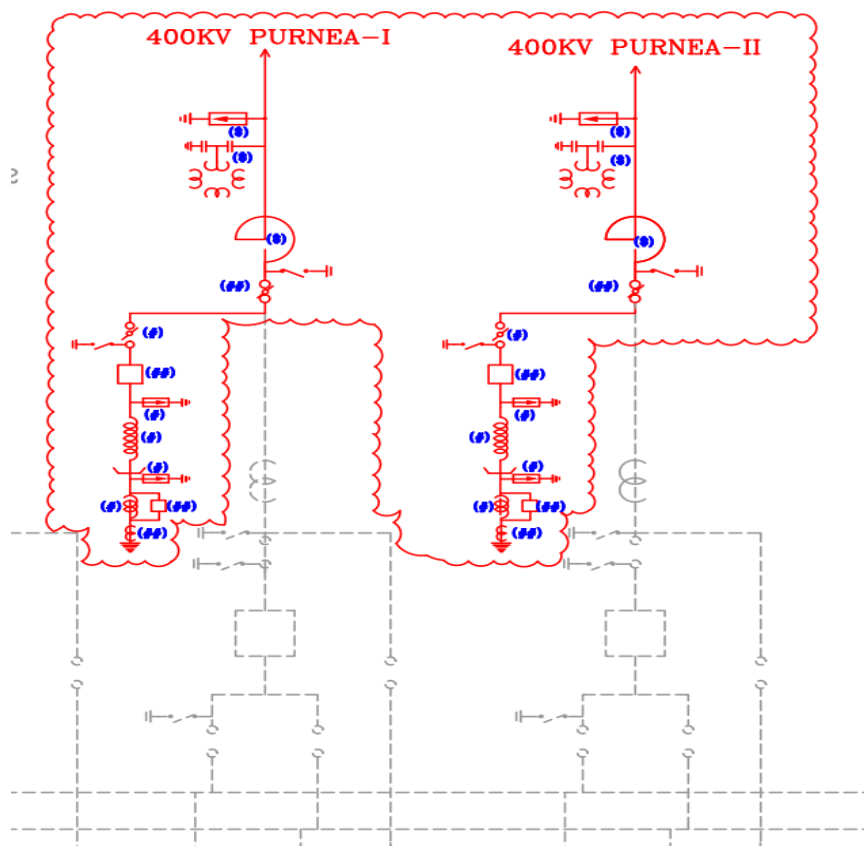
Powergrid ER-II requested for carrying out FTC of the 220kV Rajarhat-New Town-II Cable from New Town end after rectification.

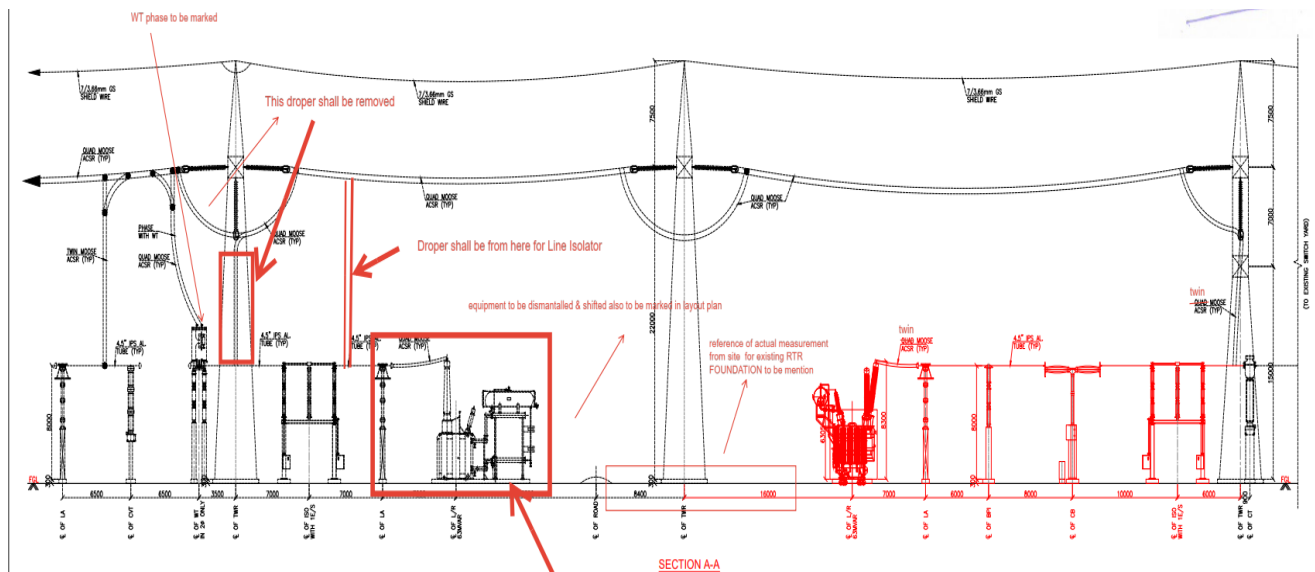
OCC Decision

- OCC noted the observations in report on no load charging of the cable.
- OCC advised WBSETCL to expedite cable repair works and reinstate all three phases to service.
- After rectification of the cable, the FTC should be carried out as proposed by Powergrid ER-II.

2.12 Modification of switchable L/R scheme for 400kV Malda-Purnea D/C at Malda SS under project ERSS-XL: Powergrid ER-II

- Under ERSS-XL scheme, existing 400 KV 63 MVAR L/R of 400kV Malda-Purnea D/C at Malda SS will be replaced.
- Further, NGR bypass arrangement along-with additional line isolator is proposed to be introduced in the system for taking the same switchable L/R as B/R whenever required.





Powergrid ER-II may explain. Members may discuss.

Deliberation in the meeting

Powergrid ER-II apprised:

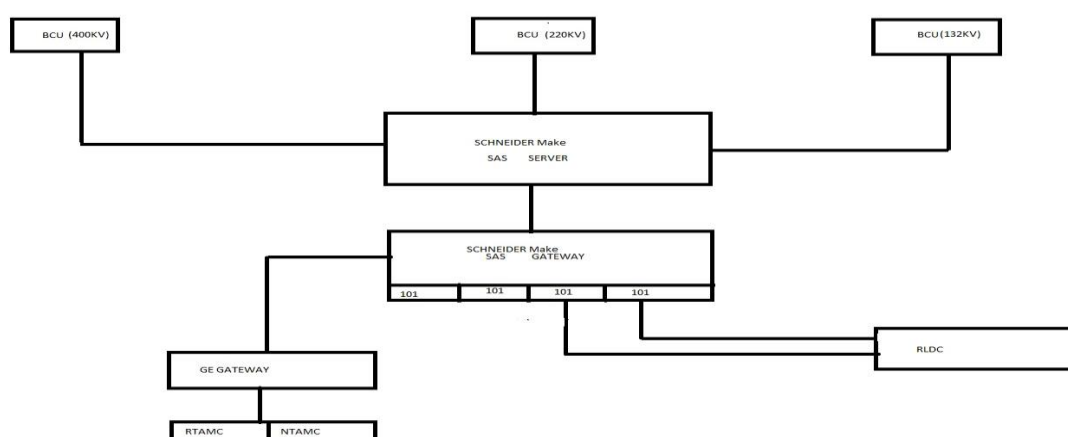
- ✓ Under ERSS-XL scheme, CTU has notified implementation of switchable LR
- ✓ Double Main Transfer (DMT) scheme being in place at Malda S/S, each feeder already has its own line isolator which separates line with Bus and presently Line Reactor is a part of line only without any switching arrangements.
- ✓ Normally in other S/S, where, one and Half CB scheme is prevailing, switchable LR of any line can be used as Bus Reactor also (Bypassing NGR & opening Line Isolator), however in this case, it is not possible to implement as existing LR is placed after Line Isolator and is a part of line only.
- ✓ Thus, in view of the existing DMT scheme at Malda S/S, deployment of bus reactor is not feasible from safety aspects as well as associated operational hazards.
- ✓ Although as per original requirement, existing fixed LR will be converted to Switchable LR.

OCC Decision

OCC noted and advised Powergrid to approach CTU.

2.13 Upgradation of Substation Automation System at Rangpo 400/220/132KV Substation: Powergrid ER-II

- 400/220/132KV Rangpo is a pooling station located at Sikkim is the most critical substation in Sikkim. Substation Automation System (Schneider Make) was commissioned at Rangpo S/s in 2014.
- Realtime data communication from POWERGRID Rangpo sub-station to ERLDC is done through 101 protocol as there is no provision of 104 protocol in the existing SAS.
- However, for NTAMC/RTAMC communication, additional gateway has been installed for reporting in 104 protocol. Present SAS Architecture is enclosed for reference.
- Now considering present, standard of communications it is prudent to go for complete 104 communication protocol & existing system required to be upgraded for Smart solution based on ethernet communication, ease of maintenance and troubleshooting, easy expandability in future etc.
- The upgradation of the present SAS is also very essential for reporting of SAS in 104 protocol for dual reporting philosophy. The life of the SAS items/IT items has been defined as 7 years as per CERC 2024-29 and the present SAS at Rangpo has already elapsed its life.



- The Cost for the above work is proposed to be booked under **ERBS-II (Schedule completion-July-2026) for all commercial purpose.**
- However, as similar LOA is already placed to M/S. Siemens for ULDC scheme, the rate/Scope will be taken from the same and also work is proposed to be carried out through LOA placed to SIEMENS for “package-Y for RTU replacement/ SAS upgradation package for Eastern region under upgradation of SCADA/RTUs/SAS in POWERGRID” where similar work is ongoing at various substations.

Powergrid ER-II may explain. Members may discuss.

Deliberation in the meeting

Powergrid ER-II apprised:

- ✓ Existing Rangpo SS was one of the first SAS based S/S among India and having comparatively older configuration of SAS architecture. Under present scope, data communication to RLDC/Remote gateways are still done through IEC 101 only and creating lot of disturbances and reliability issues.
- ✓ To address the issue, it is proposed to upgrade the existing SAS architecture with standard architecture with IEC 104 protocol. Further, he stressed upon that, already under SAS-Y package has rate/Scope of similar items executed in other SS and can be exercised. Further as per CERC tariff regulation (2024-29) vide definition of Useful Life, IT equipment/SCADA considered as 07 Years, whereas SCADA system at Rangpo is already 10 years old and commercially eligible for replacement.
- ✓ Although, as upgradation of Rangpo SAS is likely to be carried out in 2025, capitalization and commercial aspects (Tariff petitions) will not be linked with existing SAS-Y package.
- ✓ It is proposed to include necessary commercial implications for all purpose including Tariff Petition with ERBS-II package for the exact part of Rangpo SAS upgradation.

OCC Decision

- OCC accorded in-principle approval to the proposal of Powergrid ER-II
- OCC also referred the proposal to next CCM for commercial vetting.

2.14 Proposal for keeping SAS at Durgapur S/S under Package-Y as spare at SAS Lab of Rajarhat: Powergrid ER-II

- Upgradation/Replacement of SCADA/RTUs/SAS in POWERGRID Stations in Eastern Region has been carried out in package-Y by M/s. Siemens under the Project - Upgradation of SCADA/RTU/SAS in Central Sector Stations and strengthening of OPGW in Eastern Region. Under this project, SAS upgradation was envisaged at Durgapur S/S.
 - However, during detail engineering stage it is observed that merely upgrading, the communication equipment will not serve the purpose as life of other associated items for data reporting (MFTs/transducers/BCU etc) which are installed in the existing system have also come to an end and also there are many operational challenges as this is an old conventional substation. Therefore, only upgradation of SAS part will not accomplish the desired aim.
 - In view of the same, Upgradation of entire Durgapur Substation into a SAS based station (Distributed control) is being undertaken under ADDCAP 2024-2029 tariff block and it is expected that the work will be completed by March-2026.
- Further in Rajarhat Sub-station, POWERGRID, has already developed inhouse SAS Lab, which are used for training/learning/bench testing of various SAS items.
 - It is proposed to keep the SAS supplied at Durgapur Substation at SAS Lab, Rajarhat Substation, as ready spare for all Siemens make SAS system of ER-II.
 - Regular health monitoring can be done as well same can be used for training & development work.
 - Only Supply and F&I cost of the SAS item will be booked in ongoing project and the erection cost will be deleted from this project.

Powergrid ER-II may explain. Members may discuss.

Deliberation in the meeting

Powergrid ER-II submitted :

- ✓ Existing Durgapur S/S was one of the oldest SS of ER and having conventional RTU communication through IEC 101 protocol.
- ✓ As per SAS Y-Package, originally, communication protocol at Durgapur S/S was envisaged to be changed to IEC 104 through new SAS, however, going by detail engineering and various site level meetings it emerged merely changing, communication equipment/SAS equipment will not make complete system reliable. All the individual components also completed almost 35 years of service life and replacement is must.
- ✓ Further, complete upgradation of entire Durgapur S/S including SCADA is planned in ADDCAP 2024-29 under which communication with ERLDC will also be upgraded.
- ✓ Moreover, the supplied SAS at Durgapur, under subject package (Package-Y), is planned to be kept as ready spare in Rajarhat(PG) Lab for contingency in any of the Siemens make System in ER/ISTS.
- ✓ For commercial purpose, only supply portion will be capitalized in SAS-Y package and other portions will be deleted.

OCC Decision

- OCC accorded in-principle approval to the proposal of Powergrid ER-II
- OCC referred the proposal to next CCM for commercial vetting.

2.15 Update on AMR Phase-5 Completion: Powergrid ER-II

AMR Phase project was awarded in Mar-24 with scope of integration of additional 320 number of Meters in the AMR system. The schedule completion of the project scope was within Sep-24.

Scope	LOA Qty	Completed within Sep-24	Balance Qty
Integration of new SEM with AMR system	320	209	111

- All Meters that were available for AMR integration have been completed within the stipulated timeline.
- However, the remaining quantity of Meters were being informed after Sep-24 month and those have been informed to ERLDC in a phased manner.
- The requests have been coming from different constituents at different time (all after Sep-24) which is beyond the control of PGCIL/ERLDC.
- Team is working on integration of the remaining Meters, and it is expected to complete the entire scope within 31-Jan-2025, subjected to receipt of balance locations from ERLDC/Utilities.

Powergrid ER-II may explain. Members may discuss.

Deliberation in the meeting

Powergrid ER-II updated the status of AMR Phase-5 implementation as follows:

- ✓ Out of awarded quantity of total 320 meters , 275 meters have been successfully integrated in AMR and remaining meters expected to be completed by Feb 2025.
- ✓ AMR integration is being carried out in phased manner as and when requirement is being received from the constituents.

OCC Decision

- OCC advised JUSNL to expedite OPGW connectivity for facilitating AMR integration.
- All ER constituents were also advised to share the requirement of AMR integration of their meters with Powergrid/ERLDC at the earliest to expedite completion of AMR phase-5 project in Eastern region.

2.16 Conducting VAPT assessment for AMR Asset in ERLDC (in Compliance of Cyber security guidelines): Powergrid ER-II

- AMR system Data Centre hardware & software was refreshed in Jun-23. New Servers, Networking Devices and the new AMR application are currently being used in AMR system.
- ERLDC using the system for getting the SEM data on weekly basis. Current System is completely isolated from ERLDC IT Network, and the entire data communication is happening over LAN network.
- No public IP based data communication is present, and the AMR application is only accessible at ERLDC LAN Network.
- However, considering the upcoming requirements and the recent testing/POC of new type of Meters, new set of data requirements etc. it is recommended to conduct a complete VA/PT assessment of the system.
- This assessment will help to locate the necessary process improvement (If any) areas in order to make the system more robust and eliminate the security vulnerabilities (if any). The VA/PT assessment will be conducted once in every six months, as per the guidelines.

Count of the AMR Hardware as below:

Server	07	AMR Application	01
Managed Switches	02		
Firewall	01		
Router	01		

VA/PT will be conducted on both the Hardware and the Application. Cost estimation has been received from a CERT-IN certified vendor to conduct one round of VAPT. The existing AMR system Contract is valid till Mar-2026, hence VAPT will be needed for upcoming 04 cycles starting from Dec-24 month. Cost involvement will be as below:

Description	Unit Price (without GST)	No. of Cycle	Total Price (without GST)
VAPT of Hardware	40,000	04	1,60,000
Security testing of AMR Application	25,000	04	1,00,000
Total Cost of ownership (without GST)			2,60,000

PGCIL will initiate the necessary tendering (Limited tendering among CERT In certified vendors) processes subject to approval.

Powergrid ER-II may explain. Members may discuss.

Deliberation in the meeting

OCC Decision

- OCC agreed to the proposal(along with cost implications) of carrying out VAPT assessment of AMR assets in ERLDC by CERT-In enlisted vendors. This is to be done after completion of third party cybersecurity audit.
- OCC referred to CCM for commercial vetting.

2.17 List of feeders to be disconnected as a physical regulatory measure to control over drawl by constituents: ERLDC

- It has been observed that All India frequency is going below IEGC band for a considerable period with minimum frequency touching around 49.5 Hz.
- A number of measures are taken by NLDC, RLDCs and SLDCs to control overdrawal in case of sustained low frequency.
- In case non-compliance of RLDC's deviation message, physical disconnection of feeders is also undertaken as a measure of last resort to minimize the over-drawl.
- The list of feeders identified by Eastern region constituents is attached as **Annex B.2.17.1**, which was ratified in the **184th OCC(Annex B.2.17.2)** meeting. However, in the meantime, there is a change in load distribution with the growth of network interconnection and demand. There is a need to review the feeder list.

All constituents are requested to review the following-

- I. Identification of feeders (Addition/modification)
- II. Feeders-wise Load distribution.

Members may discuss.

Deliberation in the meeting

ERLDC highlighted:

- ✓ Under-frequency is observed in the grid owing to overdrawl by some constituents.
- ✓ Feeder disconnection will be the last resort to prevent under-frequency. Alert message for sustained overdrawl shall be given beforehand to the concerned constituent

OCC Decision

- All SLDCs were advised to confirm the existence as well as feasibility of disconnection for the feeders as per **Annex B.2.17**
- In case of disconnection as last resort, only radial feeders catering to non-critical loads to be considered. The feeder list to be finalized by ERLDC in consultation with ER states.

2.18 Review of AUFLS in Eastern region: ERPC

- ◆ A Task Force was constituted by NPC vide letter dated 25.08.2023 on Implementation of AUFLS and df/dt scheme under the chairmanship of Member Secretary, SRPC and comprising members from NPC, RPCs and Grid-India.
- ◆ The Task force after convening meeting on 11.09.2023 submitted its report to NPC in 14th NPC meeting on 05.02.2024, wherein certain recommendations were made.

- ♦ Accordingly, as per decision of 214th OCC meeting, a special meeting was convened on **10.07.2024** to deliberate on successful implementation of Automatic Under Frequency Load Shedding (AUFLS) in Eastern region wherein following course of action was delineated to all constituent ER states.
- ♦ **Action points:**
 - All SLDCs were instructed to shift the load quantum from Stages –III & IV to stage-I & II respectively as an interim measure till new feeders for additional load relief gets identified by individual state DISCOMs.
This must be implemented at the earliest with necessary changes in frequency settings of the existing UFRs and the same shall be reviewed in upcoming OCC meeting.
 - All SLDCs were advised to share the identified feeders list for revised load relief quantum within a month. The status shall be reviewed in monthly OCC meetings.
 - Curtailment of critical loads should be avoided. However, in stage-III and stage-IV, as it operates only in severe threat to grid stability, industrial loads may also be considered. Accordingly DVC and IPCL (having dominant industrial consumers) were urged to identify industrial feeders for load relief in stage-III and stage-IV.
 - All SLDCs were urged to expedite and ensure SCADA visibility of existing as well as newly identified feeders under AUFLS for effective supervision of load relief quantum.
- ♦ Based on submission by DVC, revised load relief quantum as follows:

(Figs in MW)

Constituent	Stage-1	Stage-2	Stage-3	Stage-4	Total
Bihar	315	379	442	442	1577
Jharkhand	87	105	122	122	437
DVC	172	207	241	241	861
Odisha	306	367	428	428	1530
West Bengal	497	597	696	696	2486
Sikkim	5	6	7	7	25
Total	1383	1660	1937	1937	6916

Constituent wise	Annual Consumption	Consumption factor	Demand met	Peak demand factor	Demand contribution
Bihar	40952	0.220	7578	0.236	0.228
Jharkhand	12391	0.067	1923	0.060	0.063
DVC	26214	0.141	3476	0.108	0.125
Odisha	41142	0.221	7104	0.221	0.221
West Bengal	65009	0.349	11868	0.370	0.359
Sikkim	526	0.003	137	0.004	0.004
Total Consumption	186234	1.000	32086	1.000	1.000

52nd TCC Decision:

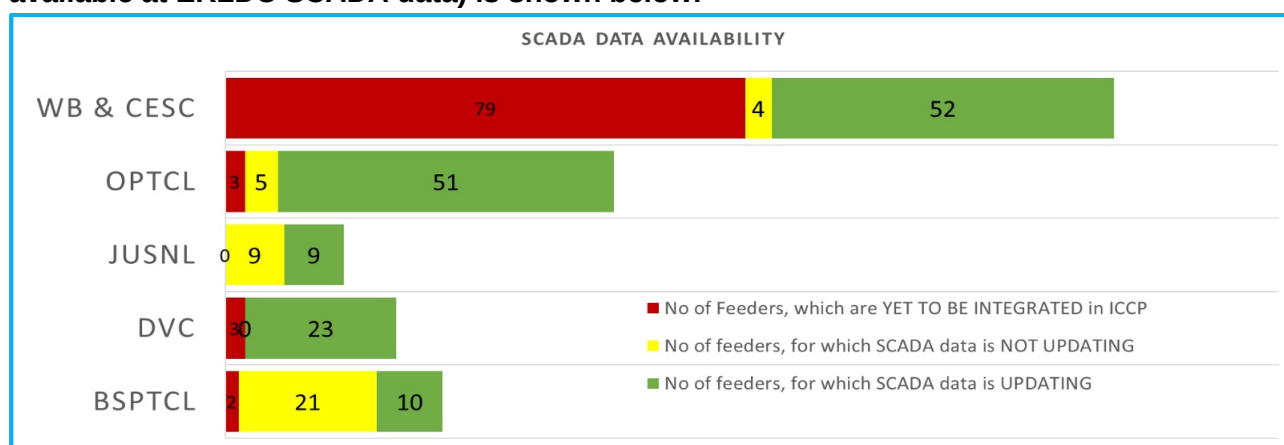
- ❖ SLDC, Odisha was directed to take up the matter with DISCOM to identify the feeder list and shifting of load at the earliest to implement AUFLS.
- ❖ Director, SLDC Odisha agreed to coordinate with concerned DISCOM and **update in next OCC.**

TCC advised all SLDCs :

- ❖ To expedite the process of implementation of AUFLS in stage I & II by shifting load quantum from stage III & IV at the earliest time possible.
- ❖ Explore the identification of new feeders to incorporate AUFLS in stage III & IV who have successfully implemented AUFLS in stage I & II by shifting load quantum from stage III & IV.
- ❖ To share the list of newly identified feeders with ERPC Secretariat within One Month for information.
- ❖ To ensure SCADA data mapping from newly identified UFR feeders at ERLDC level & In case of non-availability of SCADA data, anticipated timelines for making availability of SCADA data must be communicated for all applicable UFR feeders.

Status of shifting AUFLS stage 3 & 4 feeders to AUFLS Stage 1 and 2 and identification of additional feeders for all stages of AUFLS (as per information received by ERLDC) is attached at **Annex-B.2.18**

Latest SCADA data availability of feeders identified for AUFLS (as per information available at ERLDC SCADA data) is shown below:



All SLDCs/STUs and individual state DISCOMs may update action taken/future plan w.r.t AUFLS. Members may discuss.

Deliberation in the meeting

✓ Bihar SLDC apprised:

By end of Feb 2025, all four stages of AUFLS will be having adequate load relief quantum.

✓ DVC informed:

- All UFRs have been tested and successfully implemented in stages –I & II as per requirement.

- New feeders have been identified in stages –III & IV where UFR testing shall be completed in one month.

✓ WBSedCL updated:

New feeders for load shed in stages-III & IV have been identified and installation of UFRs will be completed by March 2025.

✓ SLDC Odisha informed:

All pending installation of UFRs across four stages of AUFLS will be completed by December 2024.

OCC Decision

OCC advised all SLDCs/STUs and concerned DISCOMs:

- To ensure availability of sufficient load relief(MW) in each of the four stages as follows:

Constituent	Stage-1	Stage-2	Stage-3	Stage-4	Total
Bihar	315	379	442	442	1577
Jharkhand	87	105	122	122	437
DVC	172	207	241	241	861
Odisha	306	367	428	428	1530
West Bengal	497	597	696	696	2486
Sikkim	5	6	7	7	25
Total	1383	1660	1937	1937	6916

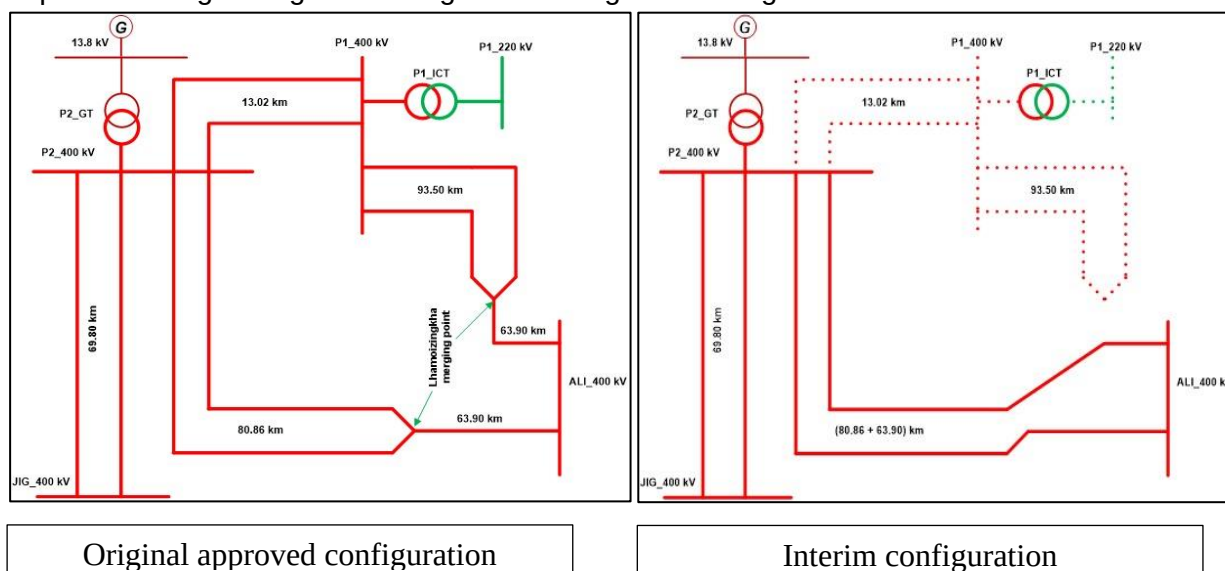
Stages-I &II, being first line of defence in AUFLS ,should be given first priority in maintaining required load relief quantum.

- All pending installation and testing of UFRs in Stages-I & II must be completed by Jan 2025.
- To identify new feeders for implementing AUFLS in stage III & IV who have successfully implemented AUFLS in stage I & II by shifting load quantum from stage III & IV. List of new identified feeders in stage III & stage IV of AUFLS must be share with ERPC by all SLDCs.
- To ensure SCADA data mapping for all newly identified UFR feeders at ERLDC level & in case of non-availability of SCADA data, anticipated timelines for making availability of SCADA data must be communicated for all applicable UFR feeders.
- To ensure periodic testing of UFRs for ascertaining their healthiness in coordination with ERLDC and submit report to ERPC/ERLDC.

The latest available AUFLS implementation status attached at **Annex-B.2.18** for reference of all ER states.

2.19 Status of Interim arrangement 400kV Alipurduar-Punatsangchu – Jigmelling: ERLDC

In the **43rd CCM** of ERPC dated **21.09.2020**, it was decided that Mangdechhu power generation would be evacuated through the interim arrangement of 400 kV Mangdechhu – Jigmelling - Punatsangchhu II – Alipurduar link. Accordingly, interim arrangement of 400kV Alipurduar - Jigmelling D/C through Punatsangchu was agreed.



Bhutan has taken shutdown of 400kV-Alipurduar(PG)-Punatsangchu-Jigmelling I & II from 10/12/24 to 25/12/24 to carry out the jumper connection and interconnection removal at Kamichu to facilitate the evacuation of upcoming Punatsangchu-II power generation. As information received from NLDC Bhutan, Punatsangchu-II (170x6 = 1020MW) has started power generation from 16/12/24 and evacuated power through 400kV Punatsangchu-Jigmelling D/C link.

Bhutan is requested to confirm the following:

1. Restoration of original network configuration plan.
2. Generation commissioning schedule.

ERLDC may elaborate. Members may discuss and Bhutan may update.

Deliberation in the meeting

ERLDC informed:

400 kV Alipurduar-Punatsangchu-Jigmelling D/c has been under shutdown since 10.12.2024 to carry out the jumper connection and interconnection removal at Kamichu to facilitate the evacuation of upcoming Punatsangchu-II power generation.

As per information received from Bhutan representative :

- ✓ *After disconnection of jumper, 400 kV Jigmelling-Punatsangchu II D/c has been charged and 2 units have started evacuating its generation through Jigmelling lines.*
- ✓ *400 kV Alipurduar-Punatsangchu II portion has not been charged yet.*
- ✓ *Two units of Punatsangchu HEP have already been synchronized with the grid. Other 4 units are expected to be synchronized by June-25.*

Powergrid ER-II sought:

- ✓ Details of longest and shortest lines from Punatsangchu along with fault level data which is required for protection setting calculation as per the new configuration(length reduced by around 65 km).
- ✓ The timeline of charging the line which is yet to be received from Bhutan.

OCC Decision

- Bhutan was advised to intimate ERLDC(India) on the timeline for restoration of original network configuration. Generation programme for Punatsangchu HEP should also be shared with ERLDC for proper planning.
- A special meeting must be convened by ERLDC/NLDC with Bhutan NLDC in this regard.

2.20 Shutdown proposal of generating units for the month of March'2025-ERPC

Deliberation in the meeting

OCC Decision

- OCC opined in favor of pre-ponement of all proposed maintenance schedules so that the respective generating units may be reinstated to service by end of March or early April 2025.
- All generating utilities were advised on carrying out maintenance activities of respective thermal generating units diligently in lean demand period to avert forced outage of the units in Summer 2025(peak demand period).

The detailed shutdown schedule of generating units (for the period: March-2025) as approved by OCC is enclosed at **Annex B.2.20**.

2.21 Deferment of Planned outage of Thermal generating units scheduled in April to June 2025: CEA

- ✓ Consequent to review meeting by Ministry of Power dated 12.12.2024 wherein minimization of planned outages during April 2025 to June 2025 was reiterated, CEA(GM division) has estimated shifting of around 1000-2000 MW outage planned in Apr-Jun 2025 to Oct-Nov 2025.
- ✓ Accordingly ER GENCOs are requested to consider postponement of following planned outages:

SYSTEM	CAPACITY (MW)	STATION NAME	UNIT NO	FROM	TO	DURATION (DAYS)	PROPOSED OUTAGE REASON
NPGC	660	NABINAGAR STPP	1	21-Jun-25	19-Aug-25	59	AOH(Boiler+Turbine + Gen)
DVC	500	BOKARO TPS `A` EXP	1	25-Mar-25	28-Apr-25	35	COH-Blr, Turb, Gen & HMI upgradation
	210	MEJIA TPS	2	1-Apr-25	28-Aug-25	149	ESP Augmentation
NTPC	500	TALCHER STPS	3	5-May-25	9-Jun-25	35	BOH

	800	DARLIPALI STPS	2	1-Apr-25	10-May-25	39	AOH (Main Turbine Thrust bearing temperature is high(110 deg C)
	210	KOLAGHAT TPS	4	27-Jun-25	21-Jul-25	24	—
WBPDCCL	500	SAGARDIGHI TPS	4	1-Apr-25	10-Apr-25	9	Boiler Lic Renewal

** As per the latest communication received from SRPC by mail, the Annual maintenance schedule for **2025-26 wrt Talcher Unit 3** has been finalised from **15-06-2025** to **19-07-2025 (35 days)**

Concerned thermal GENCOs may update.Members may discuss.

Deliberation in the meeting

- ✓ NTPC Darlipalli pleaded for availing shutdown of Unit#2 from 1st April 2025 highlighting high turbine bearing temperature as well as numerous cracks in boiler hangers.GRIDCO representative raised concern on managing intra-state demand portfolio in absence of Darlipalli Unit#2 during peak period.
- ✓ WBPDCCL agreed to defer the planned S/D of Kolaghat TPS Unit#4 to mid-July 2025.

OCC Decision

- OCC opined in favor of shifting the majority of planned outages for April-June 2025 to lean demand period in line with guidelines from Ministry of Power, Govt. of India as well as to ensure maximum resource availability in the crunch period (Summer 2025).Southern region constituents being major beneficiary in NTPC Talcher STPS Unit#3, the capacity under outage may be considered in Southern region.
- All generating utilities were advised to prepone their planned outages in March 2025 to extend maximum generation support to ER grid during Summer 2025.
- The revised schedule of planned outage for FY 2025-26 as approved by OCC is attached at **Annex-B.2.21** and the same will be shared with CEA.

2.22 Shutdown Program of Generating Units at NKSTPP: NTPC

Unit 1 Annual OH from **23.02.2025** for **35** days to address the following issues:

- Boiler overhauling and boiler license renewal
- Turbine bearing and valve inspection
- LP turbine overhauling
- Generator rotor thread out

OCC may approve.

NTPC may explain. Members may discuss.

Deliberation in the meeting

NKSTPP highlighted:

- ✓ Shutdown of Unit#1 is necessary to address the challenges pertaining to LP turbine and Generator rotor.
- ✓ For upcoming Unit#3: Boiler lightup in 1st week of January 2025 followed by steam blowing to be carried out. COD by end of March 2025.

SLDC Bihar submitted:

Bihar , having significant share allocated (45%) in North Karanpura STPP, shall be affected adversely in managing demand portfolio if shutdown availed in peak demand season.

OCC Decision

Assessing merit of the proposal, OCC approved shutdown of NKSTPP Unit#1 from 23rd February 2025 to 29th March 2025 for 35 days. At the same time, NTPC was advised to strictly abide by the timeline of unit restoration to service.

2.23 Postponement of overhauling BRBCL units: BRBCL

- Capital Overhaul of Unit-2 is scheduled from 8th Jan 2025 for 45 days. Being a major Overhaul HP, IP, LP Turbine and Generator will be opened and inspected. But in case of Unit 2 Overhaul spare Rotor will not be available at site before commencement of Rotor thread out and inspection activities as per present schedule.
- In unit 3 Capital overhauling in July 2024 crack was found in Generator Rotor during inspection and the same was replaced with spare Rotor. Therefore, it was decided that during Capital Overhauls a spare Rotor should be available at site before Rotor thread out.
- Therefore, it is requested that Unit 2 overhaul be rescheduled from 15th March 2024 onwards.
- This will give us sufficient time to arrange spare Rotor and there by timely complete the Overhauling as per Schedule

OCC may approve.

BRBCL may update. Members may discuss.

Deliberation in the meeting

OCC Decision

- OCC consented to the proposed shutdown of BRBCL generating units for 45 days, i.e from 15th March to 29th April 2025 subject to availability of rotor spare. In case of non-availability of rotor spare, the unit may be taken out of service for maintenance in July 2025.
- Being the major beneficiary, consent from Indian Railways should be taken prior to availing planned shutdown of the unit.

2.24 Proposal for Shutdown of units at Rangit HPS (3*20MW): NHPC

Planned outage requested for NHPC for generating units at Rangit HPS(3*20MW) is as follows:

Name of Power Station: RANGIT (3x20MW), SIKKIM

Sl. No.	Unit	IC (MW)	Shutdown Period		Reason
			From	To	
1	I	20	23 December 2024	26 December 2024	Replacement of MIV Seal of all the three Units
2	II	20	23 December 2024	26 December 2024	
3	III	20	23 December 2024	26 December 2024	

OCC may approve.

NHPC may explain. Members may discuss.

Deliberation in the meeting

OCC Decision

OCC granted consent to the proposed shutdown of units of Rangit HPS from 25th December 2024 for 4 days.

2.25 Annual Shutdown of 113 MW Rongnichu HEP: MBPCL

- MBPCL is anticipating the reduction in river inflow in coming months, particularly during the leanest period in February and are planning the annual maintenance activities for the Rongnichu HEP to coincide with this opportunity.
- Accordingly, the plant will undergo a complete shutdown from **15th February 2025 to 28th February 2025** to facilitate the required annual maintenance work.
- The primary activities during this period will include **the replacement of both units' runners and maintenance of the barrage gates**.

OCC may approve.

MBPCL may update. Members may discuss.

Deliberation in the meeting

OCC Decision

OCC consented to the proposed shutdown schedule of Rongnichu HEP **15th February 2025 to 28th February 2025** for carrying out annual maintenance.

2.26 Requirement of AGC Implementation Before COD: SJVN

- As per to Clause 24.2 of the IEGC: *"All thermal generating stations having a capacity of more than 200 MW and hydro generating stations having a capacity of more than 25 MW shall submit documents confirming the enablement of automatic operation of the plant from the appropriate load despatch centre by integrating the controls and tele-metering features of their system into the automatic generation control in accordance with the CEA Technical Standards for Construction and the CEA Technical Standards for Connectivity."*
- As per the Ministry of Power, Government of India, order dated 18/09/2024, power from the 2 x 660 MW BTPP has been allocated as 85% to Bihar and 15% as Unallocated (UA), effective from the date of commercial operation (COD).
- BTPP, Buxar, will transmit its power through the State Transmission Utility (STU) lines of Bihar at 400 kV and 220 kV.
- As per the Bihar Electricity Grid Code, mandate for Automatic Generation Control (AGC) for COD is not required.
- In this context, deliberation is sought on whether implementing AGC at BTPP, Buxar, is mandatory before COD based on the above provisions.

SJVN may explain. Members may discuss.

Deliberation in the meeting

ERLDC highlighted:

Though provision of AGC is mandated for CERC regulated generators, in view of increased need for reserves, AGC integration should be done for the intra-state generators also.

OCC Decision

- OCC opined that AGC implementation will not pose hindrance in COD of the Buxar TPP(SJVN) units. But at the same time, for optimal availability of reserves, SJVN was urged to implement AGC in BTPP units subject to consent from Bihar SERC.
- OCC also advised SJVN to execute COD of their 2 X660 MW plant by 1st week of March 2025.

2.27 Power Evacuation of BTPP through Buxar-Naubatpur 400 kV Lines: SJVN

- During the 206th OCC meeting held on 31.08.2023, consent was granted for the provision of startup power to the 2x660 MW Buxar Thermal Power Project (BTPP), Chausa, Buxar, through a temporary arrangement at the 400 kV through Naubatpur until the 220 kV system is operational.
- At present, BTPP, Buxar, is drawing startup power from SBPDCL via the 400 kV Buxar—Naubatpur Circuit-2, as per the temporary arrangement suggested in the 206th OCC meeting.
- The 220 kV GIS in BTPP is expected to be commissioned by January 2025, following which STPL plans to transition the startup power supply from the 400 kV line to the 220 kV line.
- In this context, STPL is seeking approval to allow evacuating the power through 400 kV Buxar—Naubatpur D/C lines till the readiness of the bay at Naubatpur for smooth Evacuation of Power.

SJVN may explain. Members may discuss.

Deliberation in the meeting

OCC Decision

- OCC advised SLDC Bihar to carry out load flow study for power evacuation and SJVN to resolve the issue of power evacuation bilaterally with BSPTCL/Bihar DISCOM.
- BSPTCL was advised to expedite in making the bay ready at Naubatpur for smooth power evacuation from Buxar TPP and submit the timeline of completion.

2.28 Power restoration plan for NKSTPP in case of total collapse of eastern region grid: NTPC

In Case of Total Collapse of Eastern Region Grid, Black Start & Restoration Procedure like Pre-define Black Start Source, Power route and associated Substations for Startup power for NTPC North Karanpura" is not available. ERLDC to provide the detailed Black Start & Restoration plan/procedure for NTPC North Karanpura.

NKSTPP and ERLDC may please explain. Members may discuss.

Deliberation in the meeting

OCC Decision

OCC advised ERLDC to finalize the necessary plan for black start, early restoration after grid collapse and route for start-up power drawl i.r.o North Karanpura STPP by 31st January 2025.

2.29 Discrepancies in SG and DC for NKSTPP: NTPC

Discrepancies in SG and DC for NKSTPP:

- SG was more than DC for **Block No.13,17,89,93 & 94** on **15.12.2024** and **Block No.89 & 95** on **14.12.2024**. OCC is requested to deliberate.

NTPC may please explain. Members may discuss.

Deliberation in the meeting

ERLDC submitted that the issue of discrepancy has been resolved in implemented schedule and North Karanpura STPP agreed to the same.

OCC Decision

OCC noted.

2.30 Periodic Testing of power system elements: ERPC

As mandated in **IEGC 2023, 40.1 & 40.2**, periodic tests shall be carried out on power system elements to ascertain the correctness of mathematical models used for simulation studies as well as ensuring desired performance during an event in the system.

Relevant portion of clause is as below:

Quote:

“

40. PERIODIC TESTING

.....

40.2 (a) The owner of the power system element shall be responsible for carrying out tests as specified in these regulations and for submitting reports to NLDC, RLDCs, CEA and CTU for all elements and to STUs and SLDCs for intra-State elements.”

40.2 (b) “All equipment owners shall submit a testing plan for the next year to the concerned RPC by 31st October to ensure proper coordination during testing as per the schedule. In case of any change in the schedule, the owners shall inform the concerned RPC in advance.

.....”

Unquote

In 217th OCC Meeting held on 24.07.2024, the matter was discussed in detail and OCC advised all the generators & owners of HVDC/FACTS devices to strictly adhere to the IEGC 2023 guidelines & submit the required testing data & plan to ERPC at the earliest (as per clause 40.2.(b).

None of the generators or owners of HVDC/FACTS devices have submitted the testing plan yet. All are requested to submit the testing schedule at the earliest.

Power Elements	System	Tests	Applicability
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Synchronous Generator	(1) Real and Reactive Power Capability assessment. (2) Assessment of Reactive Power Control Capability as per CEA Technical Standards for Connectivity (3) Model Validation and verification test for the complete Generator and Excitation System model including PSS. (4) Model Validation and verification of Turbine/Governor and Load Control or Active Power/ Frequency Control Functions. (5) Testing of Governor performance and Automatic Generation Control.	Individual Unit of rating 100MW and above for Coal/lignite, 50MW and above gas turbine and 25 MW and above for Hydro.
HVDC/FACTS Devices	(1) Reactive Power Controller (RPC) Capability for HVDC/FACTS (2) Filter bank adequacy assessment based on present grid condition, in consultation with NLDC. (3) Validation of response by FACTS devices as per settings.	To all ISTS HVDC as well as Intra-State HVDC/FACTS, as applicable

221st OCC Meeting:

- All GENCOs of ER were advised to furnish the testing plan i.r.o individual synchronous generators within a week positively.
- ERLDC was requested to maintain a consolidated database on testing schedule of all generating utilities of ER.

ERLDC had already shared a Google sheet link to update the testing schedule and circulated the link to Gencos. Link of the Google sheet is as below:

Periodic testing - Google Sheets:

<https://docs.google.com/spreadsheets/d/1m6KCkONdObMhre9-1me1kvHTEBYUdXUOISYdn5FR4fM/edit?gid=0#gid=0>

NTPC, CESC, MPL, Bihar & DVC submitted the tentative schedule for testing. All other Gencos are requested to submit schedule in the Google sheet.

All GENCOs and HVDC/FACTS owners may update.

Deliberation in the meeting

OCC Decision

- All GENCOs of ER who have not yet submitted details were advised to furnish the testing plan i.r.o individual synchronous generators in the shared google sheet within a week positively in compliance to IEGC 2023.
<https://docs.google.com/spreadsheets/d/1m6KCkONdObMhre9-1me1kvHTEBYUdXUOISYdn5FR4fM/edit?gid=0#gid=0>
- ERLDC was requested to maintain a consolidated database on testing schedule of all generating utilities of ER.

2.31 Periodic Mock Drill Exercises in areas of generation, transmission and distribution of the power sector: ERPC

In compliance to **Disaster Management Plan for Power Sector (2022)** as drafted by CEA(as per Disaster Management Act 2005) and approved by Ministry of Power (Govt. of India) as well as in order to be prepared for any eventuality, periodic mock drill exercises are to be undertaken in various areas of generation, transmission and distribution of the power sector by considering various crisis and disaster situations like an earthquake, floods etc. Depending on the vulnerability of the installations/plant, mock drills to handle such situations need to be undertaken. The utilities are also required to ensure that at least one mock drill exercise for every crisis/disaster situation to which the installation/plant is vulnerable is undertaken in each quarter. The adverse observations made on each event of Mock drill should be taken into account and it should be ensured to prevent occurrence of such undesirable events in the future.

In this regard, Secretary (Security) , Cabinet Secretariat, Govt of India has stressed on undertaking the following measures:

- ✓ Availability of details pertaining to local district authorities, revenue authorities, law enforcement, fire management authorities, etc., across the townships
- ✓ Adequate vetting of personnel/organisation responsible for township security by local law enforcement agencies.
- ✓ Regular conduct of mock drills in the townships, especially evacuation drills with
- ✓ ambulance and drills for handling major fire accidents.

Letter from Joint Secretary(MOP) to all power/energy secretaries (states) enclosed at **Annex-B.2.19**.

Till now quarterly mock drill reports have been received from NHPC and WBPDC. Mock drill reports regularly received from WBPDC. NTPC has shared mock drill reports for ER thermal generating units for Q2 of FY 2024-25

□ Action points:

As per deliberation of **1st MEETING ON REGIONAL DISASTER MANAGEMENT (EASTERN REGION)** dated **09.07.2024**(MOM at **Annex-B.2.19** :

- ♦ At least one mock drill exercise for every crisis/disaster situation to which the installation/plant is vulnerable must be undertaken in each quarter and quarterly report by the utilities to be shared with CEA for review and onward submission to Ministry of Power (Govt of India) . (Action: All thermal GENCOs (Central,IPP), all hydro generating stations, all ISTS licensees . SLDCs to coordinate with respective GENCOs,STUs and DISCOMs within their jurisdiction)
- ♦ Utilities are requested to share the experience on the mock drill exercises and scope for improvements.

221st OCC Decision:

OCC advised all the utilities to:

- Conduct periodic Mock Drills i.e. at least one mock drill exercise in each quarter to which the installation/plant is vulnerable in order to be prepared for any unforeseen eventuality.

- Share Quarterly mock drill reports with ERPC which will then be sent to CEA for review & finally report will be submitted to Ministry of Power (Govt of India). This is in compliance to Disaster Management Plan in Power sector and Disaster management Act 2005.
All concerned utilities may update action plan.

Deliberation in the meeting

OCC Decision

- ✓ OCC advised all the utilities to:
 - Conduct periodic Mock Drills i.e. at least one mock drill exercise in each quarter to which the installation/plant is vulnerable in order to be prepared for any unforeseen eventuality.
 - Share Quarterly mock drill reports with ERPC which will then be sent to CEA for review & finally report will be submitted to Ministry of Power (Govt of India). This is in compliance to Disaster Management Plan in Power sector and Disaster management Act 2005.
- ✓ Besides all generating utilities, all transmission licensees (ISTS licensee and STUs) were advised to furnish detailed mock drill reports on quarterly basis. SLDCs were requested to coordinate with respective GENCOs, STUs and DISCOMs within their jurisdiction in this regard.

2.32 Training on working principle of Energy Meters and vis-a-vis the AMR system for data retrieval: Powergrid ER-II

- In Eastern Region, all the constituents, ERLDC, ERPC have been dealing with different type of Meters, different protocols, and the subsequent automated system for Meter data collection. Also, the system is going through periodic upgradation, process improvements, various security aspects in order to be prepared for the upcoming regulations of Renewable Energy accounting.
- In this regard, PGCIL is planning to organize a training session for all the constituents of ER along with ERLDC, ERPC to give a detailed understanding of the process. The proposed date will be finalized in consultation with ERPC/ERLDC. Mainly Online training session will be conducted for various OEM,s and first will be done for M/S. Secure make meters in Jan-2025.

Members may discuss.

Deliberation in the meeting

OCC Decision

OCC commended the initiative by Powergrid ER-II and thereby recommended to conduct training on working principle of meters for knowledge sharing among the constituents.
OCC advised all the utilities to avail this opportunity.

ADDITIONAL AGENDA

2.33 Update on 3rd ICT at Rajarhat (PG): WB SLDC

- In **July 2023(28.07.2023)**, at **21st CMETS-ER** meeting West Bengal conveyed their consent for 3rd 500 MVA transformer for Rajarhat (PG) sub-station. Load pattern of last summer highly supports the need.

Concern of West Bengal SLDC:

- It is apprehended that maximum around 475 MW per ICT loading may reach in coming summer across each ICT of Rajarhat (PG) sub-station in case of nonavailability of the 3rd ICT at Rajarhat (PG) sub-stn.
- If the said ICT is not coming to service by summer 2026, then severe congestion will result across ICTs of Rajarhat(PG) as apprehended. Hence rolling on the process of procurement, transportation etc are the essence to avoid extreme time pressure during transportation, commissioning phases. Also to mention here that state Bidhansabha Election is expected in 2026.
- Prior to this also, SLDC WB has given this agenda in 212th and 220th OCC meetings to expedite the procurement process.

As per deliberation in **212th OCC** meeting:

- West Bengal SLDC representative, based on present loading pattern in and around Rajarhat(PG) ,raised deep concern on facing power crisis at Rajarhat(PG) in 2025-2026 similar to present scenario at Subhasgram(PG). He urged to lay equal emphasis on installation of 3rd ICT at Rajarhat (PG) as that of Subhasgram (PG) to thwart recurrence of similar emergent situation in future. Thus importance of regular monitoring of 3rd ICT commissioning at Rajarhat (PG) was underscored.
- Powergrid ER-II apprised the following:
 - Tender for procurement of new ICT at Rajarhat (PG) had been floated and presently in technical evaluation stage, price bid is awaited to be opened.
 - LOA shall be tentatively placed by end of March 2024.
 - Delay in planning for new ICT requirement was highlighted which in turn is leading to unwanted delays at later stage, since ICT procurement, especially for GIS like Rajarhat (PG) is undoubtedly a time-consuming affair.
 - Standard timeline for new ICT commissioning is 18 months as per CEA guidelines.

212th OCC Decision:

- OCC advised Powergrid ER-II to closely coordinate with Powergrid Corporate centre so as to expedite tendering process keeping urgent requirement of the ICT at Rajarhat (PG) in consideration.
- OCC advised Powergrid ER-II to adhere to standard timeline of 18 months(as per CEA guidelines) for 3rd ICT commissioning at Rajarhat (PG).

An update of the status of procurement process vs target time to commission may please be discussed and intimated.

West Bengal SLDC may explain and Powergrid ER-II may update. Members may discuss.

As per deliberation in **220th OCC** meeting:

West Bengal SLDC submitted:

- Regular monitoring of 3rd ICT commissioning at Rajarhat(PG) that was approved in 21st CMETS-ER, is needed .
- If the 3rd ICT doesn't get charged at Rajarhat(PG) by Summer 2026, reliable power supply scenario will get jeopardized.

Powergrid ER-II updated:

- The tender floated for procurement of 3rd ICT, as informed in 217th OCC , has been annulled and a fresh tender shall be floated soon.
- The new ICT commissioning at Rajarhat(PG) is not possible by March 2025.
- Regional spare of 500 MVA ICT is available at Maithon(PG) but the same is not possible for immediate commissioning at Rajarhat(PG) owing to highly challenging transportation process.

220th OCC Decision:

- OCC advised Powergrid ER-II to coordinate with Corporate Centre for re-initiating the procurement process of 3rd ICT at Rajarhat (PG) on priority. The 3rd ICT at Rajarhat (PG) must be put to service latest by March 2026 and accordingly planning should be carried out by Powergrid to avoid any instance of grid vulnerability.
- OCC advised Powergrid ER-II to plan the procurement and installation process for 3rd ICT commissioning at Rajarhat (PG) adhering to the standard timeline of 18 months (as per CEA guidelines).
- OCC also urged West Bengal STU to consider Long term planning of intra-state network of West Bengal in terms of augmentation or expansion (New Lakhikantapur S/S).

WB SLDC may elaborate and Powergrid ER-II may update. Members may discuss.

Deliberation in the meeting

Powergrid ER-II updated:

- ✓ *Tender for procurement of the 3rd ICT has been annulled twice and currently re-tendering is under progress.*
- ✓ *Bid opening is scheduled tentatively in Feb 2025, thereafter commissioning of the ICT will take 18 months.*
- ✓ *The 3rd ICT shall be commissioned by end of 2026 or first half of 2027.*
- ✓ *Rajarhat(PG) being a GIS, charging of the new ICT may consume a bit more time.*

WB SLDC raised concern on the delay in commissioning of 3rd ICT that may result in crisis situation in Summer 2026 as both ICTs at Rajarhat(PG) are expected to operate on full load in Summer 2025 itself.

OCC Decision

- OCC expressed serious concern on non-availability of 3rd ICT at Rajarhat(PG) as per earlier committed timeline.
- OCC advised Powergrid ER-II to coordinate with Corporate Centre for expediting the procurement process of 3rd ICT at Rajarhat (PG) on priority. The 3rd ICT at Rajarhat (PG) must be put to service latest by March 2027 and accordingly planning should be carried out by Powergrid to avoid any instance of grid vulnerability.

- OCC advised ERLDC to explore interim network arrangement to evade possible crisis situation in upcoming Summer in absence of 3rd ICT at Rajarhat(PG). ERLDC was also requested to convene a special meeting on challenges in intra-state transmission system of West Bengal.
- ERLDC was advised to put up a report on percentage utilization of ICTs in ER stations so that structured planning on optimal spares requirement may be carried out.

2.34 Restoration of 132 kV Chuzachen-Rangpo-1: ERLDC

132 kV Chuzachen-Rangpo-1 has been under breakdown since **29.11.2024**. As reported, B-ph conductor has snapped between tower number 58 and 59. ERLDC communicated to Sikkim on **12th December 2024** to update the tentative plan for restoration and yet to receive any communication. The entire generation of Chuzachen is being evacuated through the remaining **132 kV Chuzachen-Rangpo-2**.

As Chuzachen evacuation depends on other ckt, the entire generation will be affected in case of tripping of existing line. Necessary action may be taken to restore the line at the earliest. So that reliability of the chuzachen evacuation may be improved.

ERLDC may elaborate. Sikkim and Chuzachen may update.

Deliberation in the meeting

ERLDC informed:

Circuit-1 of 132 kV Chuzachen-Rangpo line has been under permanent outage since 29th November 2024 due snapping of B-phase conductor.

As a consequence, there lies possibility of total generation loss of Chuzachen HEP in case .

OCC Decision

OCC urged Govt. of Sikkim to initiate prompt action for restoration

2.35 Annual Maintenance of JLHEP(2*48MW): DANS ENERGY

We are anticipating the reduction in river inflow in the coming months, particularly during the leanest period in **February**, we have strategically planned the annual maintenance activities for Jorethang Loop HEP to coincide with the opportunity.

Accordingly, the plant will undergo a complete shutdown from **1st February 2025 to 15th February 2025** to facilitate the required annual maintenance work.

The primary activates during this period will include under water parts maintenance of **Turbine, MIV and seal beam of under sluice Gates at Barrage**.

If the maintenance work will complete earlier or requires additional time the plant will be restored accordingly. Any changes to the schedule will be communicated in advance

DANS ENERGY may update. Members may discuss.

Deliberation in the meeting

OCC Decision

OCC granted consent to the proposed shutdown of units of Jorethang Loop HEP from 1st February 2025 to 15th February 2025.

3. PART-C: ITEMS FOR UPDATE/FOLLOW-UP/INFORMATION

3.1. ER Grid performance during November 2024.

The average and maximum consumption of Eastern Region and Max/Min Demand (MW), Energy Export for the month November-2024 were as follows:

AVERAGE CONSUMPTION (MU)	MAXIMUM CONSUMPTION(MU)/ DATE	MAXIMUM DEMAND (MW)	MINIMUM DEMAND (MW)	SCHEDULE EXPORT	ACTUAL EXPORT
		DATE / TIME	DATE / TIME	(MU)	(MU)
452.1 MU	526.1 MU, 01.11.2024	25883 MW, 02.11.2024 at 17:45 Hrs.	15055 MW, 25.11.2024 at 03:48 Hrs.	5904	6067

ERLDC/ERPC may highlight the performance of the ER grid.

Deliberation in the meeting

The grid performance of ER for the month of November 2024 was highlighted.

3.2. Update on Reconductoring of ISTS lines under Eastern Region Expansion Scheme-44: ERPC

- ❖ Several 220 kV transmission lines and substations were implemented in Indian grid along with cross border lines for importing power from Chukha Hydro Electric Plant in Bhutan. The generating station was commissioned in years 1986-88 and the transmission system is now more than 35 years old. Considering the age of conductors and increase in conductor snapping incidences, reconductoring of these transmission lines has become necessary.
- ❖ The matter was deliberated in various OCC forums as well as in 52nd TCC meeting of ERPC.
- ❖ In a meeting was convened by CEA under the chairpersonship of Member (Power System) on 27-08-2024, it was decided that matter of reconductoring of cross border lines will be separately taken up with Bhutan.
- ❖ However, reconductoring of ISTS portion of 220 kV corridor viz. Alipurduar (POWERGRID) – Falakata (WBSETCL) – Birpara (POWERGRID) – Binaguri (POWERGRID) – Siliguri (POWERGRID) – Kishanganj (POWERGRID) – Dalkhola (POWERGRID) – Gazole (WBSETCL) – Malda (POWERGRID), may be taken up under ISTS. Further, reconductoring of intra-state LILO portion of Birpara (POWERGRID) – Alipurduar (POWERGRID) 220 kV D/c line at Falakata (WBSETCL) and Dalkhola – Malda 220 kV D/c line at Gazol (WBSETCL) shall be carried out by WBSETCL matching with HTLS conductor of the main ISTS line in the matching timeframe.

Name of the scheme	Implementation timeframe	Implementation mode	Implementing agency	Estimated Cost (Rs. in Cr)
ERES-44	18 months (15 months on best effort basis) from the date of allocation	RTM	Powergrid	385.77

WBSETCL works associated with reconductoring of ISTS lines

- ❖ In the NCT(National Committee on Transmission) meeting dated 23.10.2024 , the following were decided:
- ✦ WBSETCL shall reconductor their following lines sections under intra-state scheme matching with completion of ISTS scheme namely ERES-44:
- ✓ About 4 km intra-state portion of Alipurduar (POWERGRID) Falakata (WBSETCL) 220 kV D/C line at Falakata end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Falakata (WBSETCL) end commensurate with rating of HTLS (1250 A).
- ✓ About 4 km intra-state portion of Birpara (POWERGRID) - Falakata (WBSETCL) 220 kV D/C line at Falakata end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Falakata (WBSETCL) end commensurate with rating of HTLS (1250 A).
- ✓ About 2 km intra-state portion of Dalkhola (POWERGRID) - Gazole (WBSETCL) 220 kV D/C line at Gazole end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Gazole (WBSETCL) end commensurate with rating of HTLS (1250 A).
- ✓ About 2km intra-state portion of Gazole (WBSETCL) - Malda (POWERGRID) 220 kV D/C line at Gazole end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Gazole (WBSETCL) end commensurate with rating of HTLS (1250 A).
- ✦ WBSETCL will LILO the Dhalkola - Gazole 220 kV D/C line with 1250 A HTLS under their intra-state scheme for establishment of 220 kV level at their existing 132/33kV Raiganj (WBSETCL) S/S.
- ✦ ISTS licensee and WBSETCL shall coordinate for reconductoring of their respective portion of the lines matching with completion schedule of this scheme.
- It is kindly requested that WBSETCL may note the scope of works (as provided in the minutes of NCT) and coordinate with POWERGRID for matching implementation of their works.
- The **progress report** may be shared on **monthly** basis to **CEA, ERPC and CTU**.
- ◆ MOM of NCT meeting dated 23.10.2024 along with CTU letter attached at **Annex-B.3.2**

As per **221st OCC** Deliberation:

- OCC was apprised of the decision in the NCT meeting (23.10.2024) on reconductoring of 220 kV network of Chukha Transmission system (CTS).
- Powergrid ER-II intimated:

- Tendering process of reconductoring works shall commence soon and NIT will be floated within a month. Award of contract shall take around 7-8 months
- BOQ for reconductoring works is currently under progress wherein provision of TLA is included in view of increased incidents of autoreclosure and tripping in the thunder prone area of North Bengal.
- Most of the end-equipment connected to Chukha Transmission system have been upgraded.
- Shutdown requirement of the critical in-service 220 kV lines was highlighted.
 - In absence of WBSETCL representative, the modalities of reconductoring works to be carried out on part of WBSETCL matching with the timeframe of ISTS lines, could not be finalized.

221st OCC Decision:

- OCC advised Powergrid ER-II to expedite the reconductoring works of CTS as per timeline approved in 24th NCT meeting (MOM attached at **Annex-B.3.2**) and share monthly progress report of the same with CEA, CTU and ERPC.
- Powergrid was also advised to conduct a bilateral meeting with Bhutan authorities to streamline the reconductoring plan in portions of 220 kV lines within the territory of Bhutan. This has to be done prior to proceeding with the tendering process of reconductoring works.
- OCC opined that shutdown requests pertaining to reconductoring of CTS shall be treated on priority and to be approved to the best feasible extent so that reconductoring of CTS gets completed as per timeline approved in 24th NCT meeting.
- WBSETCL was advised to coordinate with Powergrid ER-II for carrying out reconductoring works on intra-state portions of 220 kV lines under CTS matching with the timeframe of ISTS lines. This is in compliance to decisions of 24th NCT meeting.

WBSETCL and Powergrid may respond. Members may discuss.

Deliberation in the meeting

Powergrid ER-II updated:

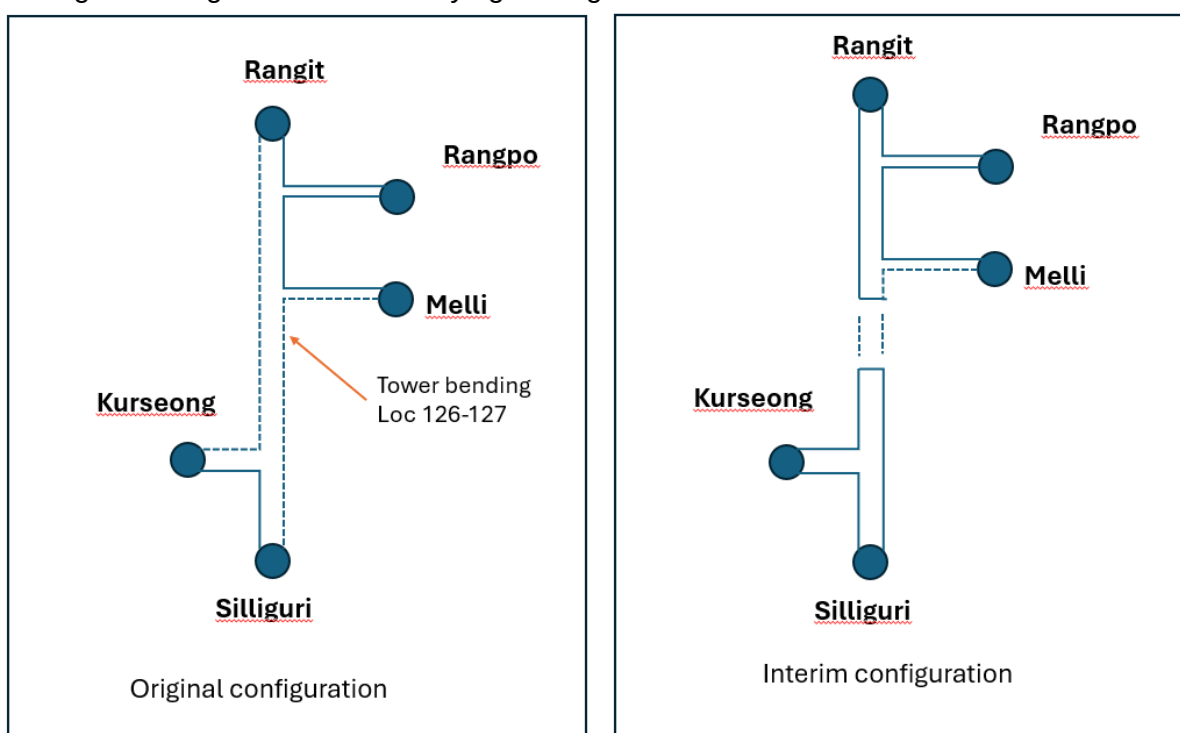
Bilateral meeting with Bhutan as well as WBSETCL on finalizing the modalities of reconductoring shall be convened by 15th January 2025. The outcome of the same will be intimated in next OCC.

OCC Decision

- *OCC advised Powergrid ER-II to expedite the reconductoring works of CTS as per timeline approved in 24th NCT meeting (MOM attached at **Annex-B.3.2**) and share monthly progress report of the same with CEA, CTU and ERPC.*
- *Powergrid was also advised to conduct a bilateral meeting with Bhutan authorities to streamline the reconductoring plan in portions of 220 kV lines within the territory of Bhutan. This has to be done prior to proceeding with the tendering process of reconductoring works.*
- *WBSETCL was advised to coordinate with Powergrid ER-II for carrying out reconductoring works on intra-state portions of 220 kV lines under CTS matching with the timeframe of ISTS lines. This is in compliance to decisions of 24th NCT meeting.*

3.3. Update on Restoration of 132kV Rangit-Kurseong & 132kV Siliguri-Melli-Rangpo lines: ERLDC

- Due to incessant rain and several landslides, towers at loc. 125-128 of **132 kV Rangit-Kurseong** and **132 kV Siliguri-Melli** got badly affected. Out of which tower at loc. 126,127 got severely damaged. Both the lines were switched on 5th October 2024 on request of PowerGrid.
- Consequently, **Kurseong** and **Melli** (Kalimpong source) are fed through single source of **Siliguri** and **Rangpo** respectively. To ensure reliable power supply at Melli & Kurseong, ERLDC conducted one meeting on **08.10.2024** (online mode) with participants from ERPC, ERLDC, West Bengal SLDC, Sikkim, Powergrid and NHPC Rangit.
- Considering the difficulties & time requirements due to hilly terrain for restoration of the said portion, temporary reconfiguration of these lines was explored to extend additional sources to Melli & Kurseong. It was decided that part of the healthy line of **132 kV Siliguri-Melli** will be reconfigured as **132 kV Siliguri-Kurseong ckt2** as a second source of Kurseong and another healthy portion of **132 kV Siliguri-Melli** will be reconfigured as **132 kV Rangit-Melli** for a second source of Melli.
- After necessary reconfiguration, **132 KV Siliguri-Kurseong-II (interim)** arrangement charged on **9th October** and **132kV-Rangit-Melli (interim)** has been charged tentatively on **22nd October**. POWERGRID intimated that it would take 15-20 Days to restore the original configuration after rectifying damaged towers.



220th OCC Decision

- OCC opined that reliability of power supply to Melli, Kurseong, adjoining Darjeeling area and power evacuation of Rangit should not be compromised.

- OCC urged Powergrid to expedite to the best feasible extent in restoration of original configuration of 132 KV D/C Siliguri-Melli & Rangit-Kurseong Lines.
- OCC requested Govt. of Sikkim to extend all possible assistance to Powergrid in resolving the ROW issues and restoring damaged towers at the earliest.
- OCC advised WBSETCL, WBSEDCL and Powergrid to mutually decide on the shutdown requests of 132 kV D/C NBU-Lebong TL & 132 kV D/C Lebong-Rammam lines

As per **221st OCC** deliberation:

- Powergrid ER-II updated that there were persistent ROW issues in tea garden adjoining the damaged tower location. Also Hill shanking observed around the affected area.
- Excessive tension of the conductor in vicinity of damaged tower no-127 due to power line crossing the span. Dismantling of Tower No-127 is under progress and would be completed by 07.12.2024.
- Soil testing and piling works are presently being carried out. Based on soil testing results, new foundation works as well as rectification of Tower No-126 shall be taken up.
- Erection works shall commence from 1st week of January 2025. Tower No-127 to be erected by 06.01.2025 while rectification of Tower No-126 to be completed by 16.01.2025.
- Re-stringing in spans :127-128 and 126-127 to be completed by 25.01.2025 and 29.01.2025 respectively.
- Modification of jumpering to make original ckt of 132 KV Siliguri-Melli Ckt and may be tentatively taken into service by 30.01.2025.
- Modification of jumpering to make original ckt of 132 KV Siliguri-Kurseong Ckt and may be tentatively taken into service by 31.01.2025.

221st OCC Decision:

- OCC requested Powergrid ER-II to expedite restoration activity of original configuration of 132 KV D/C Siliguri-Melli & Rangit-Kurseong Lines to the best feasible extent.

Concern of West Bengal SLDC/WBSETCL:

- ✓ Tower line restoration work after damage of Towers 126/127 at Soom Tea Estate, Darjeeling, Rangit-Kurseong-Siliguri (PG) normal configuration couldn't be restored in last 2 and half months. The temporary arrangement has put Kurseong sub-station on single source(Siliguri(PG)) with obvious problem of redundancy.
- ✓ On **20/01/2024**, PG has proposed for total bus shutdown at Siliguri(PG), which will lead to "No Power" status to Kurseong. Hence, the normal configuration needs to be restored before that.
- ❖ **Powergrid ER-II** has also communicated its concerns on lowering of OPGW of WBSETCL line at powerline crossing (Span:10-11) to Chief Engineer(O&M-I,WBSETCL) on **16.12.2024**.

WBSETCL and Powergrid may update. Members may discuss.

Deliberation in the meeting

Powergrid ER-II apprised:

- Short pilling for the tower 126-127 is not feasible in the area due to lack of adequate soil strength. The required soil strength is of 8-9 unit however it has been found to have below 4.5 unit.
- Thus, nearby locations have been surveyed with higher soil strength to hold the tower.
- New towers have been diverted from Berhampore and Dalkhola.
- New route for shifting of tower no-127 & 126 has been carried out as it is not possible to reassert the line in same tower foundation.
- Restoration of the said line with diverted route will be completed by 15.03.2025.

OCC Decision

- OCC advised Powergrid ER-II to expedite restoration activity of original configuration of 132 KV D/C Siliguri-Melli & Rangit-Kurseong Lines to the best feasible extent.
- OCC requested Govt. of West Bengal to extend all possible administrative assistance to Powergrid ER-II for resolving the persistent ROW issues.
- WBSETCL was advised to coordinate with Powergrid ER-II in lowering the existing OPGW of 132 kV WBSETCL lines below the span 126-127 shall be routed underground via approach cable and overhead Powergrid lines shall provide lightning protection to these 132 kV lines of WBSETCL. Letter dated 16.12.2024 from Powergrid ER-II to WBSETCL attached at **Annex-B.3.3** for initiating necessary action.

3.4. Update on actions taken to prevent repeated tripping of 132 kV Chuzachen-Rangpo D/C: ERPC

- **132 kV Chuzachen-Rangpo D/C** tripped more than **10 times** since **May'24** causing total generation loss occurred at Chuzachen HEP (110 MW) due to sequential tripping of both lines in three instances.
- In most of the trippings, **phase to phase** fault was reported with a distance of around **12 km** from **Rangpo**.
- A joint committee with members from Powergrid, Chuzachen HEP and Sikkim transmission wing, Dept. of Power (Sikkim) was constituted for joint site inspection. The committee submitted its report after visiting the site on 01.10.2024.
- Committee observations during the visit were as below:
 - ✓ Critical tree infringement and bamboo trees between loc. 27-29 along the corridor.
 - ✓ Severe infringement along with several flashover marks on the conductor and burnt trees along the corridor.
 - ✓ Less ground clearance b/w loc. 28-29 for Ckt-1 (4.1 meter instead of minimum requirement of 6.1 meter).
- **The Committee recommended two new towers to be constructed between loc. 28-29 and 35-36 (one each) and hill cutting along the periphery of tower no. 27 to improve ground clearance.**
- Considering the severity of less ground clearance and potential of damage to human life, the recommended measures need to be implemented on an immediate basis.
- As per **221st OCC** Deliberation:
In absence of Sikkim representative, the latest status could not be updated

The same matter was discussed in the 220th OCC Meeting wherein OCC advised Sikkim to expedite in implementation of Committee recommendations i.r.o increasing ground clearance by construction of a new tower(between loc. 28-29) and hill cutting (around tower no. 27). OCC also advised to update the same to ERPC/ERLDC every week. However, the update has yet not been received.

Sikkim may update. Members may discuss.

Deliberation in the meeting

OCC Decision

OCC advised Sikkim SLDC and representative from Sikkim transmission wing (Govt. of Sikkim) to attend the next OCC meeting for updating the status on actions taken as per Committee recommendations.

3.5. Update on Restriction of Talcher-Kolar HVDC Bi-pole: ERPC

- ❖ HVDC Talcher-Kolar Pole-2 was operated at reduced capacity from March 24, 2024, due to problem with the R-phase converter transformer at the Talcher end. There was no spare converter transformer at Talcher and subsequently, it was decided to shift the spare converter transformer from HVDC Kolar to Talcher (PG)
- ❖ Since April'24, either pole of HVDC blocked 5 times out of which, in 4 times the other pole went to ground return mode instead of metallic return mode resulting in overloading of 400kV Talcher-Meeramundali D/C and generation backdown was done either manually or through operation of SPS.
- ❖ Meanwhile, power order of Talcher-Kolar poles was reduced to 1500MW from 2000MW due to which other critical lines of the region were getting overloaded. **Accordingly, Talcher generation was curtailed in the range of 800-900MW during peak hours for approximately 50 days in the summer, impacting both Eastern Region (ER) and Southern Region (SR) beneficiaries.** Later on, with decrease in ambient temperature HVDC Talcher-Kolar power order was restored and generation back down was withdrawn.
- ❖ **To mitigate the risk of similar power supply challenges experienced during summer 2024, Odisha (PG) may share the replacement plan of existing Converter Transformer.**

As per 221st OCC Deliberation.:

Powergrid Odisha apprised:

- The Converter transformer has arrived at Talcher(PG) in the first week of October 2024 and presently oil filtration is under process. The same shall be ready for charging by end of November 2024.
- If this converter transformer is put into service, no spare will be available at HVDC Talcher or HVDC Kolar stations to deal with any contingency. So it is proposed to keep this converter transformer as hot spare (ready for service) that may be replaced within 5-6 days as per grid requirement.

ERLDC submitted:

If the converter transformer is kept as hot spare as proposed Powergrid Odisha, it shall lead to backing down of generation at NTPC Talcher for at least one week. Consequently NTPC Talcher as well as ER beneficiaries will be adversely affected.

221st OCC Decision

OCC recommended Powergrid Odisha to keep the existing converter transformer as spare and put the new one into service at HVDC Talcher station. Powergrid Odisha agreed to propose the same to Powergrid Corporate for obtaining necessary clearance at the earliest.

Powergrid Odisha may update. Members may discuss.

Deliberation in the meeting

ERLDC raised concern on generation backdown of NTPC Talcher and consequent challenge of demand portfolio management to be faced by ER beneficiaries in upcoming Summer 2025 if the converter transformer is not charged and put to service at HVDC Talcher station at the earliest.

Powergrid Odisha updated:

Formal clearance from Powergrid Corporate is still pending for charging the new converter transformer at HVDC Talcher station.

OCC Decision

OCC advised Powergrid Odisha to present a detailed report on the same in next OCC.

3.6. Update on Implementation of AGC in Intra-state generating units: ERLDC

- ♦ AGC is now operational at most ISGS plants across India, which together have a total installed capacity exceeding 70 GW. However, the dispatchable margin provided through AGC and Secondary Reserve Ancillary Services (SRAS) remains insufficient for maintaining frequency within the IEGC band. With the increasing penetration of renewable energy, managing frequency is expected to become more challenging in the future. Therefore, it is crucial to enhance frequency control and stability through increased participation from intra-state AGC.
- ♦ In response to this need, efforts are underway to encourage more intra-state generators to join the SRAS scheme. Feasibility reports have been prepared, and stakeholder meetings have been held with DVC, West Bengal, and Bihar to explore potential solutions and address any concerns.
- ♦ Present status of **Intra-state AGC integration** process is as follows:

SLDC/State	Generator name	Unit Capacity (MW)	Status
Bihar	Barauni unit # 8 & 9	2x250	Pending discussion between NTPC Barauni, SLDC Bihar and its DISCOM for mutually agreeing to Mechanism for recovery of one-time cost of AGC implementation and Mechanism for Sharing of gains which is to be fixed bilaterally.

DVC	Mejia-B, DSTPS and Koderma	(2x500) (2x500) (2x500)	Final procurement order was awarded to Siemens on 7th August 2024 with timeline of completion of 4 months .
West Bengal	Units of WBPDC	-	West Bengal SERC notified WBERC (Ancillary Services) regulation, 2023 dated 26th December 2023. M/s WBPDC refers to WBSERC for implementing the AGC server at WBSLDC after which plants will be connected to SLDC one by one.

As per deliberation in **52nd TCC**:

- ❖ DVC apprised that final procurement order was awarded to Siemens on 7th August 2024 for all identified six Units & it is expected that within 4 months AGC implementation will be completed.
- ❖ NTPC representative informed that NOC for implementing AGC in its Barauni unit # 8 & 9 is yet to be received from SLDC, Bihar & also discussion is pending between NTPC Barauni, SLDC Bihar and its DISCOM for mutually agreeing to cost recovery and gain sharing mechanism.
- ❖ RED, NTPC mentioned that since the need for AGC Implementation in its Barauni unit # 8 & 9 is principally agreed & for this, a formal clearance from Bihar is required. They will resolve the issue by joint meeting.
- ❖ ED, ERLDC requested SLDC, Odisha to organize a meeting with OPGC to formulate a methodology so that OPGC units can be integrated with AGC.
- ❖ SLDC, Odisha submitted that attempt has been made by OPGC but OEM has not yet responded.
- ❖ OPGC suggested to have a special meeting with M/S BHEL & SLDC, Odisha to finalize the modalities of Implementation of AGC & will update the status within one month.
- ❖ WB SLDC submitted that another meeting shall be convened with WBPDC to resolve contractual issues and decide next course of action.

TCC Decision:

- TCC appreciated efforts of DVC in initiating AGC implementation process
- SLDC Bihar and Bihar DISCOMs were advised to resolve the pending issues with NTPC bilaterally for AGC implementation at the earliest.

- SLDC Odisha was advised to organize meeting with OPGC and ERLDC to resolve AGC implementation in OPGC units.
- WB SLDC was advised to resolve contractual issues with WBPDCCL bilaterally for expediting AGC implementation.
- TCC advised all the concerned utilities to expedite the execution process & complete the AGC Implementation as early as possible.
- Status of AGC implementation to be updated regularly in OCC meetings.

As per **221st OCC** Deliberation:

- DVC updated that final procurement order for AGC has been placed.
- WBPDCCL informed that the modalities of procurement are yet to be finalized with WBSLDC.

221st OCC decision

- OCC advised NTPC and Bihar to convene joint meeting for obtaining formal clearance from Bihar i.r.o AGC implementation.
- WBPDCCL was advised to coordinate with WB SLDC and update the status in next OCC meeting.

All concerned may update the status. Members may discuss.

Deliberation in the meeting

A special meeting was held between WB SLDC and WBPDCCL wherein following key outcomes emerged:

- ✓ *Part of WB SLDC in AGC implementation is already included in the scope of ULDC Phase-III SCADA project. So no need of separate tendering from WB SLDC end.*
- ✓ *Tendering for AGC will be carried out by WBPDCCL separately.*

OCC Decision

OCC noted the proceedings and recommended for intra-state AGC integration in Bihar, West Bengal and DVC at the earliest.

3.7. LAN integration & Port Opening status of AMR Phase-5 :Powergrid ER-II

In reference to the agenda discussed previously in OCC, the following locations are yet to be connected through LAN connection to ERLDC:-

SNO	STATE	UTILITY	SUBSTATION
1	WB	WBSETCL	KLC BANTALA
2	BIHAR	BGCL	DUMRAO NEW
3	JHARKHAND	JSEB	LATEHAR (**)

It may be noted that, TCS/ERLDC team has requested many times for port opening but till now no confirmation received from respective utilities.

**** - integration request received on 19th Dec 2024.**

It is requested to enable LAN for these locations as early as possible. LAN port details may be shared with PGCIL/ERLDC.

Powergrid ER-II may explain. Members may discuss.

Deliberation in the meeting

OCC Decision

- OCC opined that LAN integration is essential for reliable and uninterrupted transfer of AMR data from all utilities to ERLDC.
- WBSETCL and BGCL were advised to confirm port opening for ensuring LAN integration at the earliest (within a week). Bihar SLDC was requested to coordinate with BGCL in this regard.

3.8. Regarding Non-Submission of Forecasting Data from States: ERLDC

The **Clause 2 of Regulation 31 of IEGC 2023** has mandated all the SLDCs to timely submit the demand estimate data to the respective RLDC and RPC.

The demand estimation data provided by SLDCs will be required in resource adequacy planning and regional load forecasts conducted by the RLDC.

Currently, the day ahead data is regularly received from all the states except Sikkim.

221ST OCC decision:

- ♦ OCC advised all SLDCs for strictly adhering to the schedule of demand estimation as mandated in IEGC 2023, timely sharing with ERLDC as well as uploading of forecasting error on their respective websites.
- ♦ OCC advised all SLDCs for strictly adhering to the schedule of demand estimation as mandated in IEGC 2023, timely sharing with ERLDC in specified format as well as uploading of forecasting error on their respective websites.
- ♦ SLDCs who are submitting day ahead forecast were advised to also share the forecasting data for their respective control areas on weekly as well as monthly basis with ERLDC.
- ♦ All SLDCs were also urged to regularly furnish resource adequacy data besides demand forecast.
- ♦ SLDC Odisha was advised to expedite implementation of the demand forecasting software and positively update the status in next OCC.

Latest Forecast receipt as 15.12.24 status is shown below:

DATA RECEIPT STATUS BY 5 th DAY FOR THE MONTH OF	JAN 25 FORECAST
Bihar	NO
DVC	YES
Jharkhand	YES
Odisha	NO
Sikkim	YES
West Bengal	NO

Weekly Resouce Adequacy:

DATA RECEIPT STATUS BY 1ST	25.11.24 TO 01.12.2024_FC	02.12.2024 TO 08.12.2024_FC	09.12.24 TO 15.12.24_FC	16.12.2024 TO 22.12.2024_FC	23.12.2024 TO 29.12.2024_FC
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WORKING DAY FOR THE WEEK OF					
Bihar	NO	NO	NO	NO	NO
DVC	NO	NO	NO	NO	NO
Jharkhand	NO	NO	NO	NO	YES
Odisha	NO	NO	NO	NO	NO
Sikkim	YES	YES	YES	YES	YES
West Bengal	NO	NO	NO	NO	NO

Weekly Forecast:

DATA RECEIPT STATUS BY 1ST WORKING DAY FOR THE WEEK OF	25.11.24 TO 01.12.2024_FC	02.12.2024 TO 08.12.2024_FC	09.12.24 TO 15.12.24_FC	16.12.2024 TO 22.12.2024_FC	23.12.2024 TO 29.12.2024_FC
Bihar	NO	NO	NO	YES	YES
DVC	YES	YES	YES	YES	YES
Jharkhand	NO	YES	NO	YES	YES
Odisha	NO	NO	NO	NO	NO
Sikkim	YES	YES	YES	YES	YES
West Bengal	NO	NO	NO	NO	NO

Resource adequacy Data Receipt Status

DATA RECEIPT STATUS BY 5 th DAY FOR THE MONTH OF	JAN 25 RA
Bihar	NO
DVC	NO
Jharkhand	YES
Odisha	NO
Sikkim	YES
West Bengal	NO

Daily Resource Adequacy:

DATE	Bihar RA	DVC RA	Jharkhand RA	Odisha RA	Sikkim RA	West Bengal RA
01-12-2024	NO	NO	NO	NO	YES	NO
02-12-2024	NO	NO	NO	NO	YES	NO
03-12-2024	NO	NO	NO	NO	YES	NO

04-12-2024	NO	NO	NO	NO	YES	NO
05-12-2024	NO	NO	NO	NO	YES	NO
06-12-2024	NO	NO	NO	NO	YES	NO
07-12-2024	NO	NO	NO	NO	YES	NO
08-12-2024	NO	NO	NO	NO	YES	NO
09-12-2024	NO	NO	NO	NO	YES	NO
10-12-2024	NO	NO	NO	NO	YES	NO
11-12-2024	NO	NO	NO	NO	YES	NO
12-12-2024	NO	NO	NO	NO	YES	NO
13-12-2024	NO	NO	NO	NO	YES	NO
14-12-2024	NO	NO	NO	NO	YES	NO
15-12-2024	NO	NO	NO	NO	YES	NO
16-12-2024	NO	NO	YES	NO	YES	NO
17-12-2024	NO	NO	YES	NO	YES	NO
18-12-2024	NO	NO	YES	NO	YES	NO
19-12-2024	NO	NO	YES	NO	YES	NO
20-12-2024	NO	NO	YES	NO	YES	NO
21-12-2024	NO	NO	YES	NO	YES	NO
22-12-2024	NO	NO	YES	NO	YES	NO
23-12-2024	NO	NO	YES	NO	YES	NO
24-12-2024	NO	NO	YES	NO	YES	NO
25-12-2024	NO	NO	YES	YES	YES	YES
26-12-2024	NO	NO	YES	YES	YES	YES
27-12-2024	NO	NO	YES	YES	YES	YES
28-12-2024	NO	NO	YES	NO	YES	YES
29-12-2024	NO	NO	YES	YES	YES	YES
30-12-2024	NO	NO	YES	YES	YES	YES

Daily Forecast:

DATE	Bihar Forecast	DVC FC	Jharkhand FC	Odisha FC	Sikkim FC	West Bengal FC
01-12-2024	NO	YES	YES	YES	YES	YES
02-12-2024	NO	YES	YES	YES	YES	YES
03-12-2024	NO	YES	YES	YES	YES	YES
04-12-2024	YES	YES	YES	YES	YES	YES
05-12-2024	YES	YES	YES	YES	YES	YES
06-12-2024	NO	YES	YES	YES	YES	YES
07-12-2024	YES	YES	YES	YES	YES	YES
08-12-2024	YES	YES	YES	YES	YES	YES
09-12-2024	NO	YES	YES	YES	YES	YES
10-12-2024	NO	YES	YES	YES	YES	YES

11-12-2024	YES	YES	YES	YES	YES	YES
12-12-2024	YES	YES	YES	YES	YES	YES
13-12-2024	YES	YES	YES	YES	YES	YES
14-12-2024	YES	YES	YES	YES	YES	YES
15-12-2024	YES	YES	YES	YES	YES	YES
16-12-2024	YES	YES	YES	YES	YES	YES
17-12-2024	YES	YES	YES	YES	YES	YES
18-12-2024	YES	YES	YES	YES	YES	YES
19-12-2024	YES	YES	YES	YES	YES	YES
20-12-2024	YES	NO	YES	NO	YES	YES
21-12-2024	NO	YES	YES	YES	YES	YES
22-12-2024	YES	YES	YES	YES	YES	YES
23-12-2024	YES	YES	YES	YES	YES	YES
24-12-2024	NO	YES	YES	YES	YES	YES
25-12-2024	NO	NO	YES	YES	YES	YES
26-12-2024	NO	NO	YES	YES	YES	YES
27-12-2024	NO	YES	YES	YES	YES	YES
28-12-2024	YES	YES	YES	NO	YES	YES
29-12-2024	YES	YES	YES	YES	YES	YES
30-12-2024	YES	YES	YES	YES	YES	YES

Hence it is again requested to all the concerned for timely submission of demand estimation data to ERLDC. This collaboration is essential for effective planning and preparedness to meet the region's electricity demands efficiently and reliably.

ERLDC may explain and all SLDCs may update. Members may discuss.

Deliberation in the meeting

OCC Decision

- OCC advised all SLDCs for strictly adhering to the schedule of demand estimation as mandated in IEGC 2023, timely sharing with ERLDC in specified format as well as uploading of forecasting error on their respective websites.
- SLDCs who are submitting day ahead forecast were advised to also share the forecasting data for their respective control areas on weekly as well as monthly basis with ERLDC.
- All SLDCs were urged to regularly furnish resource adequacy data besides demand forecast.
- SLDC Odisha was advised to expedite implementation of the demand forecasting software and positively update the status in next OCC.

3.9. Non-Submission of FRC data in stipulated timeframe: ERLDC

Adhering to IEGC clauses **30.8** and **30.10.(a)** to **30.10.(q)**, generating stations within the Eastern region are required to submit essential data to ERLDC within two days of receiving a notification regarding a reportable frequency event. Additionally, according to clause 30.10.(n), all control areas within the eastern region must assess their frequency response

characteristics and share the evaluation, along with high-resolution data, with the ERLDC. Therefore, timely submission of primary response data is crucial for compliance with the IEGC.

221st OCC decision:

All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas.

The latest data receipt status is given below: (as on **15.12.2024**)

STATIONS	03.04.2024	06.04.2024	19.04.2024	23.04.2024	02.05.2024	10.05.2024	28.05.2024	04.06.2024	04.06.2024	11.06.2024	17.06.2024	19.06.2024	16.07.2024	23.08.2024	13.09.2024	21.10.2024
	05:29	11:24	10:28	20:15	14:41	19:35	19:45	10:26	10:34	14:10	13:53	12:42	22:10	12:34	13:15	16:49
FSSTP #STG 1 & 2	Pending	Pending	Pending	Pending	Pending	Received	Pending	Received	Received	Received	Pending	Pending	Received	Received	Received	Received
FSSTP # STG 3	Pending	Pending	Pending	Pending	Pending	Received	Pending	Pending	Pending	Received	Pending	Pending	PLANT OUT	Received	Received	Received
KhSTPP #STG 1	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending
KhSTPP #STG 2	Pending	Received	Received	Received	Received	Received	Pending	Received	Received	Pending	Received	Received	Received	Received	Received	Received
TSSTP #STG 1	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received
Barh stage-1	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Received	Received	Received	Received	Received	Received 29.07	Received	Received	Received
Barh stage-2	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Received	Received	Received	Received	Received	Received	Received	Received	Received
BRBCL	Pending	Pending	Received	Received	Received	Received	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Received	Received	Received
Darlipalli	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received
North Karanpura	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received
NPGC	Received	Received	Received	Received	Received	Received	Received	Received	Received	Pending	Received	Received	Received	Received	Received	Received
GMR	Received	Received	Received	Received	Received	Received	Received	Pending	Received	Received	Received	Pending	Received	Received	Pending	Received
MPL	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received
ADHUNIK	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received
JITPL	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Pending	Pending	Pending	Received
INDBHARAT	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending
TASHIDING	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Bihar	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Jharkhand	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending
DVC	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Received	Received	Pending	Pending	Pending	Pending	Pending	Pending	Pending
OPTCL	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received	Received
WB	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending	Pending

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill the google sheet below to include the email address where notifications of reportable events should be sent.

https://docs.google.com/spreadsheets/d/1slvAOMQIEQVIMn0LnB78eKMa2sz2QYICZ-sPEpeV_jk/edit?usp=sharing

ERLDC may explain. Members may discuss.

Deliberation in the meeting

OCC Decision

All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas

3.10. Mock Black Start: ERLDC

- As per IEGC Reg. 34.3: A mock trial run of the procedure for different sub-systems including black-start of generating units along with grid forming capability of inverter-based

generating station and VSC-based HVDC black-start support at least once a year under intimation to the concerned SLDC and RLDC.

- Also, diesel generator sets and other standalone auxiliary supply source to be used for black start shall be tested on a weekly basis and the test reports are to be shared to the concerned SLDC, RLDC and NLDC on a quarterly basis.
- As per IEGC Reg. 34.4: Simulation studies are to be carried out by each user in coordination with RLDC for preparing, reviewing and updating the restoration procedures considering the following:

- (a) Black start capability of the generator;
- (b) Ability of black start generator to build cranking path and sustain island;
- (c) Impact of block load switching in or out;
- (d) Line/transformer charging;
- (e) Reduced fault levels;
- (f) Protection settings under restoration condition

As per intimation received in Winter Preparedness 2024 dated 12.11.2024 hosted by ERLDC, a tentative date was received from each user regarding the mock drill of black start of generating units under their jurisdiction. The same is listed below:

Sl. No.	Name of Hydro Station	Schedule of Mock Black Start	Tentative date as on 12.11.2024	2024 Actual Date of Test
1	U. Kolab	Jun-24	Jan-24	
2	Balimela	Jul-24	Nov-24	
3	Rengali	Jun-24	Nov-24	
4	Burla	Jul-24	Jan-24	
5	U. Indravati	May-24	N/A	Sep-24
6	Maithon	Dec-24	2nd week of Dec-24	
7	TLDP-III	Oct-24	Nov-24 – Dec-24	
8	TLDP-IV	Oct-24	Nov-24 – Dec-24	
9	Subarnarekha	Sep-2024 4th week	1st week of Dec-24	3 rd December 2024
10	Teesta-V	N/A	N/A	N/A
11	Chuzachen	Oct-24	Yet to be informed	
12	Teesta-III	N/A	N/A	N/A
13	Jorethang	Dec-2024 3rd week	Yet to be informed	
14	Tashiding	2nd week of Dec 2024	Yet to be informed	
15	Dikchu	N/A	Yet to be informed	N/A

16	Rongnichu	Mar-24	Test already conducted	18th March and 20th March 2024
17	Mangdechu		Yet to be informed	

All the users are requested to confirm dates for mock drill of black start of each generating unit. Also, the users are requested to share the data required simulation studies before the scheduled date of mock drill.

Deliberation in the meeting

OCC Decision

- OCC advised all black start capable hydro generating units of ER to update their schedule of mock black start to ERLDC at the earliest. This is in compliance to IEGC 2023 (CERC).
- OCC further opined that in case of non-receipt of further update by respective hydro generating units the proposed tentative schedule of mock black start may be considered as final. Thereafter all black start capable hydro units shall have to conduct mock black start at least once in a year as mandated in IEGC 2023.

4. PART-D: OPERATIONAL PLANNING

4.1. Anticipated power supply position during January-2024

The abstract of peak demand (MW) vis-à-vis availability and energy requirement vis-à-vis availability (MU) for the month of January 2024 is prepared by ERPC Secretariat (**Annexure D.1**) on the basis of LGBR for 2024-25 and feedback of constituents, keeping in view that the units are available for generation and expected load growth etc.

Members may update.

Deliberation in the meeting

*The updated anticipated power supply position for January 2024 is provided at **Annexure D.1**.*

4.2. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on as on 15-12-2024)

a) **Thermal Generating Stations outage report:**

SL No	STATION	STATE	AGENCY	UNIT NO	CAP ACIT Y (MW)	REASON(S)	OUTAGE DATE
1	BARAUNI TPS	BIHAR	NTPC	7	110	Poor condenser vacuum	19-Jul-2023
2	BARAUNI TPS	BIHAR	NTPC	6	110	Low vacuum	22-Jul-2023
3	MEJIA TPS	DVC	DVC	3	210	Tube leakage in economiser	14-Dec-2024
4	IB.TPS	ODISHA	OPGC	2	210	Boiler Tube Leakage	14-Dec-2024
5	Sterlite	ODISHA	SEL	4	600	Ash evacuation problem	14-Dec-2024
6	MEJIA TPS	DVC	DVC	6	250	Boiler Tube Leakage	15-Dec-2024
7	Sterlite	ODISHA	SEL	2	600	APH problem	15-Dec-2024
8	FSTPP	WEST BENGAL	NTPC	4	500	Boiler tube leakage	14-Dec-2024
9	SANTALDIH TPS	WEST BENGAL	WBPDC	6	250	Annual Overhauling	23-Nov-2024
10	HALDIA ENERGY LTD	WEST BENGAL	HEL,CESC	1	300	Yearly maintenance activities	15-Dec-2024
11	CHANDRAPURA TPS	DVC	DVC	8	250	Annual overhauling	15-Dec-2024

12	KHSTPP	BIHAR	NTPC	5	500	Annual overhauling	14-Nov-2024
13	KBUNL	BIHAR	NTPC	2	195	Capital overhauling	15-Nov-2024
14	MPL	JHARKHAND	MPL	2	525	Annual Overhauling	22-Nov-2024
15	FSTPP	WEST BENGAL	NTPC	1	200	Capital Overhauling	01-Dec-2024
16	NABINAGAR(NPGC)	BIHAR	NTPC	3	660	Annual Overhauling	06-Dec-2024

All Generating stations are requested to update expected restoration time and reason outage to ERLDC/ERPC on weekly basis in case of any change at their end.

b) Major Generating stations Out on Reserve Shutdown due to low system demand:

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	SOUTHERN	WEST BENGAL	CESC	1	67.5	Low system demand	14-Dec-2024
2	SOUTHERN	WEST BENGAL	CESC	2	67.5	Low system demand	11-Dec - 2024

c) Hydro Unit Outage Report:

S. NO	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	TEESTA STG III Hep	SIKKIM	TUL	1	200	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
2	TEESTA STG III Hep	SIKKIM	TUL	2	200		
3	TEESTA STG III Hep	SIKKIM	TUL	3	200		
4	TEESTA STG III Hep	SIKKIM	TUL	4	200		
5	TEESTA STG III Hep	SIKKIM	TUL	5	200		
6	TEESTA STG III Hep	SIKKIM	TUL	6	200		
7	DIKCHU Hep	SIKKIM	SKPPL	2	48	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023

8	TEESTA HPS	SIKKIM	NHPC	1	170	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
9	TEESTA HPS	SIKKIM	NHPC	2	170		
10	TEESTA HPS	SIKKIM	NHPC	3	170		
11	CHIPLIMA HPS / HIRAKUD II	ODISHA	OHPC	1	24	Capital Overhauling	15-Dec-2023
12	BURLA HPS/HIRAKUD I	ODISHA	OHPC	2	49.5	Capital Maintainance	26-Nov-2024

d)Long outage report of transmission lines (As on 15.12.2024):

Transmission Element / ICT	Outage From	Reasons for Outage
220/132KV 100 MVA ICT II AT LALMATIA	22.01.2019	220/132KV, 100MVA Transformer (NTPC side) is charged on 07.02.2024 from HV side on no load. Now, it is in idle charged condition
220KV-FSTPP-LALMATIA-I	21.04.2021	Two nos. of tower collapsed on 29.05.2024 near to Lalmatia GSS in the Loc. No. 246 & 247. Presently 220 kV Farakka-Lalmatia line is charged(from loc no 241 to loc 84) at 132 kV voltage level for anti-theft purpose by tapping at loc. No. 100-101.
220KV-WARIA-BIDHANNAGAR-1 & 2	08.06.2022	To control overloading of 220 kV Waria-DSTPS (Andal) D/C line
132KV-BARHI-RAJGIR-1	25.03.2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earth wire between tension tower no. 218-237 in same line.
132KV-NALANDA-BARHI(DVC)-1	25.03.2023	
400KV-RANGPO-TEESTA-V-1 & 2	04.10.2023	Tower near gantry of Teesta V powerhouse collapsed due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-TEESTA-III-RANGPO-1	04.10.2023	Hand tripped from Teesta-III end due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-TEESTA-III-DIKCHU-1	04.10.2023	
400KV-JHARSUGUDA-ROURKELA-4	01.04.2024	Reconductoring work
132KV-RANGPO-SAMARDONG-1	22-05-2024	Rangpo:Y-n fault with fault distance 0.157 km 14.562kA Samardong: NA
220KV-RAJARHAT-NEW TOWN(AA-II)-2	10-07-2024	Initially line out due to rectification of gas leakage problem from B-Ph breaker pole. Line declared under breakdown after charging attempt after return of shutdown. After that fault found in b-phase cable.
132KV-RANGPO-SAMARDONG-2	02-08-2024	132/66/11kV Samardong ss have become inaccessible due to continuous raining and

		landslides. It is very difficult for round the clock deployment of shift manpower due to road non-accessibility
400KV/220KV 315 MVA ICT 2 AT INDRAVATI.	09-09-2024	Tripped due to Over Flux protection operated
400KV/220KV 315 MVA ICT 1 AT NORTH KARANPURA	12-09-2024	Tripped on Differential protection
132KV-MADHEPURA (BH)-SAHARSA(PMTL)-1	23.09.2024	To control loading on 132kV Madhepura-Saharsa line
132KV-MELLI-SILIGURI-1	05-10-2024	S/d for inspection of tower of Loc.127 found twisted due to heavy landslide & heavy continuous rainfall in Soom Tea Garden under Darjeeling section. Line charged as 132 KV Siliguri-Melli II (Interim arrangement) at 19:20 hrs on 09-10-2024. This interim arrangement is obtained by horizontal jumpering at Loc-129 after disconnecting main jumper for both Rangit & Melli side.
132KV-RANGIT-KURSEONG-1	05-10-2024	S/d for inspection of tower of Loc.127 found twisted due to heavy landslide & heavy continuous rainfall in Soom Tea Garden under Darjeeling section. Line charged as 132 KV Siliguri-Melli II (Interim arrangement) at 19:20 hrs on 09-10-2024. This interim arrangement is obtained by horizontal jumpering at Loc-129 after disconnecting main jumper for both Rangit & Melli side..
400KV/220KV 315 MVA ICT 1 AT INDRAVATI	05-10-2024	Erection, testing & commissioning work of fire fighting system
400KV/220KV 315 MVA ICT 1 AT TSTPP	01-11-2024	Tripped on PRD protection
400KV-JHARSUGUDA-ROURKELA-2	04-11-2024	Reconductoring Works in 400 KV Sundargarh-Rourkela Ckt-2
132KV-RANGPO-GANGTOK-2	04-11-2024	Reconductoring work by HTLS Conductor
132KV-RAXAUL(NEW)-PARWANIPUR-1 & 2	14-11-2024	The loop in loop out work at Nepal side
132KV-PATRATU-PATRATU-1 & 2	16-11-2024	Diversion/Heightening of line due to inadequate clearance from under construction railway Line by PVUNL
400KV-NEW PURNEA-MUZAFFARPUR-1 & 2	21-11-2024	Facilitating Erection of New Tower on Pile foundation, De-Stringing, Dismantling of old tower & Stringing Works
400KV-BINAGURI-TALA-1	25-11-2024	Voltage Regulation, converted to s/d for AMP Work from 09.12.2024
132KV-CHUZACHEN-RANGPO-1	29-11-2024	Rangpo : B-N ,Z-1, 7.8 KA, 5.61 KM
400KV-DURGAPUR-KAHALGAON-2	30-11-2024	Shifting of line bay equipment under installation of 63 MVAR line reactor at NTPC Kahalgaon
400KV-ALIPURDUAR (PG)-PUNASANGCHUN-JIGMELING-2	02-12-2024	SD Availed by Bhutan for rectify/Replace the LA for 400kV Jigmeling _Puna_ALI-1.

400KV/220KV 500 MVA ICT 1 AT SAHARSA	05-12-2024	To attend OLTC leakage present in ICT
400KV-KHSTPP-BARH-2	07-12-2024	Upgrading of bay & line equipments
220KV-KISHANGANJ(PG)-DALKHOLA (PG)-2	08-12-2024	Replacement of isolators
220KV/132KV 160 MVA ICT 1 AT SILIGURI	08-12-2024	Replacement work of C&R
400KV-RANCHI-SIPAT-1	08-12-2024	Construction activity- NHAI Diversion work under Bharatmala Pariyojana

Transmission licensees/ Utilities are requested to update expected restoration date & work progress regarding restoration regularly to ERLDC/ERPC on monthly basis by 5th of each month so that status of restoration can be reviewed in OCC. Utilities are also requested to update outage of any elements within their substation premises like isolator/breaker to ERLDC/ERPC regularly. (Reported as per Clause 5.2(e) of IEGC)

Deliberation in the meeting

Members noted.

4.3. Commissioning of new units and transmission elements in Eastern Grid in the month of November -2024.

The details of new units/transmission elements commissioned in the month of November 2024 based on the inputs received from beneficiaries:

NEW ELEMENTS COMMISSIONED DURING November, 2024

उत्पादन इकाइयाँ / GENERATING UNITS

क्र. Sl. No.	स्थान Location	मालिक/यूनिट का नाम OWNER/UNIT NAME	यूनिट संख्या/स्रोत Unit No/Source	संकलित क्षमता (मेगावाट) Capacity added (MW)	कुल/स्थापित क्षमता (मेगावाट) Total/Installed Capacity (MW)	दिनांक DATE	टिप्पणी Remarks
1	Dikchu	Sneha kinetic power projects Pvt ltd	1	48	48	20-11-2024	Post-Restoration following Flash Flood in Sikkim on 4 th Oct 2023

आई.सी.टी./जी.टी./एस.टी / ICTs/ GTs / STs

क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	उप-केन्द्र SUB-STATION	आईसीटी संख्या ICT NO	वोल्टेज (केवी) Voltage Level (kV)	क्षमता (एमवीए) CAPACITY (MVA)	दिनांक DATE	टिप्पणी Remarks
1	Sneha kinetic power projects Pvt ltd	DIKCHU HEP	ICT 1	400KV/132KV	270 MVA	20-11-2024	Post-Restoration following Flash Flood in Sikkim on 4 th Oct 2023
2	NTPC	NTPC Kaniha	Tie Transformer-1	220KV/33KV	100 MVA	08-11-2024	For FGD Projects
3	NTPC	NTPC Kaniha	Tie Transformer-2	220KV/33KV	100 MVA	22-11-2024	For FGD Projects

प्रेषण लाइन / TRANSMISSION LINES						
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	लाइन का नाम LINE NAME	लंबाई (किमी) Length (KM)	कंडक्टर प्रकार Conductor Type	दिनांक DATE	टिप्पणी Remarks
1	BSPTCL	132 kV Chandauti (PMTL)-Barachatti ckt 1	29.251	ACSR Panther	13-11-2024	
2	BSPTCL	132 kV Chandauti (PMTL)-Barachatti ckt 2	29.251	ACSR Panther	13-11-2024	
3	WBSETCL	132KV-MALDA (PG)-MANIKCHAK-1	24.0	ACSR Panther	19-11-2024	first time idle charged from Malda end only at 18-11-2024 16:42 Hrs
4	WBSETCL	132KV-MALDA (PG)-MANIKCHAK- 2	24.0	ACSR Panther	19-11-2024	first time idle charged from Malda end only at 18-11-2024 16:46Hrs
5	BSPTCL	220KV-BANKA (PG)-Havelikharagpur-1	92.8	ACSR Zebra (32.617 kms) + HTLS Zebra(60.22Kms)	05-11-2024	
लिलो / प्रेषण लाइन की पुनर्व्यवस्था / LILO/RE-ARRANGEMENT OF TRANSMISSION LINES						
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	लाइन का नाम / लिलो पर Line Name/LILO at	लंबाई (किमी) Length (KM)	कंडक्टर प्रकार Conductor Type	दिनांक DATE	टिप्पणी Remarks
NIL						
बस/लाइन रिएक्टर / BUS/LINE REACTOR						
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks
NIL						
बस / BUS						
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks
NIL						
एच.वी.डी.सी/ए.सी फिल्टर बैंक/फैक्ट्स डिवाइस संबद्ध प्रणाली / HVDC /AC Filter bank / FACTS DEVICE associated System						
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks
NIL						
बे / BAYS						
क्र. Sl. No.	एजेंसी/मालिक Agency/Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks
1	WBSETCL	132 KV Bus Coupler Bay at MANIKCHAK	MANIKCHAK	132	19-11-2024	
2	NTPC	220KV MAIN BAY OF 100 MVA TIE TRANSFORMER-2 AT TSTPP	TSTPP	132	22-11-2024	
3	DVC	220KV MAIN BAY OF ICT-2 AT RTPS	RTPS	220	19-11-2024	400 / 220 kV ICT at RTPS charged after LILO of 220KV D/C CTPS-Kalyaneswari T/L at RTPS
4	DVC	220KV MAIN BAY OF ICT-1 AT RTPS	RTPS	220	19-11-2024	400 / 220 kV ICT at RTPS charged after LILO of 220KV D/C CTPS-Kalyaneswari T/L at RTPS

5	BSPTCL	132KV MAIN BAY OF CHANDAUTI (PMTL) -2 AT BARACHATTI	BARACHATTI	132	15-11-2024	
6	BSPTCL	132KV MAIN BAY OF CHANDAUTI (PMTL) -1 AT BARACHATTI	BARACHATTI	132	15-11-2024	
7	BSPTCL	132KV MAIN BAY OF BARACHTI-2 AT CHANDAUTI (PMTL)	CHANDAUTI (PMTL)	132	13-11-2024	
8	BSPTCL	132KV MAIN BAY OF BARACHTI-1 AT CHANDAUTI (PMTL)	CHANDAUTI (PMTL)	132	13-11-2024	
9	NTPC	220KV MAIN BAY OF 100 MVA TIE TRANSFORMER -1 AT TSTPP	TSTPP	132	08-11-2024	

Members may note.

Deliberation in the meeting

Members noted.

4.4. UFR operation during the month of November 2024.

Frequency profile for the month as follows:

MONTH	MAX	MIN	% LESS IEGC BAND	% WITHIN IEGC BAND	% MORE IEGC BAND
	(DATE/TIME)	(DATE/TIME)			
November, 2024	50.32 Hz on 02-11-2024 at 11:04 hrs	49.55 Hz on 27-11-2024 at 07:18 hrs	5.2	80.8	14.0

Hence, no report of operation of UFR has been received from any of the constituents.

Members may note.

Deliberation in the meeting

Members noted.

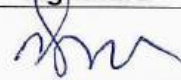
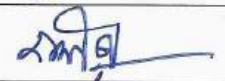
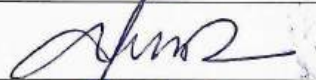
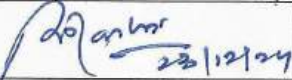
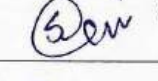
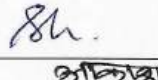
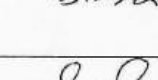
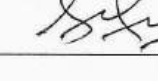
Annex-A

Participants in 222nd OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 11:00 Hrs.

Date: 23.12.2024 (Monday)

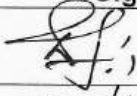
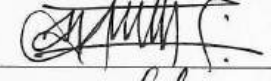
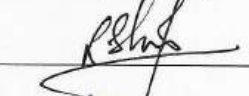
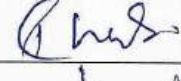
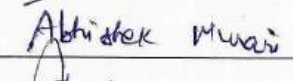
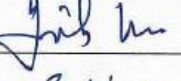
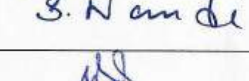
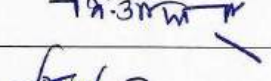
Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
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Participants in 222nd OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 23.12.2024 (Monday)

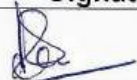
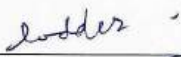
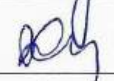
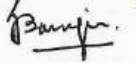
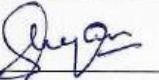
Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
16	Dawa Zangmo	AE	DGPC	+975 17707714	d.zangmo3582@drutgreen.bt	
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28	Chandan mallick	C. M.	ERLDC GRID-INDIA	9007059666	chandan.mallick@grid-india.in	
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Participants in 222nd OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 23.12.2024 (Monday)

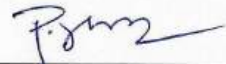
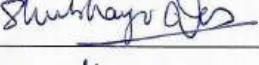
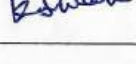
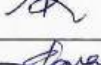
Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
31	S. K. Maheshwari	DGM(EI.)	GRIDCO	9438591100	ele.skamaheshwari@gridco.co.in	
32	S.M.S. Sahoo	DGM(Elect)	OPTCL	9438908353	ele.smsahoo@optcl.co.in	
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36	ALOK KR GHOSH	GM(OS)	WBPDCL	8336904026	mkghosh@wbpdcl.co.in	
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39	Preetam Banerjee	ACE, WBSECL	WBSECL	7003871189	preetam.banerjee@wbsecl.in	
40	S.K. Sharma	SLGM(SLDC)	SLDC, DVL	9434539423	Sanjay.Sharma@dvl.gov.in	
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44	Agniva Chatterjee	AD-I	ERPC	8100307562	agniva.cca@gov.in	
45	S. K. JAIWAL	SE	ERPC	9831919509	shyam.kejriwal@gov.in	

Participants in 222nd OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 23.12.2024 (Monday)

Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
46	Somnath Pattna	Manager	GMR	7682811501	somnathpatna295@gmail.com	
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48	P. Ghosh	DGM	ERIL	9434748263	patna.shuman@powersid.in	
49	Gulshan K. Smigra	Sr. Manager	MBPL Rongcheno HEP	8016889840	gulshan.smigra@mbpl.co.in	
50	P. P. Sridharan	DGM	APNRL	9952459520	opsridharan@apnrlpower.in	
51	Mishra Kund	Sr. Manager	SSVN/STPL	9418055536	elect.stpl@ssvn.nic.in	
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55	S. K. Panda	Manager	ERDC, WCL	6370121794	senitosh.panda@edc.gov.in	
56	PRATHAM KUMAR	E.E.	ERPC, CEA	8076873558	Prathankumarce@nic.in	
57	SHUBHAYU DAS	ASSISTANT DIRECTOR	ERPC (CEA)	8917360785	shubhayudas5@gmail.com	
58	KUMARI SWATI	"	"	7903336191	Swati cea@gov.in	
59	SOURAV BERA	Project Manager	TCS	9123801884	bera_sourav@tcs.com	
60	ABHISHEK DAS	Tech Lead	TCS	9433835015	das.abhishek2@tcs.com	

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63. D. BISWAS Sr. DGM. ERLDC 9434748041 dbiswas@grid-india.in 

ANNEX B.2.17

Odisha Feeder list for Disconnection
New feeders
ISTS feeders
220 kV Rengali(PG)-Rengali(OPTCL)
220/132 kV Baripada 160 MVA ICT
220 kV Baripada(PG)-Balsore (Odisha)
Intra-state feeders
132kV Bolangir(New)-Patnagarh S/C
132kV Bhadrak-Agarpada S/C
132kV Balasore-Soro
132kV Chandaka-Nimapara / Ranasighpur
132kV Baripada(PG)-Jaleswar/Bhograi
132kV Jajpur Rd. – Kendrapara D/C

DVC Feeder list for Disconnection
New 33kV feeders
Intra-state feeders
JSEB Barhi
JSEB Bokaro
JSEB Chas
JSEB Konar Banaso
JSEB Mugma
JSEB Giridih
JSEB Digwadih
JSEB Ganeshpur
JSEB Jamadoba
JSEB PADMA PSS RGGVY
JSEB Dugdha
JSEB Ramgarh
JSEB Karma PSS RGGVY
JSEB Kumardubi

JSEB Dumbri Banaso
JSEB Mukunda
JSEB Katras (Tilatand)
PMCH Medical College
JSEB West Bokaro (Ghato)
JSEB Sindri
JSEB Biada-Chas
JSEB Katras (Sijua)
JSEB Sendra Bansjora
JSEB Jainamore
JSEB Badjna
JSEB Hazaribagh
Jharkhand feeder list
New feeders
ISTS feeders
One 400/220 kV 315 MVA ICT Jamsedpur
220 kV Ranchi(PG)-Chandil(JUVNL)
Intra-state feeder
132kV Chandil – Golmuri D/C
132kV Dumka-Pakur S/C
33kV-Gumla-Bharno
33kV-Kamdara-Torpa
33kV-Lalmatia-Mahagama
33kV-Nouamundi-Noamundi
33kV-Kendposi-Kumardungi
33kV-Rajkharsawan-Rajkharsawan
33kV-Latehar-Manika
33kV-Japla-Md. Ganj
33kV-Daltonganj-Tukbera
33kV-Lohardaga-Kuru
33kV-Simdega-Kolebera
33kV-Japla-Japla
33kV-Kamdara-Karra

33kV-Lohardaga-Kisko
33kV-Tamar-Bundu
33kV-Simdega-Simdega
33kV-Godda-Pathargama
33kV-Chaibasa-Saraikela
33kV-Chaibasa-2-Rajnagar
33kV-Chakardharpur-CKP
33kV-Lalmatia-Mahagama
33kV-Chitra-Simla
33kV-Gumla-Ghagra
33kV-Kamdara-Basia
33kV-Chitra-Sarath
Bihar feeder list
New feeders
ISTS feeders
132 KV Sitamarhi(PG)-Runnisaipur D/C T/L
132 kv Banka(PG)-Sultanganj D/C T/L
132 Kv Banka(PG)-Banka D/C T/L
Intra-state feeder
West Bengal feeder list
New feeders
ISTS feeders
220 kV Dalkohla (PG)-Dalkohla(WB)
132 kV Malda (PG)-Malda(WB)
132 kV Birpara(PG)-Birpara(WB)
Intra-state feeder
Sikkim feeder list
New feeders
ISTS feeders

PRIORITY WISE ISTS FEEDERS TO BE DISCONNECTED IN CASE OVERDRAWAL AND WHEN FREQUENCY IS BELOW 49.4 HZ AND NO UFR RELIEF HAS BEEN OBSERVED

Odisha Feeder list for Disconnection
132 kV Baripada-Bangriposi D/C
132 kV Baripada-Bhogarai/Jaleswar D/C
220 kV Bolangir-New Bolangir D/C
West Bengal Feeder list for Disconnection
220 kV Dalkohla (PG)-Dalkohla(WB) D/C
132 kV Malda (PG)-Malda(WB) D/C
132 kV Birpara(PG)-Birpara(WB) D/C
DVC Feeder list for Disconnection
220 kV Maithon-Dhanbad D/C
220 kV Maithon-Kalyaneswari D/C
220 kV Parulia-Parulia D/C
Jharkhand Feeder list for Disconnection
220 kV Daltonganj- Gardwa D/C
220 kV Daltonganj- Chatra D/C
220 kV Maithon-Dumka D/C
Bihar Feeder list for Disconnection
132 kV Sitamarhi (PG)-Runnisaipur D/C
132 kV Banka (PG)-Sultanganj D/C
132 kV Banka (PG)-Banka D/C

ANNEX B.2.18

Constituent	Stage	MW Requirement		Shifted from other stages	MW currently installed	Installed but not tested yet	Installation pending	Remarks
		Old	New					
BSPTCL	Stage 1	126	315	-	128	0	187	Feeders identified for all 4 stages. Complete testing and installation expected by Feb'25
	Stage 2	118	379	6 MW (From St. 1)	168	0	211	
	Stage 3	153	442	15 MW (From St. 2)	167	0	275	
	Stage 4	85	442	374 MW (From other 3 stages)	390	0	52	
JUSNL	Stage 1	54	87	25 MW (From St. 2)	85	16	2	Feeder identification pending for St. 3 & 4. Testing pending for 3 feeders in St.1 & 2
	Stage 2	64	105	68 MW (From St.3 & 4)	111	13	0	
	Stage 3	35	122	39 MW (From St. 4)	39	0	83	
	Stage 4	73	122	-	0	0	122	
DVC	Stage 1	122	172	43 MW (From St.2), 69 MW (From St. 3 & 4)	173	0	0	Feeders identified for all 4 stages. Complete testing and installation expected by Jan'25
	Stage 2	145	207	64 MW (From St. 3 & 4)	209	0	0	
	Stage 3	147	241	-	0	0	241	
	Stage 4	138	241	-	0	0	241	
OPTCL	Stage 1	181	306	40 MW (From St. 2), 26 MW (From St. 3&4)	216	0	90	Feeders identified for all 4 stages. Shortfall in stage-3 of 49 MW and excess in St 1 & 4 of 64 MW and 28 MW respectively. Testing and installation of identified feeders expected by Dec'24 end.
	Stage 2	183	367	26 MW (from St. 3)	135	0	233	
	Stage 3	184	428	-	100	0	328	
	Stage 4	186	428	-	187	0	242	
West Bengal (WBSEDCL)	Stage 1	316	377	100 MW (From St. 4)	416	0	0	New feeders identification pending for St. 3 & 4
	Stage 2	284	457	176 MW (From St.3)	458	0	0	
	Stage 3	265	536	86 MW (From St.4)	153	0	383	
	Stage 4	273	536	24 MW (From St.3)	111	0	425	
West Bengal (CESC)	Stage 1	65	120	-	120	0	0	CESC- Installation and testing complete for all 4 stages
	Stage 2	90	140	-	140	0	0	
	Stage 3	125	160	-	160	0	0	
	Stage 4	120	160	-	160	0	0	

ANNEX B.2.20

Approved Maintenance Schedule of Thermal Generating Units of ER during 2024-25 for March 2025												
System	Station	Unit No.	Capacity (MW)	Period as per LGBR 2024-25		No. of Days	Period proposed in OCC		No. of Days	Whether as per LGBR or Not	Reason	Remarks
				From	To		From	To				
DVC	BTPS-A	1	500	25-12-2024	28-01-2025	35	15-03-2025	18-04-2025	35	NO	COH-Blr, Turb, Gen & HMI upgradation	Approved
	Mejia TPS	4	210	07-03-2025	31-03-2025	25	—	—	—	NO	AOH- Boiler, LPT & De-Nox burner	NOT TO AVAIL
		3	210	01-07-2024	25-12-2024	178	07-02-2025	23-03-2025	45	NO	Boiler Overhauling & ESP Augmentation	Approved
	CTPS	8	250	01-09-2024	05-10-2024	35	15-12-2024	11-01-2025	28	NO	AOH-Blr, Turb, Gen & De Nox	Approved
	KTPS	1	500	01-12-2024	28-12-2024	28	—	—	—	NO	AOH- Boiler,LPT	NOT TO AVAIL
NTPC	Barh-II	5	660	01-02-2025	07-03-2025	35	01-02-2025	07-03-2025	35	YES	Boiler + TG Bearing Inspection	Approved
	KhSTPS	2	210	05-07-2024	03-08-2024	30	10-02-2025	11-03-2025	30	NO	Boiler + Boiler RLA + Boiler Acid cleaning	Approved
	KhSTPS	3	210	10-02-2025	11-03-2025	30	—	—	—	NO	Boiler + Boiler RLA + Boiler Acid cleaning	NOT TO AVAIL
	NKSTPP	1	660	—	—	—	23-02-2025	29-03-2025	35	NO	LP Turbine + Generator Rotor Problems	Approved
BRBCL	BRBCL	2	250	08-01-2025	21-02-2025	45	15-03-2025	29-04-2025	46	NO	COH-HP,IP and LP turbine overhaul	Approved
WBPDCCL	Kolaghat TPS	5	210	07-02-2025	13-03-2025	35	07-02-2025	26-02-2025	20	YES	Boiler License renewal	Approved

NOTE: All the generating units are advised to strictly abide by the approved maintenance schedule for March'25 .

ANNEX B.2.21

REVISED OUTAGE PLAN FOR APRIL-JUNE 2025(As discussed in 222nd OCC of ERPC)

SYSTEM	CAPACITY (MW)	STATION NAME	UNIT NO	FROM	TO	DURATION (DAYS)	PROPOSED OUTAGE REASON	Remarks
NPGC	660	NABINAGAR STPP	1	1-Nov-25	30-Dec-25	59	AOH(Boiler+Turbine + Gen)	APPROVED (Deferred to Nov'2025)
BRBCL	250	BRBCL	2	15-Mar-25	29-Apr-25	45	COH-HP,IP and LP turbine overhaul	APPROVED
DVC	500	BOKARO TPS `A` EXP	1	15-Mar-25	18-Apr-25	35	COH-Blr, Turb, Gen & HMI upgradation	APPROVED
	210	MEJIA TPS	2	—	—	—	ESP Augmentation	NOT TO AVAIL
NTPC	500	TALCHER STPS	3	—	—	35	BOH	As 90% power from NTPC Talcher STPS Unit#3 is allocated to SR beneficiaries,So for the same, consent may be sought from SRPC.
	800	DARLIPALI STPS	2	—	—	—	1.AOH (Main Turbine Thrust bearing temperature is high(110 deg C) 2.Cracks in Boiler Hanger	OCC advised NTPC to ensure availability of material and complete necessary maintenance before 15th March 2025 or avail the shutdown after 15th June 2025
WBPDC	210	KOLAGHAT TPS	4	15-Jul-25	08-Aug-25	24	BOH and Boiler License Renewal	APPROVED (Deferred to Jul'2025)
	500	SAGARDIGHI TPS	4	—	—	—	Boiler License Renewal	NOT TO AVAIL

Total capacity originally prposed under outage(Apil-June 2025)	3380 MW
Revised capacity under outage (April-June 2025)	750 MW
Reduction in Outage Capacity(April-June 2025)	2630 MW

ANNEX C.3.2



भारत सरकार
Government of India
विद्युत मंत्रालय
Ministry of Power
केंद्रीय विद्युत प्राधिकरण
Central Electricity Authority
विद्युत प्रणाली योजना एवं मूल्यंकन प्रभाग- II
Power System Planning & Appraisal Division-II

सेवा में /To

As per list of Addresses

विषय: ट्रांसमिशन पर राष्ट्रीय समिति (एनसीटी) की चौबीसवीं बैठक के कार्यवृत्त - के सम्बन्ध में ।

Subject: Minutes of the 24th Meeting of National Committee on Transmission (NCT) - regarding.

महोदया (Madam) / महोदय (Sir),

The 24th meeting of the National Committee on Transmission (NCT) was held on 23rd October, 2024, at CEA, New Delhi. Minutes of the meeting are enclosed herewith.

भवदीय/Yours faithfully,


22.11.2024

(बी.एस.बैरवा/ B.S. Bairwa)
मुख्य अभियन्ता (इंचार्ज) एवं सदस्य सचिव, एन.सी.टी./
Chief Engineer (I/C) & Member Secretary (NCT)

प्रतिलिपि / Copy to:

Joint Secretary (Trans), Ministry of Power, New Delhi-110001

List of Addresses:

1.	Chairperson, Central Electricity Authority Sewa Bhawan, R.K. Puram, New Delhi – 110 066.	2.	Member (Power Systems), Central Electricity Authority Sewa Bhawan, R.K. Puram, New Delhi – 110 066.
3.	Member (Economic & Commercial), Central Electricity Authority Sewa Bhawan, R.K. Puram, New Delhi – 110 066.	4.	Director (Trans), Ministry of Power Shram Shakti Bhawan, New Delhi-110001.
5.	Sh. Lalit Bohra, Joint Secretary Room no 602, Atal Akshay Urja Bhawan Opposite CGO Complex, Gate No. 2, Lodhi Road, New Delhi – 110003	6.	Chief Operating Officer, CTUIL, Floors No. 5-10, Tower 1, Plot No. 16, IRCON International Tower, Institutional Area, Sector 32, Gurugram, Haryana - 122001.
7.	Sh. Rajnath Ram, Adviser (Energy), NITI Aayog, Parliament Street, New Delhi – 110 001.	8.	CMD, Grid Controller of India, B-9 (1 st Floor), Qutub Institutional Area, Katwaria Sarai, New Delhi – 110016
9.	Sh. Ravinder Gupta Ex. Chief Engineer CEA		

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Minutes of the 24th meeting of National Committee on Transmission (NCT)

The 24th meeting of NCT was held on 23rd October, 2024 at CEA, New Delhi. List of participants is enclosed at **Annexure-I**. Agenda wise deliberations are given below:

1 Confirmation of the minutes of the 22nd and 23rd meeting of National Committee on Transmission.

- 1.1 The minutes of the 22nd meeting of NCT held on 23.08.2024 were issued on 01.09.2024 vide CEA letter No. CEA-PS-12-13/3/2019-PSPA-II. No comments have been received on the minutes.
- 1.2 The minutes of the 23rd meeting of NCT held on 02.09.2024 were issued on 09.09.2024 vide CEA letter No. CEA-PS-12-13/3/2019-PSPA-II. No comments have been received on the minutes.
- 1.3 Members confirmed the minutes of 22nd and 23rd meetings of NCT.

2 Status of the transmission schemes noted/approved/recommended to MoP in the 22nd and 23rd meetings of NCT:

2.1 Status of new transmission schemes approved/recommended:

Sr. No	Name of the Transmission Scheme	Noted/ Recommended/ Approved	Mode of Implementation	BPC	Award/ Gazette notification
22nd NCT Meeting					
1.	Transmission system for supply of power to Green Hydrogen/Ammonia manufacturing potential in Mundra area of Gujarat under Phase-I: Part B1 scheme (3 GW at Navinal S/s)”	Recommended	TBCB	PFCCCL	Gazette Notified by MoP dated 12.09.2024
2.	Eastern Region Expansion Scheme-43 (ERES-43)	Approved	RTM	Not applicable	Informed to CTUIL vide letter dated 02.09.2024
3.	Additional Transmission System Proposed for redundant power supply to Dholera area	Approved	RTM	Not applicable	CTUIL awarded the projects to the implementing agency on 02.09.2024
4.	Transmission System for Integration of Anantapur-II REZ - Phase-I (for 4.5 GW)	Recommended	TBCB	PFCCCL	Gazette Notified by MoP dated 12.09.2024
5.	Transmission system for proposed Green Hydrogen / Green	Recommended	TBCB	RECPDCL	Gazette Notified by MoP dated 12.09.2024

Minutes of the 24th meeting of National Committee on Transmission (NCT)

Sr. No	Name of the Transmission Scheme	Noted/ Recommended/ Approved	Mode of Implementation	BPC	Award/ Gazette notification
	Ammonia projects in Tuticorin area)				
6.	Augmentation of transformation capacity by 3x500 MVA, 400/220 kV ICTs (6th - 8th) and 1x1500 MVA, 765/400 kV ICT (4th) at Bidar PS	Approved	TBCB	RECPDCL	Gazette Notified by CEA on 25.09.2024
7.	Scheme for Requirement of Additional FOTE for redundancy at AGC locations in NER: Revised	Approved	RTM	Not applicable	Informed to CTUIL vide letter dated 02.09.2024
8.	Optical Fibre Connectivity for NLDC new building, August Kranti Marg, New Delhi	Approved	RTM	Not applicable	CTUIL awarded the projects to the implementing agency on 02.09.2024
23rd NCT Meeting					
1.	Transmission System for Integration of Kurnool-IV REZ - Phase-I (for 4.5 GW)	Recommended	TBCB	RECDPCL	Gazette Notified by MoP dated 19.09.2024

2.2 Status of transmission schemes where modifications was suggested by NCT:

S. No.	Scheme where modifications was suggested	Status
1.	Modification in Transmission system for evacuation of power from Luhri Stage-I HEP	Informed to RECPDCL vide letter dated 02.09.2024
2.	Transmission system for evacuation of power from Shongtong Karcham HEP (450 MW) and Tidong HEP (150 MW)	Informed to RECPDCL vide letter dated 02.09.2024
3.	Modification in timeframe of one of the elements in the scope of "Transmission system for offshore windzone phase-1(500 MW VGF off coast of Gujrat for subzone B-3)	Informed to CTUIL vide letter dated 02.09.2024 CTUIL awarded the projects to the implementing agency on 02.09.2024
4.	Time extension for Communication Scheme "Requirement of additional FOTE of STM-16 capacity at Bhuj-II substation to cater connectivity of RE Gencos"	

3 Modifications in the earlier approved/notified transmission schemes:**3.1 Revision in SCOD of 400 kV D/C Jhatikara-Dwarka line under REZ Phase-III Part-D Phase-II scheme**

3.1.1 Representative from CTUIL stated that the implementation of the 400 kV D/C Jhatikara-Dwarka line, along with two 400 kV bays each at Jhatikara and Dwarka

under “Transmission system for evacuation of 20 GW REZ power from Rajasthan under phase-III, Part-D, Phase-II” was allocated to POWERGRID under RTM mode with completion schedule of 18 months vide MoP OM Ref. No. 15/3/2018-Trans-Part(5) dated 06.11.2023. POWERGRID vide letter dated 29.12.2023 requested an extension of the implementation timeline to at least 24 months due to technical and execution challenges and proposed changing the conductor configuration from quad to Twin HTLS on Monopole structure.

- 3.1.2 NCT in its 17th meeting held on 31.01.24 directed CTUIL to re-survey of the scheme through implementing agency so as to arrive at the optimum requirement of monopole/narrow base tower towers, and work out the revised estimated cost. Further, NCT in its 19th meeting held on 29.04.2024 approved the scope modifications in the Jhatikara – Dwarka 400 kV D/c line under Rajasthan REZ Ph-III, Part-D- Ph-II Scheme. Tentative implementation time-frame of 18 months from MOP OM-06/11/23 was unchanged. CTUIL vide letter Ref. No. CTUIL/OM/14/19 NCT dated 29.05.2024 informed that the scope of project was revised and conductor configuration was changed to Twin HTLS, However, the implementation timeline remains unchanged. POWERGRID on 06.06.2024, once again requested an extension of project timelines to at least 24 months from the fresh allocation date of 29.05.2024, instead of 18 months from original allocation date.
- 3.1.3 Subsequently, in a meeting chaired by Secretary (Power) on 01.07.2024, POWERGRID was advised to proceed with inviting tender based on 18 months’ timelines for the implementation of Rajasthan Phase-II, Part-D, Phase-II scheme. In compliance with the directives, POWERGRID floated the tender and the award is expected by November, 2024 with a project completion timeline of February 2026.
- 3.1.4 Director (SO), Grid-India stated that 765/400 kV Jhatikara ICTs and 400 kV lines from Jhatikara were N-1 non-compliant during summer of 2024. In case of further delay in 400 kV D/C Jhatikara - Dwarka line, severe constraints are expected in the existing 400 kV Jhatikara – Dwarka and 400 kV Jhatikara – Bamnoli lines with further RE capacity addition in Rajasthan etc. The loading of these lines emerged as N-1 non-compliant during high demand season of NR and the same may lead to RE curtailment in future. Requirement of any augmentation in the Delhi intra-state system also needs to be examined. Chairperson, CEA directed that the above issues shall be studied in a holistic manner in the transmission Resource Adequacy Plan of Delhi.
- 3.1.5 After deliberations, NCT approved the revised SCOD for 400 kV D/C (Twin HTLS) Jhatikara-Dwarka line under “Transmission system for evacuation of 20 GW REZ power from Rajasthan under phase-III, Part-D, Phase-II” scheme as 28th February, 2026 (31st December 2025 on best effort basis).
- 3.2 **Change in scope of Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1: 4 GW) [Sirohi/Nagaur] Complex**
 - 3.2.1 The transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1:4 GW) (Sirohi/Nagaur complex) was recommended in the 21st NCT meeting held on 06.08.2024. Subsequently, the scheme was notified by MoP vide Gazette dated

29.08.2024. The scheme involves 5x500 MVA, 400/220 kV ICTs along with 6 Nos. 220 kV line bays at Sirohi S/s for RE interconnection at Sirohi S/s. The scheme is currently under bidding by RECPDCL.

3.2.2 Representative from CTUIL stated that connectivity up to 2100 MW was agreed to be granted at Sirohi S/s. Out of this, 1400 MW was agreed to be granted at 220 kV level through 5 Nos. of 220 kV line bays and balance 700 MW was agreed to be granted at 400 kV level (1 No. bay). It is to mention that out of above 700 MW, earlier 400 MW was agreed to be granted at 220 kV level of Sirohi S/s, however due to additional application of 300 MW, considering cumulative quantum (700 MW), it was proposed to be granted at 400 kV level. Further, it is proposed to add the following transmission element as part of Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part- 1) (Sirohi Complex)

- 1 No. of 400 kV line bay at Sirohi S/s for RE interconnection

3.2.3 As total connectivity granted at Sirohi S/s on 220 kV level is 1400 MW through 5 Nos. of 220 kV line bays (out of 6 Nos.), 1 No. of 220 kV line bay which is part of the above scheme shall remain unutilised. Additionally, for RE evacuation requirement of 1400 MW at 220 kV level, through 4 Nos. of 400/220 kV ICTs (out of 5 Nos.), 1 No. of 400/220 kV ICT which is part of the above scheme shall remain unutilised. In view of the above, it is proposed to delete the following elements from Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part- 1) (Sirohi Complex)

- 1 No. of 220 kV line bay at Sirohi S/s
- 1x500 MVA, 400/220 kV ICT at Sirohi S/s along with transformer bays

3.2.4 The original cost of scheme is Rs 5027.61 Cr. With above modifications of scope, cost shall reduce only by Rs 40.75 Cr which is about (-) 0.81 % of original cost of package.

3.2.5 Representative of RECPDCL stated that NIT for the scheme was issued on 26.09.2024. The bid submission deadline is 29.11.2024 while SPV transfer is targeted in December 2024.

3.2.6 After Deliberations, NCT approved the revised scope of Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1: 4 GW) [Sirohi/Nagaur] Complex as follows:

Sl. No.	Original scope of the transmission scheme	Revised scope of the transmission scheme
1. Transmission system for immediate Evacuation of Power from Sirohi S/s (2 GW)		
1	5x500 MVA, 400/220 kV ICTs at Sirohi S/s along with transformer bays • 400/220 kV 500 MVA ICTs- 5 Nos. • 400 kV ICT bays-5 Nos. • 220 kV ICT bays- 5 Nos.	4x500 MVA, 400/220 kV ICTs at Sirohi S/s along with transformer bays • 400/220 kV 500 MVA ICTs- 4 Nos. • 400 kV ICT bays-4 Nos. • 220 kV ICT bays- 4 Nos.
2	6 Nos. 220 kV line bays at Sirohi S/s	5 Nos. 220 kV line bays at Sirohi S/s

Sl. No.	Original scope of the transmission scheme	Revised scope of the transmission scheme
	for RE interconnection • 220 kV line bays – 6 Nos.	for RE interconnection • 220 kV line bays – 5 Nos.
3	220 kV Sectionalizer bay (1 set) along with 220 kV BC (2 Nos.) bay and 220 kV TBC (2 Nos.) bay at Sirohi S/s	• 220 kV Sectionalizer bay (1 set) • 220 kV BC (2 Nos.) bay and 220 kV TBC (2 Nos.) bay
4	-	1 No. 400 kV line bays at Sirohi S/s for RE interconnection • 400 kV line bay – 1 No.
	<i>Note: There will be no change in other elements of the transmission scheme w.r.t agreed in the 21st NCT meeting/ MoP Gazette dated 29.08.2024.</i>	

3.3 Bid process for selection of Bidder as Transmission Service Provider (TSP) to establish “Augmentation of transformation capacity at Bhuj-II PS (GIS)” and “Transmission system strengthening to facilitate evacuation of power from Bhadla/Bikaner complex”

3.3.1 The transmission scheme “Augmentation of transformation capacity at Bhuj-II PS (GIS)” was agreed in the 16th meeting of National Committee on Transmission held on 30.11.2023 under TBCB route with estimated cost of Rs. 428 crores and implementation timeframe of 21 months. Gazette was notified on 23.01.2024 with PFCCCL as BPC. The RFP for the transmission scheme was issued on March 29, 2024. RFP bid submission originally scheduled on May 31, 2024 had been extended to August 20, 2024 on request from the bidders. Out of the two bidders who purchased the RFP documents for the subject transmission scheme, only one bidder i.e. Power Grid Corporation of India Limited submitted the bid on August 20, 2024. The bid was again extended and on August 27, 2024 also only one bidder i.e. Power Grid Corporation of India Limited submitted the bid. As there was only one bid, PFCCCL vide letter dated 04.09.2024 sought the guidance from MoP on the matter. MoP vide letter dated 09.10.2024 referred the matter to National Committee on Transmission (NCT) for deliberation in the next NCT meeting and submit the recommendations to the Ministry.

3.3.2 Another transmission scheme “Transmission system strengthening to facilitate evacuation of power from Bhadla/Bikaner complex)” was approved in the 19th meeting of National Committee on Transmission held on 29.04.2024 under TBCB route with estimated cost of Rs. 198.75 crores and implementation timeframe of 18 months. Gazette was notified on 18.06.2024 with PFCCCL as BPC. The RfP for the subject transmission scheme was issued on August 01, 2024. RFP bid submission was originally scheduled on October 04, 2024. RfP documents for the subject transmission scheme have been purchased by only one bidder, i.e. Power Grid Corporation of India Limited. Accordingly, bid submission for the project has already been extended three

(03) times and latest date of bid submission is 25th October, 2024. Subsequently, no bid has been received on the due date.

3.3.3 CTU stated that above strengthening schemes are getting delayed due to above bid issues.

3.3.4 As per clause 9.6 of "Tariff based Competitive-bidding Guidelines for Transmission Service" issued by Ministry of Power (MoP) on August 10, 2021, there have to be minimum two qualified bidders for conducting the bid process.

3.3.5 It was mentioned that in case of single bid is received, clarity is not available in TBCB Guidelines. Members opined that to save time of rebidding, certain provisions need to be worked out. It was discussed that whether the project can be awarded in case of single bid if the cost discovered is of the level of estimated tariff.

3.3.6 After deliberations, following was agreed:

A. For the project costing less than Rs. 500 crore

- a) In case, no bid or single bid is received, the BPC may extend the bid process for 7 more days. After the extended period, the BPC may open the bid. If single bid is received, steps given in subsequent paragraphs need to be followed.
- b) Based on estimated cost by the Cost Committee constituted by CEA and the norms provided in CERC tariff regulations, the levelised tariff for the project shall be calculated by the BPC.
- c) If the quoted transmission charges by the bidder are lower than levelised tariff estimated by the BPC, the bidder may be declared as successful bidder and Letter of Intent (LoI) may be issued.
- d) In case, the quoted transmission charges by the bidder are higher than the estimated levelised tariff by the BPC, the bidding process may be annulled and matter may be referred by the BPC to the Government. The Government may give exemption to the transmission project from competitive bidding and allocate the same under Regulated Tariff Mechanism (Section-62 of the Electricity Act, 2003).
- e) If no bids are received, even after extension, the bidding process may be annulled by BPC and the matter may be referred by the BPC to the Government. The Government may give exemption to the transmission project from competitive bidding and allocate the same under Regulated Tariff Mechanism (Section-62 of the Electricity Act, 2003).

B. For the project with estimated cost of more than Rs. 500 crore, if only one bid is received, BPC may not open the bid and refer the matter to the Government. In case no bids are received, the bid process would be annulled and the matter shall be referred to the Government.

C. For making the enabling provisions, "Tariff based Competitive-bidding Guidelines for Transmission Service-2021" needs to be modified.

3.4 Change in the implementation timeframe of Transmission System for evacuation of RE power from Raghanesda area of Gujarat – 3 GW under Phase-I

- 3.4.1 Representative from CTUIL stated that Transmission System for evacuation of RE power from Raghanesda area of Gujarat – 3 GW under Phase-I was recommended in the 20th NCT meeting held on 25.06.2024 under TBCB route with PFCCL as BPC and implementation time frame of 30 months from SPV transfer. Applications for cumulative 3050 MW linked with Ph-I 3 GW scheme have been received at Raghanesda S/s, out of 3050 MW applications, start date of connectivity required for 4 nos. of applications (connectivity quantum of 1150 MW) is from Dec'26 (JSW Neo Energy Limited: 400 MW & ACME: 400 MW) to Mar'27 (Sunsure Solarpark RJ One Pvt. Ltd.: 350 MW).
- 3.4.2 CTUIL proposed to change the implementation timeframe of the subject scheme from 30 months from SPV transfer to 24 months from SPV transfer so that the substation can come up earlier matching with requirement of above RE developers.
- 3.4.3 Representative of PFCCL informed that the RfP was issued on 14.09.2024 while the bid submission date is 19.11.2024.
- 3.4.4 After deliberations, it was decided that the implementation timeline of Transmission System for evacuation of RE power from Raghanesda area of Gujarat – 3 GW under Phase-I may be kept as 30 months and need not be changed.

4 New Transmission Schemes:

4.1 Eastern Region Expansion Scheme-44 (ERES-44)

- 4.1.1 Representative of CTUIL stated that several 220 kV transmission lines and substations were implemented in Indian grid along with cross border lines for importing power from Chukha Hydro Electric Plant in Bhutan. The generating station was commissioned in years 1986-88 and the transmission system is now more than 35 years old. Considering the age of conductors and increase in conductor snapping incidences, reconductoring of these transmission lines has become necessary. The matter was also deliberated in ERPC forum.
- 4.1.2 As the system involved cross border links also, a meeting was convened by CEA under the chairpersonship of Member (Power System) on 27-08-2024, wherein it was decided that matter of reconductoring of cross border lines will be separately taken up with Bhutan. However, reconductoring of ISTS portion of 220 kV corridor viz. Alipurduar (POWERGRID) – Falakata (WBSETCL) – Birpara (POWERGRID) – Binaguri (POWERGRID) – Siliguri (POWERGRID) – Kishanganj (POWERGRID) – Dalkhola (POWERGRID) – Gazole (WBSETCL) – Malda (POWERGRID), may be taken up under ISTS. Further, reconductoring of intra-state LILO portion of Birpara (POWERGRID) – Alipurduar (POWERGRID) 220 kV D/c line at Falakata (WBSETCL) and Dalkhola – Malda 220 kV D/c line at Gazol (WBSETCL) shall be carried out by WBSETCL matching with HTLS conductor of the main ISTS line in the matching timeframe.

- 4.1.3 Director (SO), Grid-India stated that it is necessary that the intra-state portion of lines (under the jurisdiction of WBSETCL) is also re-conducted in the matching time-frame of that of the ISTS portion. Without the reconductoring of the intra-state portion, the benefits of reconductoring of the ISTS part cannot be realized. He further suggested that as intra-state portion is quite less compared to the inter-state portion, the intra-state part may also be re-conducted under ISTS at the cost of WBSETCL in matching timeframe.
- 4.1.4 It was suggested that reconductoring of Intra state portion of WBSETCL by an ISTS licensee may lead to commercial complications, therefore, reconductoring of intra state LILO portion may be carried out by the owner of the asset i.e. WBSETCL in matching timeframe.
- 4.1.5 After deliberations, NCT approved Transmission scheme “Eastern Region Expansion Scheme-44 (ERES-44))” as mentioned below:

4.1.5.1 Summary of the scheme is given below:

Sl. No.	Name of the scheme and tentative implementation timeframe	Estimated Cost (₹ Cr)	Remarks
1.	Eastern Region Expansion Scheme-44 (ERES-44) Tentative implementation timeframe: 18 months (15 months on best effort basis) from the date of allocation of project	385.77	Approved under RTM through POWERGRID

4.1.5.2 Detailed scope of the scheme is given below:

Sl. No.	Scope of the Transmission Scheme	Capacity (MVA) / Circuit km (ckm) / Nos.
(i)	Reconductoring of ISTS portion of Alipurduar (POWERGRID) – Falakata (WBSETCL) 220 kV D/c line with HTLS conductor of ampacity 1250 A	54 ckm
(ii)	Reconductoring of ISTS portion of Falakata (WBSETCL) – Birpara (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A	54 ckm
(iii)	Reconductoring of Birpara (POWERGRID) – Binaguri (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A	160 ckm
(iv)	Reconductoring of Binaguri (POWERGRID) – Siliguri (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A	12 ckm
(v)	Reconductoring of Siliguri (POWERGRID) – Kishanganj (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A	216 ckm
(vi)	Reconductoring of Kishanganj (POWERGRID) – Dalkhola (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A	62 ckm
(vii)	Reconductoring of ISTS portion of Dalkhola (POWERGRID) – Gazole (WBSETCL) 220 kV D/c line with HTLS conductor of ampacity 1250 A	195 ckm
(viii)	Reconductoring of ISTS portion of Gazole (WBSETCL) –	33 ckm

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Sl. No.	Scope of the Transmission Scheme	Capacity (MVA) / Circuit km (ckm) / Nos.
	Malda (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A	
(ix)	Upgradation of associated 220 kV bay equipment at Alipurduar (POWERGRID)	Replacement of Wave Traps of Alipurduar (POWERGRID) – Falakata (WBSETCL) 220 kV D/c line commensurate with rating of HTLS.
(x)	Upgradation of associated 220 kV bay equipment at Birpara (POWERGRID)	Replacement of Wave Traps of Falakata (WBSETCL) – Birpara (POWERGRID) and Birpara (POWERGRID) – Binaguri (POWERGRID) 220 kV D/c lines commensurate with rating of HTLS.
(xi)	Upgradation of associated 220 kV bay equipment at Binaguri (POWERGRID)	Replacement of Wave Traps of Birpara (POWERGRID) – Binaguri (POWERGRID) and Binaguri (POWERGRID) – Siliguri (POWERGRID) 220 kV D/c lines commensurate with rating of HTLS.
(xii)	Upgradation of associated 220 kV bay equipment at Siliguri (POWERGRID)	Replacement of Wave Traps of Binaguri (POWERGRID) – Siliguri (POWERGRID) and Siliguri (POWERGRID) – Kishanganj (POWERGRID) 220 kV D/c lines commensurate with rating of HTLS.
(xiii)	Upgradation of associated 220 kV bay equipment at Dalkhola (POWERGRID)	Replacement of Wave Traps of Kishanganj (POWERGRID) – Dalkhola (POWERGRID) and Dalkhola (POWERGRID) – Gazole (WBSETCL) 220 kV D/c lines commensurate with rating of HTLS.
(xiv)	Upgradation of associated 220 kV bay equipment at Malda (POWERGRID)	Replacement of Wave Traps of Gazole (WBSETCL) – Malda

Sl. No.	Scope of the Transmission Scheme	Capacity (MVA) / Circuit km (ckm) / Nos.
		(POWERGRID) 220 kV D/c line commensurate with rating of HTLS.
(xv)	Supply and installation of OPGW along with terminal equipment at both ends of Siliguri (POWERGRID) – Kishanganj (POWERGRID) 220 kV D/c (HTLS) line	108 km

Note:

(a) WBSETCL shall reconductor their following lines sections under intra-state scheme matching with completion of ISTS scheme namely ERES-44:

- About 4 km intra-state portion of Alipurduar (POWERGRID) – Falakata (WBSETCL) 220 kV D/c line at Falakata end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Falakata (WBSETCL) end commensurate with rating of HTLS (1250 A).
- About 4 km intra-state portion of Birpara (POWERGRID) – Falakata (WBSETCL) 220 kV D/c line at Falakata end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Falakata (WBSETCL) end commensurate with rating of HTLS (1250 A).
- About 2 km intra-state portion of Dalkhola (POWERGRID) – Gazole (WBSETCL) 220 kV D/c line at Gazole end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Gazole (WBSETCL) end commensurate with rating of HTLS (1250 A).
- About 2km intra-state portion of Gazole (WBSETCL) – Malda (POWERGRID) 220 kV D/c line at Gazole end with HTLS conductor of ampacity 1250 A along with necessary upgradation of associated 220 kV bay equipment at Gazole (WBSETCL) end commensurate with rating of HTLS (1250 A).

(b) WBSETCL will LILO the Dhalkola – Gazole 220 kV D/c line with 1250 A HTLS under their intra-state scheme for establishment of 220 kV level at their existing 132/33kV Raiganj (WBSETCL) S/s.

(c) ISTS licensee and WBSETCL shall coordinate for reconductoring of their respective portion of the lines matching with completion schedule of this scheme.

4.2 Transmission system for Evacuation of Power from RE Projects in Rajgarh (1500 MW) SEZ in Madhya Pradesh-Phase III

4.2.1 Representative of CTUIL stated that 2.5 GW REZ potential has been identified at Rajgarh (MP).

- i. Phase-I of 1.5 GW involves establishment of Pachora PS with 3x500 MVA 400/220 kV ICTs and Pachora PS – Bhopal 400 kV D/c line which has been implementation by M/s G R Infraprojects Ltd. (Commissioned).
- ii. Phase-II (1 GW) involves ICT augmentation (4th, 5th & 6th) Pachora PS along with Pachora PS – Ujjan (MPPTCL) 400 kV D/c line which is presently under implementation by M/s G R Infraprojects Ltd. with SCOD of 14.02.2026.

4.2.2 He further stated that in view of applications received for cumulative capacity of ~4000 MW at Pachora PS till July-2024, it was found prudent to expand the substations to its full capacity so as to accommodate applications being received

beyond 2.5 GW at Pachora PS. Out of 1508 MW applications received for Rajgarh Ph-III system, RE projects for 1321 MW have been agreed for grant with start date of March 2027. For additional 187 MW applications received in July 2024 are under process and shall also be granted with start date of March 2027.

- 4.2.3 To evacuate the power from these areas, CTUIL proposed a transmission scheme broadly consisting of augmentation at Pachora PS and Pachora PS – Rajgarh(PG) 400 kV D/c line. CTUIL also mentioned that with the augmentation at Pachora PS, the substation will be closed for further connectivity.
- 4.2.4 After deliberations, NCT recommended the transmission scheme “Transmission system for Evacuation of Power from RE Projects in Rajgarh (1500 MW) SEZ in Madhya Pradesh-Phase III” to enable evacuation of upto 4000 MW Power from RE Projects in Rajgarh (i.e. 1500 MW beyond 2500 MW) SEZ in Madhya Pradesh as mentioned below:

4.2.4.1 Summary of the scheme is given below:

Sl. No.	Name of the scheme and tentative implementation timeframe	Estimated Cost (₹ Cr)	Remarks
1.	Transmission system for Evacuation of Power from RE Projects in Rajgarh (1500 MW) SEZ in Madhya Pradesh-Phase III Tentative implementation timeframe: Implementation timeframe of elements at Sl. No. 1, 2a, 3 & 4 shall be 24 months from date of SPV Transfer & for element at Sl. No. 2b shall be 31.03.2028 in matching timeframe of RE generator (Purvah Green Power Pvt. Ltd.: 297 MW)	1079	Recommended under TBCB route with RECPDCL as BPC

4.2.4.2 Detailed scope of the scheme is given below:

Sl. No.	Scope of the Transmission Scheme	Capacity
1.	Creation of New 220 kV Bus Section (3rd) with 220 kV Bus Sectionalizer and 400/220 kV, 3x500 MVA ICT augmentation (7th, 8th & 9th) at Pachora PS terminated on 220 kV Bus Section (3rd)	500 MVA 400/220 kV ICT – 3 Nos. 400 kV ICT bay – 3 Nos. (on Section-II) 220 kV ICT bay – 3 Nos. (on Section-III) 220 kV Bus Sectionalizer bays – 1 set 220 kV BC & TBC – 1 Nos. each
2.	2a. 3 Nos. 220 kV line bays for RE interconnection on Bus Section (3rd)	3 Nos. on Sec-III
	2b. 1 Nos. 220 kV line bay for RE Interconnection of Purvah Green Power Pvt. Ltd. on Bus Section (3rd)	1 No. on Sec-III
3.	Pachora PS – Rajgarh(PG) 400 kV D/c line (Quad ACSR/ AAAC/ AL59 Moose equivalent) along with associated line bays at both ends and 50 MVA Switchable Line Reactors (Sw LR) on each ckt at both ends	Line length: 180 km. 400 kV line bays: 4 Nos. (2 at Rajgarh(PG) & 2 at Pachora PS) 420 kV, Switchable Line Reactors (Sw LRs): 4 Nos. (2 at Rajgarh(PG) & 2 at Pachora PS) Switching equipment for 400 kV line reactor – 4 Nos. (2 at Rajgarh(PG) & 2 at Pachora PS)
4.	Installation of 1x125 MVAR, 420 kV bus	125 MVA, 420 kV Bus reactor – 1 Nos.

Sl. No.	Scope of the Transmission Scheme	Capacity
	reactor at Pachora PS (400 kV Bus Section-II)	400 kV Bus reactor bay: 1 Nos.

4.3 Transmission system for Evacuation of Power from RE Projects in Neemuch (1000 MW) SEZ in Madhya Pradesh-Phase II

4.3.1 Representative of CTUIL stated that applications for cumulative capacity of 1970 MW has been received at Neemuch PS till July 24, it was found prudent to expand the substations to its full capacity so as to accommodate applications being received beyond 1 GW at Pachora PS. CTUIL proposed transmission scheme consisting of augmentation a Neemuch PS, creation of 400/220 kV Handiya substation, Neemuch PS – Pachora PS 400 kV D/c line, Pachora PS – Handiya 400 kV D/c line and LILO of Khandwa(PG) – Itarsi(PG) 400 kV D/c (Twin Moose) line at Handiya S/s etc. CTUIL also mentioned that with the augmentation at Neemuch PS, the substation will be closed for further connectivity.

4.3.2 After deliberations, NCT recommended the scheme “Transmission system for Evacuation of Power from RE Projects in Neemuch (1000 MW) SEZ in Madhya Pradesh-Phase II” to enable Evacuation of upto 2000 MW Power from RE Projects in Neemuch (i.e. 1000 MW beyond 1000 MW) SEZ in Madhya Pradesh as mentioned below:

4.3.2.1 Summary of the scheme is given below:

Sl. No.	Name of the scheme and tentative implementation timeframe	Estimated Cost (₹ Crore)	Remarks
1.	Transmission system for Evacuation of Power from RE Projects in Neemuch (1000 MW) SEZ in Madhya Pradesh-Phase II Implementation timeframe: 24 months from date of SPV transfer	2393	Recommended under TBCB route with PFCCL as BPC

4.3.2.2 Detailed scope of the scheme is given below:

Sl.No.	Scope of the Transmission Scheme	Capacity
1.	Creation of New 220 kV Bus Section-II at Neemuch PS with Augmentation of transformation capacity by 3x500 MVA, 400/220 kV ICTs (3 rd , 4 th & 5 th) at Neemuch S/s along with associated bays.	500 MVA 400/220 kV ICT – 3 Nos. 400 kV ICT bay – 3 Nos. 220 kV ICT bay – 3 Nos. (on Sec-II) 220 kV Bus Sectionalizer bays – 1 set 220 kV BC & TBC – 1 Nos. each
2.	4 Nos. 220 kV Line bays at Neemuch PS for RE interconnection	220 kV Bays – 4 Nos. on Sec-II
3.	Neemuch PS – Pachora PS 400 kV D/c line (Quad ACSR/ AAAC/ AL59 Moose equivalent) along associated Line bays and 50 MVar Switchable Line Reactor (Sw LR) on each ckt at both ends	Line length: 190km. 400 kV Line bays: 4 Nos. (2 at Neemuch PS & 2 at Pachora PS) 420 kV, Switchable Line Reactors (Sw LR): 4 Nos. (2 at Neemuch PS & 2 at Pachora PS) Switching equipment for 400 kV

		line reactor – 4 Nos. (2 at Neemuch PS & 2 at Pachora PS)
4.	Establishment of 2x500 MVA, 400/220 kV S/s at Handiya alongwith 2x125 MVar 420 kV Bus Reactors Future provision (space for): ➤ 400 kV line bays along with switchable line reactors– 6 Nos. (Sec-II) ➤ 400/220 kV ICT along with bays - 4 Nos. (1 Nos. on Sec-I & 3 Nos. on Sec-II) ➤ 400 kV Bus Reactor along with bays: 2 Nos. (Sec-II) ➤ 220 kV line bays: 8 Nos. (on Sec-II) ➤ 400 kV Sectionalization bay: 1 set ➤ 220 kV Sectionalization bay: 1 set ➤ 220 kV TBC & BC: 1 Nos.	400/220 kV ICTs: 2 Nos. 400 kV ICT Bays: 2 Nos. 220 kV ICT Bays: 2 Nos. 400 kV Line bays: 6 Nos. 220 kV line bays for MPPTCL – 8 Nos. 125 MVar, 420 kV Bus reactor – 2 Nos. 400 kV Bus reactor bay: 2 Nos. 220 kV TBC bay – 1 Nos. 220 kV BC bay – 1 Nos.
5.	Pachora PS – Handiya 400 kV D/c line (Quad ACSR/ AACR/ AL59 Moose equivalent) along with associated bays at Pachora PS end and 50 MVar Switchable Line Reactor (Sw LR) on each ckt at both ends	Line length: 190 km. 400 kV bays: 2 Nos. (at Pachora PS) 420 kV, Sw LR: 4 Nos. (2 at Handiya & 2 at Pachora PS) Switching equipment for 400 kV line reactor – 4 Nos. (2 at Handiya & 2 at Pachora PS)
6.	LILO of Khandwa(PG) – Itarsi(PG) 400 kV D/c (Twin Moose) line at Handiya S/s	LILO route length : 22 km (88 ckm) The Khandwa(PG) – Itarsi(PG) 400 kV D/c line is of Twin Moose configuration and LILO shall be of similar conductor configuration
7.	Installation of 1x125 MVAR, 420 kV bus reactor (2 nd) at Neemuch PS	125 MVar, 420 kV Bus reactor – 1 Nos. 400 kV Bus reactor bay: 1 Nos.
<i>Note: TSP of Neemuch & Pachora PS shall provide space for above scope of work</i>		

4.4 North Eastern Region Expansion Scheme-XXI Part-B (NERES-XXI Part-B)

4.4.1 Representative from CTUIL stated that the existing 132 kV Badarpur (POWERGRID) switching station was commissioned in 1999 and shall be completing 25 years in service by 2024. POWERGRID the owner of the substation has informed that they are facing issues in O&M of the switching station and to improve the reliability it would be prudent to upgrade the switching station from single main and transfer bus scheme to double main transfer bus scheme by converting from Air Insulated Switchgear (AIS) to Gas Insulated Switchgear (GIS).

4.4.2 Further, towards adoption of new technology in the Indian Grid, it was proposed that the upgradation could be carried out as Green GIS instead of conventional GIS owing to the following benefits:

- Green GIS is a new technology in which Sulfur Hexafluoride (SF₆) gas is not used and this technology is being adopted by several countries in the world.
- This would help in the reduction of usage of Green House Gas and would be a step towards achieving sustainable development targets.

- 4.4.3 The scheme was taken up for deliberations in the 15th meeting of NCT held on 25-08-2023, wherein it was decided to review the scheme subsequently. The scheme was thereafter discussed in the 16th meeting of NCT held on 30-11-2023, it was decided to defer the scheme at present and take it up after additional discussions on new technology such as major benefits of Green GIS, availability of Green GIS vendors in India, additional cost implication (conventional GIS vis-à-vis Green GIS) etc.
- 4.4.4 Director (SO), Grid-India suggested that instead of going for green GIS for complete station, some portion of the station (limited number of bays) may be considered for green GIS. Also, stations in other regions where green GIS might be more suitable due to environmental conditions may also be considered as potential candidates for green GIS.
- 4.4.5 After deliberations, it was decided that a committee with members from CEA, CTUIL and POWERGRID to be constituted to survey the green GIS literature, technical aspects, undertake visit of the substation, exploring possible solutions etc. The committee shall submit its recommendations within 06 months.
- 4.5 Upgradation of ± 800 kV, 6000 MW Raigarh-Pugalur HVDC system for enhancement of reverse power capacity upto 6000 MW from existing 3000 MW**
- 4.5.1 Raigarh-Pugalur ± 800 kV, 6000 MW HVDC system is capable of transferring 6000 MW of power from Raigarh to Pugalur. However, its reverse power capacity i.e. Pugalur to Raigarh is 3000 MW.
- 4.5.2 Representative of CTUIL stated that enhancement of reverse power capacity upto 6000 MW from existing 3000 MW has been approved in 52nd SPRC meeting held on 03.08.2024 at an indicative cost of Rs 1000 Cr (including cost of system studies) and required AC system strengthening at Pugalur (estimated cost of Rs 400 crores) & Raigarh (estimated cost of Rs 1800 Crores)
- 4.5.3 Further, SRPC vide letter dated 02.09.2024 recommended CTUIL to take up the matter to NCT at the earliest.
- 4.5.4 Director (SO), Grid-India stated that until the adequate AC system is available on both ends, the HVDC capacity of 6000 MW in reverse direction can't be utilized even after the proposed HVDC upgradation.
- 4.5.5 CTU stated that as confirmed by POWERGRID vide e-mail dated 10.10.2024, there is no space available at both Raigarh (Kotra) and Dharamjaygarh S/s for augmentation of AC system to enable reverse power flow on Raigarh- Pugalur HVDC beyond 3000 MW. AC System augmentation at Raigarh (Kotra) S/s in WR is not possible and if at all reverse flow of more than 3000 MW is required with N-1 compliance, it would require an elaborate exercise of shifting certain Thermal generating stations from Raigarh (Kotra) S/s to a new substation which would be cumbersome and shall entail significant costs as well as consent from thermal generating stations which are already connected at Raigarh (Kotra) S/s. Further, it was also informed that with the reversal

of HVDC from 3000 MW to 6000 MW, no enhancement in TTC/ATC between SR-NEW Grid is expected looking into the very less sensitivity of the HVDC in TTC/ATC. .

- 4.5.6 After deliberations, it was decided that a committee will be formed comprising members from CEA, CTUIL, POWERGRID, Grid-India, SRPC and WRPC. The committee shall carry out comprehensive study and propose comprehensive plan including AC system strengthening in Southern Region and Western Region along with the total scheme cost. The comprehensive plan may again be put up to SRPC before bringing it to NCT.

4.6 Augmentation of transformation capacity at KPS3 (GIS) S/s under Khavda Phase-V Part B3 scheme

- 4.6.1 Representative from CTUIL stated that KPS3 S/s with 3x1500 MVA ICTs on Section-I is under implementation by M/s KPS3 Transmission Ltd. (Subsidiary of POWERGRID). 1x1500 MVA Addl. ICT at Section-I is also being implemented by M/s POWERGRID (under RTM) under Khavda Ph-IV Part E3 scheme. Further, Section-II of KPS3 is being established by M/s Khavda IV A Power Transmission Ltd. (Subsidiary of Adani Energy Solutions Ltd.) with SCOD of Aug-26 with 3x1500 MVA ICTs and 3 Nos. 400 kV bays at Bus Section-II for RE interconnection. Out of above bays, 2 bays had been allocated to M/s SRPL (1250 MW) & NHPC (600 MW).
- 4.6.2 Applications for cumulative 3290 MW have been received at KPS3 (Sec-II) which require 3 Nos. 400 kV bays. 1 No. 400 kV bay is already being implemented under Khavda Phase-IV Part A scheme. Balance 2 nos. 400 kV bays along with addl. 1x1500 MVA ICT at KPS3 (Sec-II) are required to be implemented after considering N-1 compliance of already planned 765/400 kV ICTs at KPS3 (Sec-II). The proposed 765/400 kV ICT & 400 kV bays would facilitate immediate injection of power at KPS3 (Section-II). CTUIL also informed that the substation will be closed for further applications.
- 4.6.3 NCT directed CTUIL to explore the possibility for Battery Energy Storage System (BESS) for optimizing transmission infrastructure at Khavda as well as other RE potential Zones. Consideration of Storage may facilitate integration of additional RE Capacity.
- 4.6.4 After deliberations, NCT approved the transmission scheme “Augmentation of transformation capacity at KPS3 (GIS) S/s under Khavda Phase-V Part B3 scheme” under RTM mode as follows

4.6.4.1 Summary of the scheme is given below:

Sl. No.	Name of the scheme and tentative implementation timeframe	Estimated Cost (₹ Crore)	Remarks
1.	Augmentation of transformation capacity at KPS3 (GIS) S/s under Khavda Phase-V Part B3 scheme	252	Approved under RTM through M/s Khavda IV A Power

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	Implementation timeframe: 24 months from the date of allocation		Transmission Ltd. (Subsidiary of Adani Energy Solutions Ltd.) (i.e. TSP of KPS3 (Sec-II))
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4.6.4.2 Detailed scope of the scheme is given below:

S.N.	Scope of the Transmission Scheme	Capacity
1.	Augmentation of transformation capacity at KPS3(GIS) by 1x1500 MVA, 765/400 kV ICT on Bus section-II (8 th) along with 1 No. 400 kV line bay for termination of 1 st ckt out of 400 kV D/c line being implemented by AGEL (Appl. No. 2200000953) for 1530 MW	• 765/400 kV ICT – 1 (1x1500 MVA) • 765 kV ICT bay – 1 (+1 no. bay for dia completion with Switchable Line Reactor (SLR) provision in future bay) on Bus section-II • 400 kV ICT bay – 1 (+ 1 no. bay for dia completion and termination of the proposed Line for RE interconnection) on Bus section-II
2.	1 Nos. 400 kV line bay on KPS3 400 kV Bus Section-II for termination of 2 nd ckt out of 400 kV D/c line being implemented by AGEL (Appl. No. 2200000953) for 1530 MW	400 kV line bays – 1 no. (+ 1 no. bay for dia completion with the provision to terminate future 400/220 kV ICT)
<i>Note: TSP of KPS3 (GIS) shall provide space for above scope of work.</i>		

4.7 Supply and Installation of additional Fiber Optic Test Equipment (FOTE) and Ethernet cards at Automatic Generation Control (AGC) and Critical Nodes of SR Region.

4.7.1 Representative of CTUIL stated that as per CEA, Manual of Communication Planning in Power System Operation 2022, CTU for high availability requirements for Power System Communication, redundancy with route diversity for critical links shall be maintained. Additional FOTE and redundant Ethernet ports are required at all AGC operated generating stations, in view of resource disjoint and criticality of AGC operation for grid operation purpose.

4.7.2 After deliberations, NCT approved the communication scheme “Supply and Installation of additional FOTE and Ethernet cards at AGC & Critical Nodes of SR Region” under RTM mode as follows

4.7.2.1 Summary of the scheme is given below:

SI No.	Name of the scheme and tentative implementation timeframe	Estimated Cost (₹ Cr)	Remarks
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1.	Supply and Installation of additional FOTE and Ethernet cards at AGC & Critical Nodes of SR Region Tentative implementation timeframe: 12 months from date of allocation of project	1.02	Approved under RTM through POWERGRID
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4.7.2.2 Detailed scope of the scheme is given below:

- (i) Supply and installation of 3 nos. FOTE with STM 16 capacity at following locations:

SI.No.	Station Name	No of FOTE	Remark
1.	Simhadri-1	1	For AGC purpose
2.	NP KUNTA	1	Shortage of ports
3.	NLC-TPS-2 Stage 1	1	For AGC purpose
Total		3	

- (ii) Supply and installation of 10 Nos. ethernet cards at following locations:

SI.No.	Station Name	Ethernet Cards required
1	Ramagundam -I	2
2	Ramagundam -II	
3	Ramagundam -III	
4	NTPC Vallur	2
5	NTPL Tuticorin	1
6	NTPC Kudgi	2
7	NLC - TPS 2 Exp	1
8	NLC - TPS 1 Exp	1
9	NNTPS new Neyveli	1
Total		10

4.8 Requirement of Additional FOTE at various ISTS nodes in ER due to exhaustion of existing capacity

4.8.1 Representative of CTUIL stated that the transmission scheme “Requirement of Additional FOTE at various ISTS nodes in ER” with capacity utilisation of approximately 90% and above and few other important stations is required.

4.8.2 After deliberations, NCT approved the communication scheme “Requirement of Additional FOTE at various ISTS nodes in ER due to exhaustion of existing capacity” under RTM mode as follows

4.8.2.1 Summary of the scheme is given below:

SI No.	Name of the scheme and tentative implementation timeframe	Estimated Cost (₹ Cr)	Remarks
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1.	Requirement of Additional FOTE at various ISTS nodes in ER due to exhaustion of existing capacity Tentative implementation timeframe: 12 months from date of allocation of project	9.78	Approved Under RTM through POWERGRID
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4.8.2.2 Detailed scope of the scheme is given below:

- a) Conversion of 13 nos. STM 16 FOTE to STM 64 FOTE
- b) Conversion of 4 nos. STM 4 FOTE to STM 16 FOTE by utilizing four (4) Nos. FOTEs freed from upgradation of STM 16 FOTE to STM 64 FOTE

S. No.	Node Name(with approx 90% capacity exhausted)	Upgradation/replacement required	Detail of Card/Equipment required for upgradation	Estimated Cost (Rs.)
1	Kasba	STM 16 to STM 64	Existing Equipment Cannot be upgraded. New STM 64 SDH Equipment Required	74 Lakhs
2	ERLDC	STM 16 to STM 64		74 Lakhs
3	Jeerat	STM 16 to STM 64		74 Lakhs
4	Subhashgram	STM 16 to STM 64		74 Lakhs
5	Farakka	STM 16 to STM 64		74 Lakhs
6	Kahalgaoon	STM 16 to STM 64		74 Lakhs
7	Saharsa	STM 16 to STM 64		74 Lakhs
8	Binaguri	STM 16 to STM 64		74 Lakhs
9	Purnea	STM 16 to STM 64		74 Lakhs
10	Kishenganj	STM 16 to STM 64		74 Lakhs
11	Sasaram	STM 16 to STM 64		74 Lakhs
12	AB380 Repeater	STM 16 to STM 64		74 Lakhs
13	Allahabad	STM 16 to STM 64		74 Lakhs
Total Cost for conversion of 13 nos. of STM 16 equipment to STM 64:A				9.62 Cr
14	Gaya	STM 4 to STM 16	Upgradation to be done by utilizing four Nos. of STM 16 equipment freed in above list after upgradation to STM 64.	4 Lakhs
15	Essar Chandwa	STM 4 to STM 16		4 Lakhs
16	Darbhang(KPTL)	STM 4 to STM 16		4 Lakhs
17	Arrah	STM 4 to STM 16		4 Lakhs
Total Cost for conversion of 04 Nos. of STM 4 equipment to STM 16: B				16 Lakhs
Total Cost for conversion of 13 Nos. of STM 16 to STM 64 and 04 nos. of STM4 equipment to STM 16: A+B				9.78 Cr

4.9 Deployment of FOTE (SDH Equipment) and amplifier solutions at Alipurduar S/s end for OPGW based communication and Teleprotection for 400 kV lines from Punatsangchhu-II Hydroelectric Project (PHEP-II), Punatsangchhu-I Hydroelectric Project (PHEP-I) and Jigmeling of Bhutan to Alipurduar, India

4.9.1 Representative of CTUIL stated that to ensure the accurate coordination of devices between SDH at one end i.e Alipurduar, India and MPLS-TP at other end i.e., Punatsangchhu-II, as well as to cater to cybersecurity issue of the Indian Grid, the proposed scheme for Alipurduar S/s end needs to be implemented.

4.9.2 After deliberations, NCT approved the communication scheme “Deployment of FOTE (SDH Equipment) and amplifier solutions at Alipurduar S/s end for OPGW based communication and Teleprotection for 400kV lines from PHEP-II, PHEP-I and Jigmeling of Bhutan to Alipurduar, India” under RTM mode as follows

4.9.2.1 Summary of the scheme is given below:

SI No.	Name of the scheme and tentative implementation timeframe	Estimated Cost	Remarks
1.	Deployment of FOTE (SDH Equipment) and amplifier solutions at Alipurduar S/s end for OPGW based communication and teleprotection for 400kV lines from PHEP-II, PHEP-I and Jigmeling of Bhutan to Alipurduar, India Tentative implementation timeframe: 6 months from date of allocation	₹ 65 lakhs	Approved Under RTM through POWERGRID

4.9.2.2 Detailed scope of the scheme is given below:

- a) One (1) set of STM-4 SDH equipment along with panel supporting minimum five directions with MSP (Multiplex Section Protection 1+1) & equipped with E1 and Ethernet interfaces.
- b) Six (6) sets of 175 km Amplifiers solutions: 2 directed towards Punatsangchhu-II (PHEP-II), 2 directed towards Punatsangchhu-I (PHEP-I) and 2 directed towards Jigmeling.

Note: POWERGRID to coordinate with Bhutan ends while procuring the equipment to avoid any non-compatibility issues

5 Grid-India Presentation on Performance of the National Grid in Q1 and Q2 of FY 2024-25

5.1.1 Representative of Grid-India made a presentation on performance of the National Grid in Q1 and Q2 of FY 2024-25. Copy of presentation is attached at Annex-II. Major points highlighted during the presentation are given in subsequent paragraphs.

5.1.2 **Large number of generation loss events in NR RE Complex:** NLDC representative informed that there have been more than 55 events between January 2022 and September 2024 involving RE generation loss of more than 1000 MW. Around 13 such generation loss events (>500 MW) have occurred between April – September 2024.

One of the major reasons for these grid events has been the non-compliance of the RE plants against CEA's standards. The details of the non-compliance and measures being taken by Grid-India to address the issues were explained by NLDC representative in the meeting.

5.1.3 **Oscillations in NR RE complex and issues with performance of STATCOMs:** NLDC representative informed that low frequency oscillations (3-4 Hz) are being observed in NR RE complex on a regular basis. The oscillations start during morning hours with increase in solar generation and are mainly observed in voltage and reactive power. The non-standardization of PPC delays is one of the potential reasons for these oscillations.

He further informed that currently 05 Nos. STATCOMs are operational in NR RE complex. These STATCOMs have been installed to provide the fast dynamic reactive power support to the grid during any contingency. The response time of the installed STATCOMs shall be less than 30 ms as per their control manuals.

However, following observations in the performance of STATCOMs (as visible in DR as well as PMU plots) in NR RE complex has been observed:

- High response time (sluggish response) during faults leading to injection of reactive power post clearance of fault
- Automatic gain reduction by stability controller of the STATCOMs on hunting detection, possibly leading to enhancement in magnitude of oscillations

The problem of amplification of oscillations with reduction of STATCOM gain in voltage control mode has been analyzed with measurement (TFR) data and simulation results for a large data set. Detailed analysis in this regard was also shared with CEA, CTUIL, POWERGRID and OEM.

It is observed that oscillations damp out when the STATCOMs are being operated in Fixed 'Q' mode. This mode of operation, however, defeats the purpose of providing STATCOMs for fast dynamic reactive power support.

The performance issues being observed in the current STATCOMs necessitate proper tuning of the STATCOM controllers. Also, necessary modifications in the RfP of upcoming STATCOMs is required based on the current experience.

NCT directed that a committee may be constituted with members from CEA, CTUIL, Grid-India, Power Grid and all the STATCOM OEMs to deliberate the performance

related issues and the possible remedial measures. The matter to be coordinated by GM Division, CEA in consultation with PS Wing, CEA.

5.1.4 17th June 2024 Load Loss Event in Northern Region and Reactive Power Planning for Bulk Loads (Electrolyzers and Data Centers): Director (SO), Grid-India explained the 17th June 2024 grid event of simultaneous tripping of both bipoles of the +/-800 kV HVDC Champa (WR) – Kurukshetra (NR) link led to a substantial load reduction (~16.5 GW) in the northern region. The event started with the tripping of the aforementioned HVDC link and triggered a series of events starting from the sudden voltage drop across the stations in the Northern region and subsequent stalling and tripping of certain portion of load.

He further informed that the Ministry of Power constituted a Committee under the Chairmanship of Member (GO&D), CEA to analyze the event. The committee, in its suggestions, recommended the planning of suitable dynamic reactive power compensation near load centers.

He further stated that the event (especially the stalling of load) was replicated in simulation studies with proper load modelling. As the load behavior is changing, the existing philosophy for modelling of loads in the planning and operational studies need to be reviewed. Without proper load modelling in the studies, it would be difficult to capture such phenomenon in the studies. The standards for protection settings of loads, especially 1-ph and 3-ph motor loads, also needs to be reviewed.

Further, as a large quantum of electrolyzer and data center load is also envisaged to be connected at ISTS level in near future, it is important that adequate reactive power compensation is planned nearby such ISTS load feeding stations.

He suggested that the following activities may be taken up on priority:

- Study of load behavior and consideration of same in the simulation studies
- Planning of dynamic reactive power compensation at both inter-state and intra-state level near major load centers
- Review of the standards specifying standards for protection settings of loads, especially 1-ph and 3-ph motor loads
- Planning of suitable reactive power compensation at large ISTS stations being planned for feeding large electrolyzer and data center loads

Chairperson, CEA directed that the reactive power planning study at all the major load centers shall be taken up on priority. Grid India was advised to identify important BIS standards/committees in this regard and CEA/Stakeholders may take up with BIS.

He further directed that two separate committees comprising of members from CEA, CTUIL and Grid-India may be constituted for comprehensive study of performance, control strategy, reactive power requirements etc. of electrolyzer and data center loads respectively so that suitable reactive power compensation could be planned. The characteristics and load pattern/behaviour of electrolyzers would be coordinated by ET&I Division, CEA and of data centres would be coordinated by PDM&LF Division, CEA.

5.1.5 Evacuation of large quantum of RE under T-GNA: NLDC representative informed that a large quantum of RE generation (~5700 MW) is being evacuated under T-GNA due to the delay in the commissioning of associated transmission system. There is possibility of certain RE curtailment if the commissioning of the associated transmission system is not expedited.

5.1.6 Flexibility and Ramping Requirement: NLDC representative informed that persistent high frequency was observed in the India's grid during solar hours on few days in the month of August 2024. The high RE generation and the limited flexibility to further reduce the thermal generation to accommodate the RE generation was one of the major factors for this high frequency operation. Further, as the thermal generating units are required during non-solar hours, these units cannot be taken out of service during the high frequency operation period.

Another challenge is being faced in meeting the ramping requirement during evening hours where the flexibility requirements have increased significantly due to the increasing demand ramp up coupled with the simultaneous decline in solar generation. There is an urgent requirement of fast ramping resources in the grid to meet the flexibility requirements in coming days.

5.1.7 Constraints in Maharashtra system during high export from Southern Region: Director (SO), Grid-India stated that the congestion is being faced in export of power from SR during high RE periods. To relieve the congestion, 765 kV Narendra – Pune D/C was planned with commissioning schedule of July 2024. However, the line is delayed and revised date of commissioning is December 2024.

He further stated that even after commissioning of 765 kV Narendra – Pune D/C, constraints in western Maharashtra would still remain in export of power from SR. There is an urgent requirement for expediting the planned transmission system augmentation in western Maharashtra area.

Chairperson, CEA directed that the augmentation works in Maharashtra and other critical areas shall be monitored on priority.

Summary of the deliberations of the 24th meeting of NCT held on 23rd October, 2024**I. Modification in the earlier approved/notified transmission schemes:****1. Revision in SCOD of 400 kV D/C Jhatikara-Dwarka line under REZ Phase-III Part-D Phase-II scheme**

NCT approved the revised SCOD for 400 kV D/C (quad) Jhatikara-Dwarka line under “Transmission system for evacuation of 20 GW REZ power from Rajasthan under phase-III, Part-D, Phase-II” scheme as 28th February, 2026 (31st December 2025 on best effort basis).

2. Change in scope of Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1: 4 GW) [Sirohi/Nagaur] Complex

NCT approved the revised scope of Transmission system for evacuation of power from Rajasthan REZ Ph-V (Part-1: 4 GW) [Sirohi/Nagaur] Complex as follows:

Sl. No.	Original scope of the transmission scheme	Revised scope of the transmission scheme
1. Transmission system for immediate Evacuation of Power from Sirohi S/s (2 GW)		
1	5x500 MVA, 400/220 kV ICTs at Sirohi S/s along with transformer bays • 400/220 kV 500 MVA ICTs- 5 Nos. • 400 kV ICT bays-5 Nos. • 220 kV ICT bays- 5 Nos.	4x500 MVA, 400/220 kV ICTs at Sirohi S/s along with transformer bays • 400/220 kV 500 MVA ICTs- 4 Nos. • 400 kV ICT bays-4 Nos. • 220 kV ICT bays- 4 Nos.
2	6 Nos. 220 kV line bays at Sirohi S/s for RE interconnection • 220 kV line bays – 6 Nos.	5 Nos. 220 kV line bays at Sirohi S/s for RE interconnection • 220 kV line bays – 5 Nos.
3	220 kV Sectionalizer bay (1 set) along with 220 kV BC (2 Nos.) bay and 220 kV TBC (2 Nos.) bay at Sirohi S/s	• 220 kV Sectionalizer bay (1 set) • 220 kV BC (2 Nos.) bay and 220 kV TBC (2 Nos.) bay
4	-	1 No. 400 kV line bays at Sirohi S/s for RE interconnection • 400 kV line bay – 1 No.
	<i>Note: There will be no change in other elements of the transmission scheme w.r.t agreed in the 21st NCT meeting/ MoP Gazette dated 29.08.2024.</i>	

II. ISTS Transmission schemes, costing between Rs 100 Crore to Rs 500 Crore, approved by NCT:**1. The transmission schemes approved by NCT under RTM route is given below:**

Sl. No.	Name of Transmission Scheme	Implementation Mode	Implementation timeframe	Estimated Cost (₹ Cr)
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1.	Eastern Region Expansion Scheme-44 (ERES-44)	RTM through POWERGRID	18 months (15 months on best effort basis) from the date of allocation	385.77
2.	Augmentation of transformation capacity at KPS3 (GIS) S/s under Khavda Phase-V Part B3 scheme	RTM through M/s Khavda IV A Power Transmission Ltd. (Subsidiary of Adani Energy Solutions Ltd.) (i.e. TSP of KPS3 (Sec-II))	24 months from the date of allocation	252

The broad scope of above schemes are given below

Sl. No.	Name of Scheme & Tentative implementation timeframe	Broad Scope
1.	Eastern Region Expansion Scheme-44 (ERES-44) Implementation timeframe: 18 Months (15 months on best effort basis) from the date of allocation of project	1. Reconductoring of ISTS portion of Alipurduar (POWERGRID) – Falakata (WBSETCL) 220 kV D/c line with HTLS conductor of ampacity 1250 A (54 ckm) 2. Reconductoring of ISTS portion of Falakata (WBSETCL) – Birpara (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A (54 Km.) 3. Reconductoring of Birpara (POWERGRID) – Binaguri (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A (160 ckm) 4. Reconductoring of Binaguri (POWERGRID) – Siliguri (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A (12 ckm) 5. Reconductoring of Siliguri (POWERGRID) – Kishanganj (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A. (216 ckm) 6. Reconductoring of Kishanganj (POWERGRID) – Dalkhola (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A (62 ckm) 7. Reconductoring of ISTS portion of Dalkhola (POWERGRID) – Gazole (WBSETCL) 220 kV D/c line with HTLS conductor of ampacity 1250 A (195 ckm) 8. Reconductoring of ISTS portion of Gazole (WBSETCL) – Malda (POWERGRID) 220 kV D/c line with HTLS conductor of ampacity 1250 A (33 ckm) 9. Upgradation of associated 220 kV bay equipment at

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		Alipurduar (POWERGRID) 10. Upgradation of associated 220 kV bay equipment at Birpara (POWERGRID) 11. Upgradation of associated 220 kV bay equipment at Binaguri (POWERGRID) 12. Upgradation of associated 220 kV bay equipment at Siliguri (POWERGRID) 13. Upgradation of associated 220 kV bay equipment at Dalkhola (POWERGRID) 14. Upgradation of associated 220 kV bay equipment at Malda (POWERGRID) 15. Supply and installation of OPGW along with terminal equipment at both ends of Siliguri (POWERGRID) – Kishanganj (POWERGRID) 220 kV D/c (HTLS) line (108 km)
2.	Augmentation of transformation capacity at KPS3 (GIS) S/s under Khavda Phase-V Part B3 scheme Implementation timeframe: 24 months from the date of allocation	1. Augmentation of transformation capacity at KPS3(GIS) by 1x1500 MVA, 765/400 kV ICT on Bus section-II (8th) along with 1 Nos. 400 kV line bay for termination of 1st ckt out of 400 kV D/c line being implemented by AGEL (Appl. No. 2200000953) for 1530MW 2. 1 No. 400kV line bay on KPS3 400 kV Bus Section-II for termination of 2nd ckt out of 400 kV D/c line being implemented by AGEL (Appl. No. 2200000953) for 1530 MW Note: TSP of KPS3 (GIS) shall provide space for above scope of work.

(Detailed scope as approved by 24th NCT and subsequent amendments thereof)

III. ISTS Transmission schemes, costing greater than ₹ 500 Crore, recommended by NCT to MoP:

The ISTS transmission schemes recommended by NCT to MoP are given below:

Sl. No.	Name of Transmission Scheme	Implementation Mode	Tentative Implementation timeframe	BPC	Estimated Cost (₹ Crs)
1.	Transmission system for Evacuation of Power from RE Projects in Rajgarh (1500 MW) SEZ in Madhya Pradesh-Phase III	TBCB	Implementation timeframe of elements at Sl. No. 1, 2a, 3 & 4 shall be 24 months from date of SPV transfer & for element at Sl. No. 2b shall be 31.03.2028 in	RECPDC L	1079

Minutes of the 24th meeting of National Committee on Transmission (NCT)

			matching timeframe of RE generator (Purvah Green Power Pvt. Ltd.: 297 MW)		
2.	Transmission system for Evacuation of Power from RE Projects in Neemuch (1000 MW) SEZ in Madhya Pradesh-Phase II	TBCB	24 months from date of SPV transfer	PFCCCL	2393

The broad scope of the above ISTS schemes to be notified in Gazette of India is as given below:

Sl. No.	Name of Scheme & Tentative implementation timeframe	Broad Scope	Bid Process Coordinator
1.	Transmission system for Evacuation of Power from RE Projects in Rajgarh (1500 MW) SEZ in Madhya Pradesh-Phase III Implementation timeframe: Implementation timeframe of elements at Sl. No. 1, 2a, 3 & 4 shall be 24 months from date of SPV transfer & for element at Sl. No. 2b shall be 31.03.2028 in matching timeframe of RE generator (Purvah Green Power Pvt. Ltd.: 297 MW)	- Creation of New 220 kV Bus Section (3rd) with 220 kV Bus Sectionaliser and 400/220 kV, 3x500 MVA ICT augmentation (7th, 8th & 9th) at Pachora PS terminated on 220 kV Bus Section (3rd) 3 Nos. 220 kV line bays for RE Interconnection on Bus Section (3rd) 1 Nos. 220 kV line bay for RE Interconnection of Purvah Green Power Pvt. Ltd. on Bus Section (3rd) - Pachora PS – Rajgarh(PG) 400 kV D/c line (Quad ACSR/ AAAC/ AL59 Moose equivalent) along with associated line bays at both ends and 50 MVAR Switchable Line Reactors (Sw LR) on each ckt at both ends (180 Km) - Installation of 1x125 MVAR, 420 kV bus reactor at Pachora PS (400 kV Bus Section-II)	RECPDCL
2.	Transmission system for Evacuation of Power from RE Projects in Neemuch (1000 MW) SEZ in Madhya Pradesh-Phase II	- Creation of New 220 kV Bus Section-II at Neemuch PS with Augmentation of transformation capacity by 3x500 MVA, 400/220 kV ICTs (3rd, 4th & 5th) at Neemuch S/s along with associated bays. 4 Nos. 220 kV Line bays at Neemuch	PFCCCL

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	Implementation timeframe: 24 months from date of SPV transfer	PS for RE interconnection 3. Neemuch PS – Pachora PS 400 kV 400 kV D/c line (Quad ACSR/ AAAC/ AL59 Moose equivalent) along with associated line bays and 50 MVar Switchable Line Reactors (Sw LR) on each ckt at both ends (190 km) 4. Establishment of 2x500 MVA, 400/220 kV S/s at Handiya with 2x125MVar 420 kV Bus Reactors Future provision (space for): • 400 kV line bays along with switchable line reactors– 6 Nos. (Sec-II) • 400/220 kV ICT along with bays - 4 Nos. (1 Nos. on Sec-I & 3 Nos. on Sec-II) • 400 kV Bus Reactor along with bays: 2 Nos. (Sec-II) • 220 kV line bays: 8 Nos. (on Sec-II) • 400 kV Sectionalization bay: 1 set • 220 kV Sectionalization bay: 1 set • 220 kV TBC & BC: 1 Nos. 5. Pachora PS – Handiya 400 kV 400 kV D/c line (Quad ACSR/ AAAC/ AL59 Moose equivalent) along with associated bays at Pachora PS end and 50 MVar Switchable Line Reactors (Sw LR) on each ckt at both ends (190 km) 6. LILO of Khandwa(PG) – Itarsi(PG) 400 kV D/c (Twin Moose) line at Handiya S/s (22 km) 7. Installation of 1x125 MVAR, 420 kV bus reactor (2nd) at Neemuch PS <i>Note: TSP of Neemuch & Pachora PS shall provide space for above scope of work</i>	
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(Detailed scope as approved by 24th NCT and subsequent amendments thereof)

IV. ISTS communication schemes approved by NCT:

Minutes of the 24th meeting of National Committee on Transmission (NCT)

Sl. No.	Name of Transmission Scheme	Implementation Mode	Implementation timeframe	Estimated Cost (₹)
1	Supply and Installation of additional FOTE and Ethernet cards at AGC & Critical Nodes of SR Region	RTM through POWERGRID	12 months from the date of allocation	1.02 Cr
2	Requirement of Additional FOTE at various ISTS nodes in ER due to exhaustion of existing capacity	RTM through POWERGRID	12 months from date of allocation	9.78 Cr
3	Deployment of FOTE (SDH Equipment) and amplifier solutions at Alipurduar S/s end for OPGW based communication and Teleprotection for 400 kV lines from PHEP-II, PHEP-I and Jigmeling of Bhutan to Alipurduar, India	RTM through POWERGRID	6 months from date of allocation	65 Lakhs

(Detailed scope as approved by 24th NCT and subsequent amendments thereof)

Annexure-I**List of participants of the 24th meeting of NCT****CEA:**

1. Sh. Ghanshyam Prasad, Chairperson, CEA & Chairman, NCT
2. Sh. Ajay Talegaonkar, Member (E&C)
3. Sh. A.K. Rajput, Member (Power Systems)
4. Sh. Ishan Sharan, Chief Engineer (PSPA-I)
5. Sh. Y.K. Swarnkar, Chief Engineer (PSPM)
6. Sh. B.S. Bairwa, Chief Engineer (I/C) (PSPA-II)
7. Sh. Rahul Raj, Director (PSPA-II)
8. Sh. B.S. Meena, Director (PSPM)
9. Sh. Pranay Garg, Deputy Director (PSPA-II)
10. Sh. Manish Maurya, Deputy Director (PSPA-II)
11. Sh. Manish Kumar Verma, Assistant Director (PSPA-II)

MoP:

1. Sh. Om Kant Shukla, Director (Trans.)

MNRE:

1. Sh. Himanshu Prabhakar, Under Secretary

SECI:

1. Sh. Vineet Kumar, DGM
2. Sh. R.K. Agarwal, Consultant

CTUIL:

1. Sh. P C Garg, COO
2. Sh. Ashok Pal, Deputy COO
3. Sh. K K Sarkar, Sr GM
4. Sh. P.S. Das, Sr GM
5. Sh. Rajesh Kumar, Sr GM
6. Sh. Anil Kumar Meena, GM
7. Sh. Kashish Bhambhani, GM
8. Sh. Bhaskar Wagh, DGM
9. Sh. Pratyush Singh, DGM
10. Sh. Venkatesh Gorli, Chief Manager
11. Sh. Anupam Kumar, Manager

GRID India:

1. Sh. Rajiv Porwal, Director (SO)
2. Sh. Rahul Shukla, Chief Manager
3. Sh. Priyam Jain, Chief Manager
4. Sh. Raj Kishan, Deputy Manager
5. Sh. Gaurab Dash, Deputy Manager

RECPDCL

1. Sh. Satyabhan Sahoo, GM (Tech)

PFCCL

1. Sh. Deepak Kumar, AM

Expert Member

1. Sh. Ravinder Gupta, Ex Chief Engineer, CEA

POWERGRID

1. Sh. Anand Shankar, CGM
2. Sh. Sanjeev Kr. Chaudhary, Sr. GM
3. Sh. YKPN Singh, GM



पावरग्रिड
POWERGRID

ANNEX C.3.3

पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड

(भारत सरकार का उद्यम)

POWER GRID CORPORATION OF INDIA LIMITED

Ref. ER-II/KOL/AM/

Date:16.12.2024

To,
The Chief Engineer, Transmission (O&M)-1
10th Floor, Bidyut Bhaban, Salt Lake
WBSETCL, Kolkata-700091

Kind Attention:- Sh. Tapas Chakraborty, CE (O&M) /TR-1.

Subject:- Lowering of existing OPGW between Span 10-11 of 132 NBU-Lebong & 132 KV Rammam-Lebong TL at Darjeeling (Near Soom Tea Garden Area).

Sir,

In reference to continuous deformation and hill sinking at Soom Tea Garden area near Darjeeling, location of existing ISTS line namely, 132 KV Siliguri-Melli & 132 KV Siliguri-Kurseung are affected heavily. The matter has been discussed in detail with higher ups of WBSETCL/ERLDC/ERPC and for contingency 132 KV Siliguri-Kurseung-II & 132 KV Rangit-Melli formed by passing location 126/127.

132 KV D/C Siliguri-Melli & Rangit-Kurseung TL span 126-127 is the overhead power line crossing of 132 D/C KV NBU-Lebong & 132 KV Rammam-Lebong TL in span Tower No.-10-11 (WBSETCL Line).

Now after affected tower dismantling and subsequent completion of survey (Around 01 meter level difference observed due to hill sinking at Loc-126, reference photograph enclosed), it is found that for reinstatement of the towers at same location, to achieve bottom conductor clearance, existing OPGW of WBSETCL line for subject span, required to be lowered and to be converted to UGFO cabling.

Subject matter also has been discussed in detail at 221st OCC, wherein POWERGRID deliberated the issue with difficulties faced at subject location. OCC opined for putting effort from all corners to return the ISTS connectivity to original configuration at the earliest.

In view of above, it is requested to kindly make necessary arrangements for conversion of existing OPGW for the subject span of Lebong connectivity to UGFO cabling. POWERGRID will extend all necessary help for execution of the work and request WBSETCL also to kindly look into the matter such that original grid connectivity could be established at the earliest.

From POWERGRID site level, Sh. T. Baruah/DGM/9434748299, Siliguri shall be the nodal point for all local level communication, while Sh. Santanu Rudrapal/Ch.Manager/9434735848/ULDC, shall co-ordinate with your communication wing as required/instructed by your good office.

Thanking you.

Yours Faithfully

Partha Ghosh
16.12.2024

(Partha Ghosh)

DGM/RHQ/AM, ERTS-II

CC:-

1. MS/ERPC for kind information.
2. ED/ER-II, CGM/AM/ER-II for kind information.
3. Director (Operation)/WBSETCL for kind information.
4. ED/ERLDC for kind information.

पूर्वी क्षेत्र पारेषण प्रणाली-II, (क्षेत्रीय मुख्यालय)/ Eastern Region Transmission System-II, (Regional Headquarter)

सीफ-17, एकक्शन एरिया- 1 सी, न्यू टाउन, कोलकता-700156, दुरभाष-2324 2840/2850

CF-17, Action Area-1C, New Town, Kolkata-700156, Tel: 2324 2840/2850

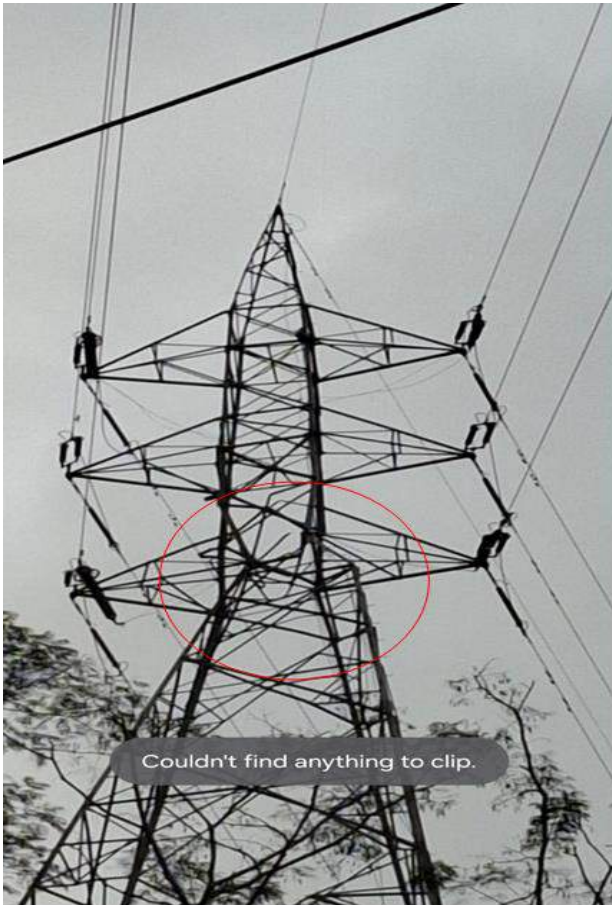
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Corporate Office: "Saudamini", Plot No.-2, Sector-29, Gurugram-122001 (Haryana), Phone No.: 0124-2571700-719

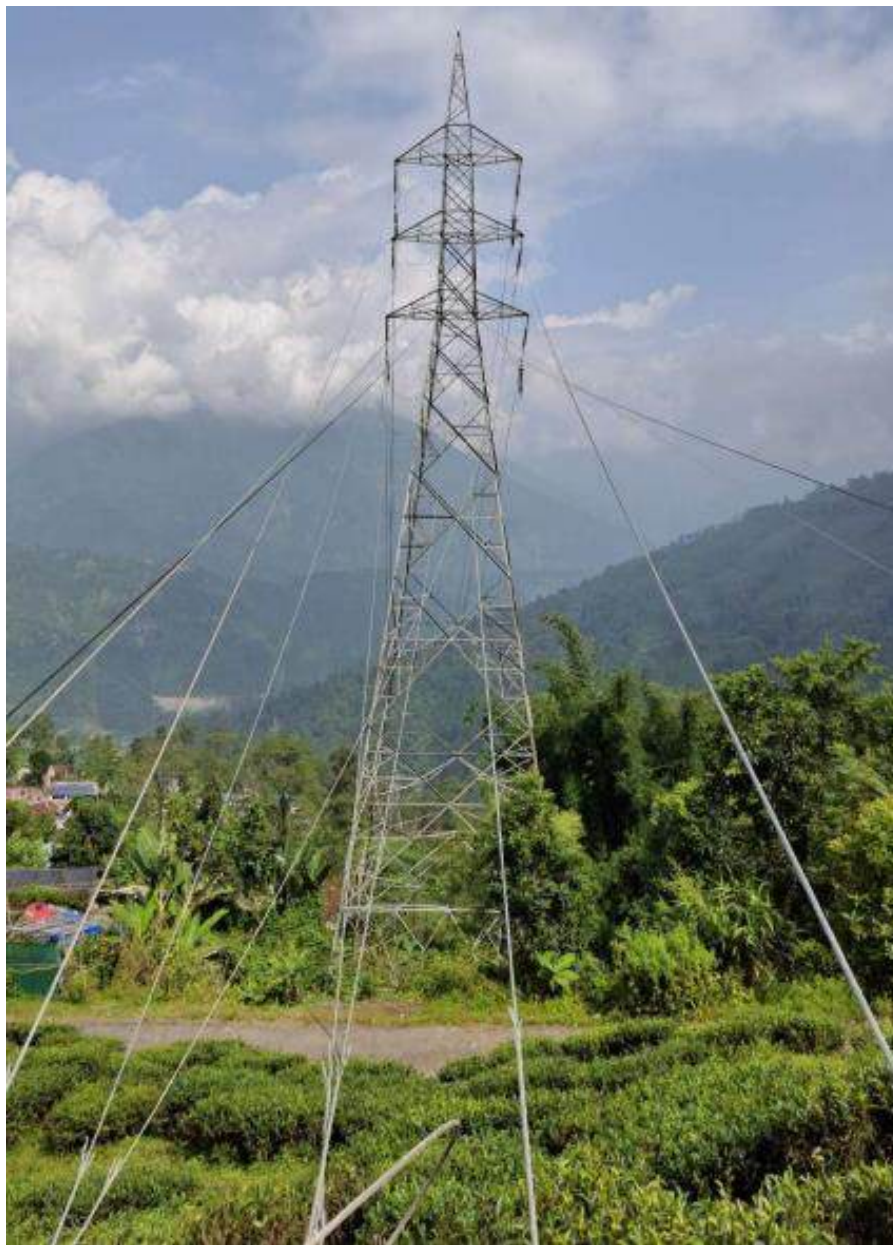
पंजीकृत कार्यालय: बी-9, कुतुब इंस्टीट्यूशनल एरिया, कटवारीया सराय, नई दिल्ली-110016, दुरभाष: 011-26560112, 1211, 26564812, 892 सीआइएन: L40101DL1989GOI038121

Registered Office: B-9, Qutab Institutional Area, Katwaria Sarai, New Delhi-110016, Phone No.: 011-26560112, 1211, 26564812, 892, CIN: L40101DL1989GOI038121

Photographs of Tower No-126 & 127 and nearby areas:



Tower No-127 (DC+25)



Tower No-126 (DC+25)



Hill Shrinking & Landslide near tower 126 & 127

Annexure D.1

Updated Anticipated Peak Demand (in MW) of ER & its constituents for January 2025

1	BIHAR	Demand (MW)	Energy Requirement (MU)
	NET MAX DEMAND	6244	3329
	NET POWER AVAILABILITY- Own Sources	488	399
	Central Sector+Bi-Lateral	6760	4716
	SURPLUS(+)/DEFICIT(-)	1005	1786
2	JHARKHAND		
	NET MAXIMUM DEMAND	2060	1157
	NET POWER AVAILABILITY- Own Source	394	208
	Central Sector+Bi-Lateral+IPP	1352	816
	SURPLUS(+)/DEFICIT(-)	-314	-133
3	DVC		
	NET MAXIMUM DEMAND	3150	2205
	NET POWER AVAILABILITY- Own Source	5500	3327
	Central Sector+MPL	200	165
	Bi- lateral export by DVC	2500	1746
	SURPLUS(+)/DEFICIT(-) AFTER EXPORT	50	-459
4	ODISHA		
	NET MAXIMUM DEMAND (OWN)	4400	2753
	NET MAXIMUM DEMAND (In Case of CPP Drawal of 900 MW(peak) and average drawl of 700 MW)	5300	3274
	NET POWER AVAILABILITY- Own Source	2331	2771
	Central Sector	2072	1427
	SURPLUS(+)/DEFICIT(-) (OWN)	3	1445
	SURPLUS(+)/DEFICIT(-) (I(In Case of CPP Drawal of 900 MW(peak) and average drawlm of 700 MW)	-897	924
5	WEST BENGAL		
	WBSEDCL		
5.1	NET MAXIMUM DEMAND	6285	3651
	NET MAXIMUM DEMAND (Incl. Sikkim)	6288	3655
	NET POWER AVAILABILITY- Own Source (Incl. DPL)	4830	2822
	Central Sector+Bi-lateral+IPP&CPP+TLDP	2634	1502
	EXPORT (To SIKKIM)	5	4
	SURPLUS(+)/DEFICIT(-) AFTER EXPORT	7462	669
5.2	CESC		
	NET MAXIMUM DEMAND	1370	706
	NET POWER AVAILABILITY- Own Source	460	385
	IMPORT FROM HEL	540	258
	TOTAL AVAILABILITY OF CESC	1000	643
	SURPLUS(+)/DEFICIT(-)	-370	-63
	WEST BENGAL (WBSEDCL+CESC+IPCL) (excluding DVC's supply to WBSEDCL's command area)		
	NET MAXIMUM DEMAND	7655	4357
	NET POWER AVAILABILITY- Own Source	7658	3207
	CS SHARE+BILATERAL+IPP/CPP+TLDP+HEL	3175	1760
	SURPLUS(+)/DEFICIT(-) BEFORE WBSEDCL'S EXPORT	3178	610
	SURPLUS(+)/DEFICIT(-) AFTER WBSEDCL'S EXPORT	3174	606
6	SIKKIM		
	NET MAXIMUM DEMAND	135	65
	NET POWER AVAILABILITY- Own Source	44	56
	Central Sector	300	166
	SURPLUS(+)/DEFICIT(-)	209	156
	EASTERN REGION		
	NET MAXIMUM DEMAND	23644	13717
	NET MAXIMUM DEMAND ((In Case of CPP Drawal of 800 MW(peak) and average drawl of 700 MW)	24544	14019
	BILATERAL EXPORT BY DVC (Incl. Bangladesh)	2346	1746
	EXPORT BY WBSEDCL TO SIKKIM	5	4
	EXPORT TO B'DESH & NEPAL OTHER THAN DVC	642	478
	NET TOTAL POWER AVAILABILITY OF ER	26884	17272
	(INCLUDING CS ALLOCATION +BILATERAL+IPP/CPP+HEL)		
	SURPLUS(+)/DEFICIT(-)	247	1326
	SURPLUS(+)/DEFICIT(-) (In Case of CPP Drawal for Odisha)	-653	1024